

# **OEC — Outdoor Emergency Care 6th Edition Full Test Bank**

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822 Questions | 34 Sections

*National Ski Patrol*

*Answer letter is shown right-aligned below each question.*

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## Section 1 — History & Legal

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21 Questions

**1. The course called "Outdoor Emergency Care" was created by the National Ski Patrol to:**

- A) create a standard of care for ski patrollers and others involved in outdoor emergency care.
- B) provide a standard of training for OEC technicians and others involved in outdoor recreation.
- C) respond to a request from Minnie Dole to develop training for patrollers working at ski resorts.
- D) create a worldwide standard of care for anyone providing outdoor emergency care.

*Ans: B*

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**2. The person who first organized a volunteer "ski patrol" and then was asked to organize a national patrol was:**

- A) Roger Langely.
- B) Roland Palmedo.
- C) Minnie Dole.
- D) Warren Bowman.

*Ans: C*

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**3. The 10th Mountain Division was founded:**

- A) through a request from the U.S. president to Minnie Dole.
- B) after Minnie Dole convinced the U.S. Army that a winter warfare unit would be valuable.
- C) by members of the 87th Mountain Infantry Regiment, many of whom were skiers.
- D) through the Norwegian Army, which had a coldweather unit during World War I.

*Ans: B*

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**4. The first emergency responder textbook created exclusively for ski patrollers was titled:**

- A) Winter Emergency Care.
- B) Outdoor Emergency Care.
- C) Ski Safety and First Aid.
- D) Safety on the Snow.

*Ans: A*

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**5. Dr. Warren Bowman is considered the father of Outdoor Emergency Care because he:**

- A) was the first program director for the National Ski Patrol.
- B) co-authored the first ski patrol manual with the American Red Cross.
- C) founded the National Ski Patrol as a separate entity.
- D) created the concept of "intermediate" medical care for ski patrollers.

*Ans: D*

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**6. The 6th Edition of Outdoor Emergency Care:**

- A) must be learned in its entirety through the book and online learning before taking the OEC technician evaluation.
- B) contains the baseline knowledge and skills identified by the U.S. Department of Transportation curriculum for EMTs in all settings.
- C) contains the knowledge and skills identified by the NHSTA for Emergency Medical Responder training plus additional information and training for patrollers.
- D) can only be taught in the winter so that students can practice the skills taught in the course.

*Ans: C*

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**7. In order to maintain OEC certification, you must:**

- A) complete the assigned refresher course each year that covers one-third of the curriculum every year during the three year refresher cycle.
- B) complete a refresher course that covers the key objectives of the entire curriculum every year.
- C) complete a refresher course that covers the key objectives of the entire curriculum every three years.
- D) demonstrate CPR and AED skills every two years.

*Ans: A*

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**8. Good Samaritan laws generally:**

- A) protect a ski patroller from being sued in every circumstance.
- B) may apply to people who do not receive compensation for helping people.
- C) are the same in every state.
- D) provide enough protection so that you need not to know whether your mountain has insurance.

*Ans: B*

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**9. Although regulations may differ from state to state, the general laws that may protect OEC technicians from liability while providing emergency care to a patient are known as:**

- A) emergency rescue protection laws.
- B) Good Samaritan laws.
- C) volunteer rescue liability laws.
- D) first rescuer liability laws.

*Ans: B*

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**10. You have just finished your shift, and while driving home still wearing your patrol jacket, you come upon a car collision in which people appear to be injured. You recognize that if you approach the collision wearing your patrol jacket, members of the public may have a reasonable expectation that you will provide care. This expectation is known as:**

- A) the absence of negligence.
- B) professional ethics.
- C) moral obligation.
- D) the doctrine of public reliance.

*Ans: D*

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**11. While you are alone and caring for an injured skier in the aid room, a fellow patroller calls and tells you that your own young child was just injured in the terrain park. Concerned for your child, you leave the patient's care unfinished and run to the terrain park. Your action could be viewed as:**

- A) a breach of ethics.
- B) your duty to act.
- C) denial of care.
- D) abandonment.

*Ans: D*

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**12. The harming of an individual by not performing up to the technician's standard of training is called:**

- A) negligence.
- B) assault.
- C) gross negligence.
- D) standard of care breach.

*Ans: A*

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**13. A situation in which an OEC technician leaves a patient before the patient's care is transferred to another qualified individual is considered:**

- A) negligence.
- B) breach of duty.
- C) abandonment.
- D) assault.

*Ans: C*

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**14. You are skiing down your favorite slope and notice a small group gathered around a middleaged man sitting on the ground. You note that the boarder is holding his wrist and appears in pain. He agrees to have you examine him but then refuses any more help. You explain the possible consequences of not splinting the wrist and of the boarder trying to get down the mountain without help. This explanation of the risk of refusing care is provided in order to:**

- A) demonstrate that you know how to treat a wrist injury.
- B) diminish your risk of being accused of abandonment.
- C) reduce your risk of an accusation of battery.
- D) help the patient understand why you are about to put the splint on anyway.

*Ans: B*

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**15. Which of the following statements concerning "standard of training" is true?**

- A) Acting within the "standard of training" is fundamental to reducing an OEC technician's legal risk.
- B) "Standard of training" is the same thing as "standard of care."
- C) "Standard of training" includes the mountain area's protocols.
- D) "Standard of training" includes your state's laws and procedures.

*Ans: A*

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**16. Which of the following is not considered a form of patient consent?**

- A) Informed consent
- B) Minor consent
- C) Implied consent
- D) Absolute consent

*Ans: D*

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**17. Upon finding an unconscious patient at the scene of a bad accident, your next step should be to:**

- A) assume "implied consent" and begin to examine and treat the patient.
- B) delay your examination and ask other patrollers to respond as witnesses.
- C) delay your examination until you can locate a spouse or a friend of the patient.
- D) begin your assessment but not provide any treatment until another patroller arrives.

*Ans: A*

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**18. The type of consent that is based on an appreciation and understanding of the facts, implications, and possible future consequences of an action is known as:**

- A) implied consent.
- B) educated consent.
- C) expressed consent.
- D) informed consent.

*Ans: D*

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**19. Which of the following descriptions is the best example of expressed consent?**

- A) A 37-year-old man who is found unresponsive
- B) A 52-year-old skier who comes to the aid room and asks for help
- C) A 19-year-old male who appears to be intoxicated and lets you examine him
- D) A man who asks you to help his unresponsive 29-year-old diabetic wife

*Ans: B*

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**20. What type of consent would apply for an unresponsive 28-year-old male victim of a motorcycle collision?**

- A) Expressed consent
- B) Implied consent
- C) Informed consent
- D) Assumed consent

*Ans: B*

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**21. You arrive at the scene of an accident where a 22-year-old female skier has hit a tree and has a large bleeding cut on her head. When you ask her if she needs help, she refuses care and says she just needs to sit for a while. Noticing the amount of bleeding, you calmly express your concern for her, but she continues to say she doesn't want help. You hesitate to touch her because you are concerned that if you do, she could accuse you of:**

- A) assault.
- B) battery.
- C) breach of duty.
- D) duty to act.

*Ans: B*

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## Section 2 — EMS Systems

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### 14 Questions

**1. A network of specially trained personnel, equipment, facilities, and other resources that respond to medical emergencies—regardless of cause, location, or the patient's ability to pay—best describes:**

- A) a community wellness program.
- B) a health care system.
- C) an emergency care system.
- D) a public health system.

*Ans: C*

**2. Which of the following levels of emergency personnel emphasizes the provision of immediate lifesaving treatment and stabilization to critically ill or injured patients while waiting for additional emergency response?**

- A) Emergency medical technician
- B) Emergency medical responder
- C) Advanced emergency medical technician
- D) Critical care responder

*Ans: B*

**3. An OEC technician meets or exceeds the NHTSA national training requirements of:**

- A) an emergency medical technician.
- B) an advanced emergency medical technician.
- C) a paramedic.
- D) an emergency medical responder.

*Ans: D*

**4. Most emergency care systems deploy personnel to emergencies using a stratified approach based on the patient's condition or anticipated needs. Which of the following lists presents the normal order of deployment of responders?**

- A) EMT, AEMT, EMR, paramedic
- B) EMR, AEMT, paramedic, EMT
- C) EMR, EMT, AEMT, paramedic
- D) AEMT, paramedic, EMT, EMR

*Ans: C*

**5. Which of the following is not one of the four nationally recognized prehospital emergency care provider levels?**

- A) Emergency medical technician
- B) Advanced emergency medical technician
- C) Outdoor emergency care technician
- D) Emergency medical responder

*Ans: C*



**6. Many emergency care systems deploy personnel to emergencies using a tiered approach. OEC technicians are most likely deployed in which tier?**

- A) Tier 1
- B) Tier 2
- C) Tier 3
- D) Tier 4

*Ans: A*

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**7. Critical care providers who bring advanced life support equipment, therapies, and interventions to the scene are typically deployed in what tier?**

- A) Tier 1
- B) Tier 2
- C) Tier 3
- D) Tier 4

*Ans: D*

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**8. The seamless delivery of high-quality emergency medical care as patient transitions from initial contact with an EMT through definitive treatment best describes:**

- A) medical direction.
- B) continuity of care.
- C) integration of health services.
- D) online medical control.

*Ans: B*

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**9. Which of the following choices is not an example of direct medical oversight?**

- A) Telephone communication
- B) Physical presence of a physician
- C) Written protocols
- D) Video conferencing

*Ans: C*

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**10. Which of the following characteristics is not a characteristic of indirect medical control?**

- A) Communication by radio
- B) Use for expediting patient care
- C) Common use for "preauthorizing" routine and life-saving procedures
- D) Provision through the use of protocols

*Ans: A*

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**11. Guidelines developed at the local level by emergency personnel familiar with an area's medical needs, available resources, system capabilities, and local standard of care are:**

- A) protocols.
- B) policies.
- C) advanced procedures.
- D) medical attributes.

*Ans: A*

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**12. Written instructions that indicate what should be done in a given situation are best described as:**

- A) regulations.
- B) protocols.
- C) indirect medical oversight.
- D) direct medical oversight.

*Ans: B*

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**13. The reason it is important that OEC technicians participate in quality improvement programs in services in which they work or volunteer is:**

- A) to guarantee that the public receives quality emergency medical care.
- B) to identify changes that, if needed, will ensure the future care a patient receives in is accordance with local protocols and guidelines.
- C) to ensure that individuals making false calls for help are prosecuted.
- D) to identify problem employees and create a corrective action plan for them.

*Ans: B*

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**14. Which of the following methods is the best method for ensuring that the treatment rendered by OEC technicians meets or exceeds customer expectations and national education standards for emergency medical personnel?**

- A) Public education
- B) Prevention training
- C) Continuity of care
- D) Quality improvement

*Ans: D*

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## Section 3 — Rescuer Safety

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12 Questions

**1. Sometimes the most important thing in a rescue situation is to:**

- A) hydrate injured patients.
- B) take care of yourself.
- C) contact local government.
- D) check vehicle fuel levels.

*Ans: B*

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**2. Mental preparedness comes best if you:**

- A) have a negative attitude to prepare for worst cases.
- B) have a neutral attitude to remain calm.
- C) have expectations of difficulties.
- D) have a positive attitude to help deal with difficulties.

*Ans: D*

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**3. During a very strenuous rescue, you may need to eat:**

- A) hourly.
- B) when you feel dizzy.
- C) only when feeling hungry.
- D) with other rescuers.

*Ans: A*

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**4. The best direction of moisture transfer in the base layer is:**

- A) outward.
- B) from head to feet.
- C) from feet to head.
- D) inward.

*Ans: A*

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**5. The best strategy to maintain warmth in the cold winter months is to wear:**

- A) clothing in one layer; multiple-purpose outerwear.
- B) clothing in two layers; a base to retain heat and a water and wind-repelling outer layer.
- C) clothing in three layers; a base to retain heat, a middle insulating layer, and a water and windrepelling outer layer.
- D) clothing primarily made of cotton.

*Ans: C*

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**6. Which of the following SPF ratings would provide the greatest level of protection?**

- A) 6
- B) 15
- C) 30
- D) 50

*Ans: D*

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**7. Some vector-borne illnesses are transmitted to humans via:**

- A) eating partially cooked foods.
- B) deer ticks.
- C) contact with human feces.
- D) sneezing and coughing.

*Ans: B*

---

**8. Wearing clean gloves, an impervious gown, and goggles while cleaning a long-board contaminated with blood helps protect the OEC technician from which form of potential infectious disease transmission?**

- A) Airborne
- B) Indirect contact with an object contaminated with pathogens
- C) Ingestion
- D) Direct contact with a human with infectious pathogens

*Ans: B*

---

**9. Pathogens from a scene can harm rescuers if they:**

- A) invade the body.
- B) pass between victims.
- C) escape the scene.
- D) are UV-durable.

*Ans: A*

---

**10. The practice of protecting yourself from exposure to body fluids based on the assumption that all patients are potentially infectious is referred to as:**

- A) standard precautions.
- B) exposure control.
- C) personal protective equipment.
- D) body fluid precautions.

*Ans: A*

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**11. Which one of the following techniques is one of the most effective techniques that OEC technicians can use to prevent the spread of infection?**

- A) Disinfecting their equipment
- B) Wearing gloves
- C) Washing their hands
- D) Getting immunizations

*Ans: C*

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**12. When caring for a patient who was stabbed with a knife in the ski area's bar (a crime scene situation), it is important for the OEC technician to do all of the following to support "chain of custody" except:**

- A) do not remove, move, or otherwise disturb anything in the environment, except as is absolutely necessary to provide critical care to patient.
- B) wrap the knife found on the floor next to the patient in plastic and hand it to the EMTs so they can show the length of the knife to the Emergency Department staff.
- C) notify law enforcement immediately.
- D) let law enforcement personnel handle any materials that could be used as evidence.

*Ans: B*

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## Section 4 — ICS & MCI

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25 Questions

**1. Which one of the following statements about the Incident Command System is true?**

- A) Each EMS agency can use its own codes and terms.
- B) There is a separate commander for EMS, fire, and law enforcement personnel.
- C) Each individual responder is accountable to two supervisors.
- D) ICS operates with standardized terminology to avoid confusion.

*Ans: D*

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**2. Of the five functional areas within the Incident Command System (ICS) structure, the first to be established should be:**

- A) operations section.
- B) incident command.
- C) planning section.
- D) logistics section.

*Ans: B*

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**3. The primary function of the incident commander is to:**

- A) not expand command to any other agency
- B) provide overall leadership at an incident.
- C) obtain supplies needed to care for all patients.
- D) ensure that radio communications are established.

*Ans: B*

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**4. The incident commander has assigned you to be the operations section chief. In this role, you are responsible for all of the following except:**

- A) reducing immediate hazards.
- B) restoring the scene to normalcy.
- C) participating in the planning process.
- D) determining incident objectives and strategy.

*Ans: D*

---

**5. Which of the following tasks is not a responsibility of the incident commander?**

- A) Triageing the injured patients
- B) Establishing the incident command post
- C) Determining incident objectives and strategies
- D) Providing overall management of the incident

*Ans: A*

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**6. All of the following are functional areas of the incident command system structure except:**

- A) planning section.
- B) operations section.
- C) transportation section.
- D) finance/administration section.

*Ans: C*

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**7. In the incident command system, the operations section is responsible for:**

- A) collecting, assessing, and distributing incident-related data.
- B) executing the strategy of the incident action plan.
- C) providing financial management.
- D) providing support for all functional areas.

*Ans: B*

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**8. A formal, organized method for managing an incident, regardless of its cause, size, scope, or complexity, is called:**

- A) a multi-agency coordination system.
- B) a federal incident management system.
- C) an incident command system.
- D) an emergency strike team.

*Ans: C*

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**9. In the incident command system, which of the following sections is responsible for maintaining all incident reports except for financials?**

- A) Logistics section
- B) Operations section
- C) Records section
- D) Planning section

*Ans: D*

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**10. The person responsible for maintaining equipment and ensuring that facilities meet specified needs is the:**

- A) operations chief.
- B) logistics chief.
- C) maintenance chief.
- D) planning chief.

*Ans: B*

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**11. In the incident command system, the section responsible for maintaining injury, death, and damage documentation as well as maintaining reimbursement records is the:**

- A) planning section.
- B) operations section.
- C) logistics section.
- D) finance/administration section.

*Ans: D*

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**12. Which one of the following triage tag indicates the lowest priority for care, "expectant," for a patient at a multiple-casualty incident using the ID-ME triage categories?**

- A) A yellow tag
- B) A red tag
- C) A black/gray tag
- D) A green tag

*Ans: C*

---

**13. You are transporting a patient at a multiplecasualty incident. The patient has a yellow triage tag. This tag means:**

- A) treatment of the patient could be delayed.
- B) the patient has no injuries.
- C) the patient should be with the "walking wounded."
- D) the patient has a head injury with altered mental status.

*Ans: A*

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**14. The most widely used triage categorization system in the world is "ID-ME." This acronym represents the four specific triage categories used by NATO forces, National Disaster Life Support, public safety agencies, and search and rescue groups. The ID-ME triage categories are:**

- A) immediate, delayed, maximum, and exceptional.
- B) immediate, delayed, minimal, and expectant.
- C) immediate, detain, minimal, and expectant.
- D) immediate, delayed, minimal, and exceptional.

*Ans: B*

---

**15. At a multiple-casualty incident that is using the STARTTriage method, you are brought a patient with a red tag tied to his wrist. You should recognize which one of the following?**

- A) Immediate care and transport of the patient are necessary.
- B) Delayed care and transport of the patient are permissible.
- C) The patient has a minor wrist injury.
- D) The patient is deceased.

*Ans: A*

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**16. You are at an outside skiing exhibition when a spectator bleacher collapses. Several injured people are walking around with various injuries. Which of the following instructions is most appropriate for these ambulatory patients?**

- A) "Walk to the area with the medical providers."
- B) "Sit down here so I can do a quick assessment on you."
- C) "Put this yellow tag on your wrist and go to the ticket booth at the bottom of the slope."
- D) "Leave the ski area and go to the hospital. Others are hurt more seriously than you."

*Ans: A*

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**17. There is a fight involving approximately 20 bar patrons. Weapons were used, and there are varying degrees and types of injuries. The first person on scene has started assessing a person lying on the floor with blood covering his shirt. Quick assessment reveals him to be breathing at 24 times per minute. Which of the following actions should be done next using the START triage method?**

- A) Assist ventilation with a BVM.
- B) Check for a radial pulse or capillary refill.
- C) Attend to the next patient.
- D) Place a yellow tag on his wrist.

*Ans: B*

---

**18. At the scene of a multiple-casualty incident, you are presented with a patient who is not breathing. According to the Simple Triage and Rapid Transport (START) system, which of the following actions should you take next?**

- A) Place a red tag on the patient.
- B) Start assisting ventilation with a BVM.
- C) Open the airway.
- D) Place a black tag on the patient.

*Ans: C*

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**19. You are triaging using the Simple Triage and Rapid Transport (START) system, and you find a patient who is breathing 18 times per minute and has a radial pulse. Which one of the following actions should you take next?**

- A) Check to see if the patient can follow simple commands.
- B) Check the patient's blood pressure.
- C) Move the patient to the treatment area.
- D) Place a yellow tag on the patient.

*Ans: A*

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**20. In using the Simple Triage and Rapid Transport (START) system of triage, which one of the following patients should have a yellow tag applied before being moved to the treatment area?**

- A) A male with a respiratory rate of 8, no radial pulse, and unresponsive mental status
- B) A female with a leg fracture who has a respiratory rate of 22, and a palpable radial pulse and who squeezes your fingers when instructed to do so
- C) A female with gasping respirations, no radial pulse, and unresponsive mental status
- D) A male with a respiratory rate of 40 and a palpable radial pulse who squeezes your fingers when instructed to do so

*Ans: B*

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**21. A teen involved in a multiple-casualty incident is found lying on the ground with an obvious deformity to his left thigh. Using the START system for triage, you note that he is not breathing. Which one of the following should you do next?**

- A) Tag the patient as black; then move on to the next patient.
- B) Provide the patient 30 seconds of assisted ventilation.
- C) Open the patient's airway; next check for a radial pulse.
- D) Give the patient a red tag; next move him to the treatment area.

*Ans: C*

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**22. While in charge of triage at a scene involving 30 patients exposed to carbon monoxide, you find a young female who is not breathing. After you open her airway and find that breathing does not return, you should:**

- A) provide her 15 seconds of assisted ventilation.
- B) place a black tag on her and move on to the next patient.
- C) provide her 15 seconds of CPR and recheck for a radial pulse.
- D) apply oxygen and check her mental status.

*Ans: B*

---

**23. When you are placed in charge of the treatment unit at the scene of a multiple-casualty incident, which of the following four patients will you treat first?**

- A) A confused 69-year-old male with a respiratory rate of 40 and a rapid and weak radial pulse
- B) An elderly female patient with a femur fracture and a yellow tag on her wrist
- C) A 21-year-old male who goes into cardiac arrest
- D) A 13-year-old boy who walked to the treatment area complaining of a severe headache

*Ans: A*

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**24. In the acronym "START," the letter "S" stands for:**

- A) Selective.
- B) Safe.
- C) Simple.
- D) Sophisticated.

*Ans: C*

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**25. When properly trained in the START system, an OEC technician should be able to complete a patient assessment in less than:**

- A) 90 seconds.
- B) 45 seconds.
- C) 60 seconds.
- D) 30 seconds.

*Ans: D*

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## Section 5 — Patient Movement

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19 Questions

**1. To lessen one's chances of injury while lifting and moving a patient, one should:**

- A) roll the patient.
- B) combine good lifting and moving techniques.
- C) use a long-axis drag to avoid lifting.
- D) avoid manual moves and instead use mechanical devices.

*Ans: B*

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**2. To maintain proper alignment when moving or lifting a heavy object, the weight of the object must be:**

- A) offset using shoulder and chest strength.
- B) distributed to the lower back.
- C) evenly transferred to the legs.
- D) transferred to the forearms and biceps.

*Ans: C*

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**3. When a person exhibits good body mechanics and lifts properly, the spine is:**

- A) slightly bent in the upper thoracic region.
- B) straight at the thoracic region and bent in the lumbar region.
- C) bent at no more than 15 degrees.
- D) straight and in an aligned position.

*Ans: D*

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**4. Extremity lifts may not be tolerated by:**

- A) obese and young patients.
- B) elderly patients and patients with respiratory disorders.
- C) patients with visual impairments.
- D) patients with ankle injuries.

*Ans: B*

---

**5. Which of the following instructions is most important for the use of good body mechanics in a power lift?**

- A) Maintain a wide stance (at least 24 inches).
- B) Keep your back muscles loose and your head bent forward.
- C) Straighten your legs to lift.
- D) Squat down and keep your knees outward at a 45-degree angle.

*Ans: C*

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**6. Like a direct ground lift, a \_\_\_\_\_ lift is used to raise patients who are lying on their back.**

- A) LEAN
- B) BEAN
- C) MEAN
- D) SEAN

*Ans: B*

---

**7. With a BEAN lift, the patient is:**

- A) assisted to a standing position.
- B) rolled onto a long board.
- C) assisted into a chair.
- D) raised for transfer onto a backboard.

*Ans: D*

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**8. An urgent move is required when:**

- A) a patient is in severe pain.
- B) the rescuer and the patient must move to a safer location.
- C) a crowd of people has formed around you and the patient.
- D) the patient complains of being cold on a 10- degree day.

*Ans: B*

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**9. To use an urgent move called the shoulder drag, you should:**

- A) grab one arm at the shoulder and drag the patient to a safe position.
- B) stand behind the patient and grab the patient's belt while their shoulders are against your chest.
- C) stand upright behind the patient, bend over, and then grab the patient under the armpits and drag.
- D) grab the patient's clothing at the shoulders, support the head, and then bend your knees and drag the patient.

*Ans: D*

---

**10. One of the most common nonurgent moves for OEC technicians to use when alone is the:**

- A) human crutch.
- B) chair carry.
- C) fore and aft carry.
- D) back carry.

*Ans: A*

---

**11. A basket stretcher is a good method for transporting a patient:**

- A) out of the backcountry.
- B) on an LSB in any terrain.
- C) with a fractured femur.
- D) who is extremely heavy.

*Ans: A*

---

**12. The first step in placing a patient in the toboggan for transport is:**

- A) deciding how to get the patient and equipment off the hill.
- B) deciding whether or not to put the patient's injury uphill.
- C) deciding the fastest route of transport.
- D) deciding who will be in the handles of the toboggan during transport.

*Ans: B*

---

**13. The uphill positioning principle has exceptions. For patients with breathing difficulty, the patient should be placed either seated or uphill depending on other injuries. The uphill concept allows easier breathing because:**

- A) the ribs have greater flexibility of movement in the uphill position.
- B) air flows more easily through a patient's nose compared to a head-downhill position.
- C) it calms patients by enabling them to see where they are going during transport.
- D) of less pressure from the abdominal contents pushing upwards on the thorax.

*Ans: D*

---

**14. A woman who is six months pregnant and experiencing back pain should be placed on a backboard and then placed in a toboggan in which of the following positions?**

- A) In a supine position
- B) On her left side
- C) Sitting up
- D) With her head downhill

*Ans: B*

---

**15. An appropriate landing zone (LZ) for safely landing a helicopter is an open area that is approximately:**

- A) 75 feet x 100 feet.
- B) 100 feet x 100 feet.
- C) 100 feet x 200 feet.
- D) 200 feet x 200 feet.

*Ans: B*

---

**16. During helicopter landing or takeoff, most protocols only allow the \_\_\_\_\_ to be in the LZ.**

- A) OEC technician and the patient
- B) landing coordinator
- C) patrol director
- D) risk manager

*Ans: B*

---

**17. After a landing helicopter is safely on the ground, one may approach the aircraft only when signaled by the:**

- A) pilot or flight crew chief.
- B) landing coordinator.
- C) fire department chief.
- D) senior police officer on the scene.

*Ans: A*

---

**18. When approaching most helicopters, whether running or not, the safest way is to do so from the:**

- A) front.
- B) back.
- C) left side.
- D) right side.

*Ans: A*

---

**19. Which of the following instructions for approaching a running rescue helicopter is most dangerous?**

- A) Remain in an upright position and approach from the rear.
- B) Remain low.
- C) Remove any loose clothing.
- D) Remain in sight of the pilot at all times.

*Ans: A*

---

## Section 6 — Anatomy

---

51 Questions

**1. The study of human and animal structures at the gross and microscopic levels is known as:**

- A) physiology.
- B) homeostasis.
- C) biology.
- D) anatomy.

*Ans: D*

---

**2. The study of mechanical, physical, and biochemical functions of humans is known as:**

- A) endocrinology.
- B) biology.
- C) anatomy.
- D) physiology.

*Ans: D*

---

**3. Which one of the following anatomical statements is true?**

- A) The hand is proximal to the elbow.
- B) The shoulder is distal to the hand.
- C) The hip is distal to the knee.
- D) The knee is proximal to the ankle.

*Ans: D*

---

**4. After examining his patient, your partner tells you that he believes the patient has hyperflexed the knee. What has happened if a patient has hyperflexed a knee?**

- A) The patient has straightened the knee beyond its normal range of movement.
- B) The patient has bent the knee backward beyond its normal range of movement.
- C) The patient has pushed the kneecap too far to the right or left.
- D) The patient has straightened the knee into a normal position.

*Ans: B*

---

**5. You arrive at an accident scene and find a patient lying on the ground with her right arm straight out to the side (at a 90-degree angle to her body). You ask her if she is able to pull the arm toward her body, but she cannot. You would document this as an inability to \_\_\_\_\_ her right arm.**

- A) adduct
- B) abduct
- C) hyperflex
- D) hyperextend

*Ans: A*

---



**6. The term used to describe the action of taking an extremity away from the midline of the body is:**

- A) abduction.
- B) flexion.
- C) adduction.
- D) extension.

*Ans: A*

---

**7. The wrist is:**

- A) proximal to the elbow.
- B) lateral to the elbow.
- C) distal to the elbow.
- D) medial to the elbow.

*Ans: C*

---

**8. You arrive at the scene where a patient has a possible fracture of the left hip. You note that she is lying on her back and her left leg is turned outward. You would document this position as:**

- A) supine with no rotation of the left lower extremity.
- B) lateral recumbent with dorsal rotation of the left leg.
- C) supine with external rotation of the left lower extremity.
- D) supine with pronation of the left leg.

*Ans: C*

---

**9. You apply and use an AED on a patient lying on his back. His pulse returns, and he is breathing adequately. What position is he in?**

- A) Right lateral recumbency
- B) Supine
- C) Left lateral recumbency
- D) Prone position

*Ans: B*

---

**10. Which one of the following descriptions best describes the normal anatomic position?**

- A) Supine with the arms at the sides and the palms facing downward
- B) Standing with the arms down at the sides and the palms facing forward
- C) Prone with the arms upward and the palms facing upward
- D) Upright and facing away from you, with the arms raised and the legs straight

*Ans: B*

---

**11. A man lying in the prone position is:**

- A) lying on his left side, knees slightly bent.
- B) lying face down.
- C) lying face up on his back.
- D) lying on his back with his legs elevated higher than his head.

*Ans: B*

---

**12. You are assisting in the care of an elderly patient complaining of shortness of breath. You place the patient on 15L of oxygen via a nonrebreather mask and place him in a high Fowler's position. This means that you will place this patient on his:**

- A) left side, with his left arm over his head.
- B) back, with his upper body elevated at 90 degrees from the waist.
- C) back, with his head down and his legs elevated higher than the head.
- D) back, with his upper body elevated at 45 degrees from the waist.

*Ans: B*

---

**13. Blockage of which of the following prevent air flow into and out of the lungs?**

- A) The right clavicle
- B) The nasopharynx
- C) The trachea
- D) The esophagus

*Ans: C*

---

**14. Which of the following statements best describes the function of the respiratory system?**

- A) Transporting oxygen throughout the body
- B) Providing nutrients to cells
- C) Releasing oxygen from the body
- D) Providing the body oxygen

*Ans: D*

---

**15. The structure containing the vocal cords is the:**

- A) pharynx.
- B) larynx.
- C) trachea.
- D) sternum.

*Ans: B*

---

**16. The leaf- shaped flap that helps prevent food from entering the lower respiratory system is called:**

- A) the epiglottis.
- B) the trachea.
- C) the pharynx.
- D) a bronchiole.

*Ans: A*

---

**17. Which one of the following lists represents a correct sequence for the passage of air into the lungs?**

- A) Nose, bronchi, larynx, trachea, pharynx
- B) Larynx, esophagus, trachea, bronchi, alveoli
- C) Epiglottis, trachea, bronchi, alveoli, nasopharynx
- D) Mouth, pharynx, trachea, bronchi, alveoli

*Ans: D*

---

**18. Under normal circumstances, carbon dioxide is excreted from the body by what system?**

- A) The intestinal tract
- B) The urinary tract
- C) The respiratory system
- D) The integumentary system

*Ans: C*

---

**19. Which of the following systems is not a body system?**

- A) The endocrine system
- B) The thoracic system
- C) The skeletal system
- D) The respiratory system

*Ans: B*

---

**20. The diaphragm separates the thoracic cavity from the:**

- A) pelvic cavity.
- B) spinal cavity.
- C) abdominal cavity.
- D) heart.

*Ans: C*

---

**21. When the diaphragm and intercostal muscles relax, which one of the following occurs?**

- A) Diastolic pressure increase
- B) Systolic pressure increases
- C) Inspiration
- D) Exhalation

*Ans: D*

---

**22. Which one of the following events causes an individual to take a breath?**

- A) The intercostal muscles relax.
- B) The chest cavity decreases in size.
- C) The diaphragm contracts.
- D) Pressure in the chest increases.

*Ans: C*

---

**23. The primary function of the heart is:**

- A) transferring oxygen to the cells.
- B) pumping blood throughout the body.
- C) oxygenating blood in the lungs.
- D) maintaining the volume of blood.

*Ans: B*

---

**24. Which of the following structures brings oxygendepleted blood to the right atrium?**

- A) The right ventricle
- B) The pulmonary artery
- C) The venae cavae
- D) The aorta

*Ans: C*

---

**25. Which chamber of the heart is responsible for pumping blood to the lungs?**

- A) The left atrium
- B) The left ventricle
- C) The right ventricle
- D) The right atrium

*Ans: C*

---

**26. Following chemotherapy, a low white cell count increases a patient's risk for which of the following?**

- A) Infection
- B) Bleeding
- C) High blood pressure
- D) Seizures

*Ans: A*

---

**27. The liquid that blood cells and nutrients are suspended in is called:**

- A) platelet fluid.
- B) hemoglobin.
- C) plasma.
- D) lymph.

*Ans: C*

---

**28. The left atrium:**

- A) receives blood from the veins of the body.
- B) receives blood from the pulmonary veins.
- C) pumps blood to the lungs.
- D) pumps blood to the body.

*Ans: B*

---

**29. The nervous system is subdivided into which two main parts?**

- A) The brain and spinal cord
- B) The spinal cord and peripheral nerves
- C) The brain and extremity nerves
- D) The central and peripheral nervous system

*Ans: D*

---

**30. A patient has suffered damage to his brainstem. As a result of this injury, which of the following signs might you expect to see?**

- A) Sweating
- B) Abnormal respirations
- C) Pinpoint pupils
- D) Muscle spasms

*Ans: B*

---

**31. The pressure exerted on the inside walls of arteries when the left ventricle contracts is called:**

- A) systolic pressure.
- B) contraction pressure.
- C) diastolic pressure.
- D) ventricle pressure.

*Ans: A*

---

**32. Cerebrospinal fluid provides three beneficial properties. Which of the following is one of those properties?**

- A) Conducts electrical impulses
- B) Provides cooling to the cerebellum
- C) Helps regulate the endocrine system
- D) Helps cushion the brain

*Ans: D*

---

**33. As a patient gives her medical history to you, which of the following would you relate to the endocrine system?**

- A) Removal of the gallbladder
- B) Failure of the kidneys
- C) Removal of the thyroid
- D) Heart failure

*Ans: C*

---

**34. The thyroid glands, adrenal glands, pituitary glands, and gonads are part of the:**

- A) nervous system.
- B) endocrine system.
- C) hormonal system.
- D) respiratory system.

*Ans: B*

---

**35. A patient has suffered a burn to the skin. Based on the functions of the skin, to which of the following conditions is the patient most susceptible?**

- A) Infection
- B) Fluid overload
- C) Heart problems
- D) Endocrine disorders

*Ans: A*

---

**36. Which of the following statements regarding the integumentary system is true?**

- A) The skin protects the body against bacteria and other pathogens.
- B) The skin contains four layers: the epidermal, dermal, subcutaneous, and nerve layers.
- C) The epidermis is the thickest and most important layer of the skin.
- D) The dermal layer is the outermost layer of the skin and contains sensory nerves.

*Ans: A*

---

**37. Which of the following is the largest organ of the body?**

- A) The heart
- B) The liver
- C) The skin
- D) The spleen

*Ans: C*

---

**38. What body system provides support and structure to the body?**

- A) The endocrine system
- B) The nervous system
- C) The skeletal system
- D) The exocrine system

*Ans: C*

---

**39. The terms occipital, frontal, and parietal refer to what part of the body?**

- A) The chest
- B) The cranium
- C) The pelvis
- D) The spine

*Ans: B*

---

**40. A new patroller tells you that he injured a tendon above his patella three years ago. You recognize that this injury involves a structure that:**

- A) connects muscles to the patella.
- B) connects the cartilage in the knee.
- C) attaches the patella to the femur.
- D) attaches a ligament to the knee.

*Ans: A*

---

**41. Which of the following groups of bones could be involved in a patient with a "broken leg"?**

- A) Acetabulum, calcaneus, carpals
- B) Femur, tibia, fibula
- C) Orbit, maxillae, mandible
- D) Radius, ulna, humerus

*Ans: B*

---

**42. The lower jaw is also called the:**

- A) mandible.
- B) zygoma.
- C) maxilla.
- D) mastoid.

*Ans: A*

---

**43. The bones of the upper extremities include the:**

- A) humerus and radius.
- B) humerus and calcaneus.
- C) phalanges and tibia.
- D) radius, ulna, and tarsals.

*Ans: A*

---

**44. Which of the following lists identifies the regions of the spinal column from superior to inferior?**

- A) Cervical, lumbar, thoracic, sacral, and coccyx
- B) Coccyx, lumbar, thoracic, cervical, and sacral
- C) Thoracic, lumbar, cervical, coccyx, and sacral
- D) Cervical, thoracic, lumbar, sacral, and coccyx

*Ans: D*

---

**45. Which of the following is a function of the skeletal system?**

- A) Producing blood cells
- B) Securing the abdominal organs in place
- C) Providing structure to blood vessels
- D) Preventing the body from infection

*Ans: A*

---

**46. A patient experiencing thoracic spine pain has pain in his:**

- A) neck.
- B) upper back.
- C) lower back.
- D) tailbone.

*Ans: B*

---

**47. You are working on a patient who has skied into a tree and are told that there is a frontal hematoma. From that, you expect to find the injury where?**

- A) The left arm
- B) The thigh
- C) The skull
- D) The chest

*Ans: C*

---

**48. A patellar fracture affects which part of the body?**

- A) The tibia
- B) The hip
- C) The elbow
- D) The knee

*Ans: D*

---

**49. A female patient suffers from a muscular disease and cannot walk. Based on this fact, you should recognize which of the following muscle types is affected?**

- A) Skeletal muscle
- B) Involuntary muscle
- C) Cardiac muscle
- D) Smooth muscle

*Ans: A*

---

**50. The pelvic cavity contains which of the following organs?**

- A) The bladder and the kidneys
- B) The kidneys and the spleen
- C) The pancreas and the bladder
- D) The rectum and the reproductive organs

*Ans: D*

---



**51. The spleen is part of which body system?**

- A) The cardiovascular system
- B) The endocrine system
- C) The gastrointestinal system
- D) The lymphatic system

*Ans: D*

---

## Section 7 — Assessment

---

44 Questions

**1. Which of the following lists presents the order in which an OEC technician should assess a patient in a field setting?**

- A) Primary assessment, reassessment, secondary assessment
- B) Scene size-up, secondary assessment, primary assessment
- C) Scene size-up, primary assessment, secondary assessment
- D) Scene size-up, history, secondary assessment, primary assessment

*Ans: C*

**2. Which of the following statements would indicate that an OEC technician is ready to begin the first phase of a patient assessment?**

- A) "I have placed an oral airway in the patient."
- B) "The scene is safe and appears to be free of hazards."
- C) "Blood pressure is 124/80 mmHg."
- D) "Can you tell me why you called for help?"

*Ans: B*

**3. Which of the following things does not indicate a possible safety issue during a scene size-up?**

- A) Downed wires
- B) Ice or mud
- C) Wild animals
- D) ABCDs

*Ans: D*

**4. You have been dispatched to an 89-year-old female with an unspecified complaint. When assessing this patient, which one of the following will the OEC technician do as quickly as possible to develop a better understanding of the emergency?**

- A) Obtain the patient's vital signs and current medications
- B) Gather a medical history, including information on allergies
- C) Contact medical direction for advice
- D) Determine if the chief complaint is medical (NOI) or trauma related (MOI)

*Ans: D*

**5. You are interviewing a prospective candidate for the position of OEC technician with your service. During the interview, you ask the applicant to describe the purpose of the primary assessment. Which of the following best describes that purpose?**

- A) To establish a chief complaint and gather an extensive medical history
- B) To determine if the patient's vital signs are stable
- C) To identify and treat life-threatening conditions
- D) To perform a comprehensive exam focused on the patient's chief complaint

*Ans: C*

**6. You have been dispatched for an elderly male complaining of shortness of breath. When should you start the process of forming a general impression about this patient?**

- A) While you and your partner are approaching him
- B) After completing a primary assessment
- C) As soon as you obtain his chief complaint
- D) After you have taken his vital signs

*Ans: A*

---

**7. Forming a general impression is done before which phase of patient assessment?**

- A) Reassessment
- B) Secondary assessment
- C) Primary assessment
- D) Communication and documentation

*Ans: C*

---

**8. You arrive at the scene of a fall, where a 42-yearold woman fell backward off a stepladder while cleaning windows. She is lying on the ground, complaining of pain to her ankle. She tells you, "If I had just been more careful and moved the ladder instead of reaching, this never would have happened!" Based on this information, which of the following can you conclude?**

- A) She has an open airway, is breathing, and has circulating blood to her brain.
- B) Her pulse rate is within normal limits.
- C) She does not require rapid transport.
- D) She does not have any other injuries.

*Ans: A*

---

**9. A 36-year-old patient on the AVPU scale who has overdosed on an unknown drug is breathing and will not open his eyes, even when his name is loudly called. Which one of the following should the OEC technician do next?**

- A) Check the patient's vital signs
- B) Perform a shoulder pinch
- C) Assess the patient for a radial pulse
- D) Identify what drug the patient took

*Ans: B*

---

**10. The "P" in "AVPU" stands for what?**

- A) Potential
- B) Pain
- C) Possible
- D) Probe

*Ans: B*

---

**11. If you have to clap your hands loudly in a patient's face to get some response, what letter is scored on AVPU?**

- A) A
- B) V
- C) P
- D) U

*Ans: B*

---

**12. Which describes a patient scoring U on AVPU?**

- A) Lurches in response to loud noise
- B) Jerks when a light shines directly at the pupil
- C) Moans when a painful stimulus is applied
- D) Does not respond to pain

*Ans: D*

---

**13. You are by the side of a patient who is unresponsive and has a history of heart failure. When assessing the airway, which one of the following observations best indicates an open airway?**

- A) You can hear normal respirations and see the chest rise and fall.
- B) She has a pulse of 80 per minute.
- C) Her mouth is open and you can hear gurgling sounds.
- D) You can see that her tongue has fallen back where it appears to be blocking the airway.

*Ans: A*

---

**14. How should an OEC technician best determine the adequacy of a patient's breathing during a primary assessment?**

- A) Determine the patient's mental status.
- B) Compare the patient's blood pressure and pulse rates.
- C) Assess the patient's ability to raise their shoulders.
- D) Look for the rise and fall of the patient's chest.

*Ans: D*

---

**15. Which of the following statements indicates that an OEC technician correctly understands capillary refill?**

- A) "Capillary refill is a reliable sign of perfusion in adults, but not as reliable in infants and children."
- B) "Flushed skin in an adult is a normal finding when assessing capillary refill."
- C) "A capillary refill of four or more seconds in a patient of any age indicates that the patient is well oxygenated."
- D) "For a capillary refill test to be normal, it must be assessed at room temperature and be less than two seconds."

*Ans: D*

---

**16. A chief complaint is defined as the:**

- A) findings from a primary survey.
- B) primary reason the person is seeking medical care.
- C) findings from a secondary survey.
- D) findings from a SAMPLE interview.

*Ans: B*

---

**17. A 44-year-old male was on a ladder cutting limbs from a tree when he fell. He is found to be unresponsive with normal respirations. His breathing is adequate at a rate of 12 per minute and his carotid pulse is strong at 88 per minute. During the primary assessment and initial management, which one of the following actions will the OEC technician attempt to complete?**

- A) Get a medical history from a family member
- B) Start positive pressure ventilation with a bagvalve mask
- C) Place an oral airway and put a cervical collar on the patient
- D) Perform a jaw-thrust maneuver to open the airway

*Ans: A*

---

**18. An OEC technician should obtain the medical history of an alert and oriented trauma patient prior to the:**

- A) primary assessment.
- B) scene size-up.
- C) secondary assessment.
- D) reassessment.

*Ans: C*

---

**19. An OEC technician identifies the medications a patient takes during which portion of a patient assessment?**

- A) The primary assessment
- B) The OPQRST exam
- C) The chief complaint
- D) The medical history

*Ans: D*

---

**20. Signs and symptoms can be identified using the:**

- A) SAMPLE acronym.
- B) DCAP-BTLS mnemonic.
- C) OPQRST mnemonic.
- D) AVPU mnemonic.

*Ans: A*

---

**21. Which of the following findings is a sign, not a symptom?**

- A) An open leg fracture
- B) An upset stomach
- C) A sharp headache
- D) Chest pain

*Ans: A*

---

**22. Which of the following findings is a symptom?**

- A) A deep laceration
- B) Hives resulting from an allergic reaction
- C) High blood pressure
- D) Ringing in the ears

*Ans: D*

---

**23. Which of the following responses is an appropriate response for the letter P in the OPQRST mnemonic for a patient with abdominal pain?**

- A) "It hurts worse when I take a deep breath."
- B) "My pulse feels as though it is racing."
- C) "I have a past history of asthma."
- D) "I ate pizza about two hours ago."

*Ans: A*

---

**24. What letter in a SAMPLE interview indicates that a patient has a history of seizures?**

- A) S
- B) A
- C) M
- D) P

*Ans: D*

---

**25. What letter in a SAMPLE interview indicates that a patient had pancakes for breakfast two hours ago?**

- A) M
- B) P
- C) L
- D) E

*Ans: C*

---

**26. As you assess a patient for a medical emergency, the patient states the pain is in their chest up to the shoulder and down the left arm. To what letter of OPQRST would this apply?**

- A) O
- B) P
- C) Q
- D) R

*Ans: D*

---

**27. What letter of the OPQRST mnemonic reflects a medical emergency patient's report that difficulty in breathing is worse while attempting to walk?**

- A) O
- B) P
- C) Q
- D) R

*Ans: B*

---

**28. What letter of the OPQRST mnemonic reflects a medical emergency patient's report that difficulty in breathing became worse about two hours ago?**

- A) Q
- B) R
- C) S
- D) T

*Ans: D*

---

**29. During assessment of a responsive medical patient, you gather a medical history using the memory aid SAMPLE. To obtain information related to M, which one of the following questions should you ask?**

- A) "Do you have any past medical problems?"
- B) "Are you allergic to any medications?"
- C) "Are you currently taking any medications?"
- D) "Can you point to where it hurts the most?"

*Ans: C*

---

**30. The finding that a patient's skin is warm, pink, and dry during a primary assessment suggests:**

- A) possible shock.
- B) possible fever.
- C) poor oxygenation.
- D) normal circulation.

*Ans: D*

---

**31. A secondary assessment consists of which of the following steps?**

- A) Hands-on physical exam and vital signs
- B) Applying an AICD, the on-going exam, and vital signs
- C) Medical history, trauma exam, and vital signs
- D) The on-going exam, detailed physical exam, and vital signs

*Ans: A*

---

**32. DCAP-BTLS is mnemonic to assist you in remembering:**

- A) a trauma patient's vital signs.
- B) what to assess while inspecting and palpating a patient.
- C) medications, allergies, and the patient's medical history.
- D) the order in which to assess a patient only during a primary physical exam.

*Ans: B*

---

**33. Which of the following actions should you take during a secondary assessment of a 45-year-old female with a severe headache?**

- A) Perform a physical exam and obtain vital signs
- B) Reassess vital signs, open the airway, and obtain OPQRST information
- C) Continue emergency care, obtain the chief complaint, and gather a SAMPLE history
- D) Assess the head, reevaluate vital signs, and continue emergency care

*Ans: A*

---

**34. A secondary assessment (including vital signs) should take about how long?**

- A) 30 seconds
- B) 1 minute
- C) 2-5 minutes
- D) 6-8 minutes

*Ans: C*

---

**35. The D in DCAP-BTLS stands for:**

- A) Dislocation.
- B) Damage.
- C) Deformity.
- D) Dyspnea.

*Ans: C*

---

**36. An easy way for an OEC technician to remember what to look for in assessing injuries in a trauma patient is the:**

- A) acronym SAMPLE.
- B) mnemonic APVU.
- C) mnemonic DCAP-BTLS.
- D) mnemonic OPQRST.

*Ans: C*

---



**37. As you assess a patient using the DCAP-BTLS mnemonic, the letter that denotes edema of the ankle is:**

- A) C.
- B) S.
- C) P.
- D) T.

*Ans: B*

---

**38. Your patient is 5 years old and has fallen, resulting in a scraped knee. What letter indicates the scraped knee in the DCAP-BTLS mnemonic?**

- A) D
- B) L
- C) A
- D) S

*Ans: C*

---

**39. You are performing a secondary assessment on the unrestrained adult passenger of a motor vehicle that rolled several times at a high rate of speed. The patient has a skull laceration and is responsive to painful stimuli and in a state of hypoperfusion. When assessing the head, which one of the following is appropriate?**

- A) Cleaning a scalp laceration
- B) Checking the pupils with a penlight
- C) Performing a blind finger sweep to clear the airway
- D) Applying pressure to a skull laceration

*Ans: B*

---

**40. During assessment, a patients eyes should move how?**

- A) In unison
- B) Upward
- C) Slowly
- D) With dilating pupils

*Ans: A*

---

**41. Which one of the following signs should an OEC technician detect during a secondary assessment of a critically injured patient?**

- A) Decreased level of responsiveness
- B) Weak carotid pulse and clammy skin
- C) Inadequate respiratory effort
- D) Bruising and tenderness to the abdomen

*Ans: D*

---

**42. A complete set of vitals is taken at the:**

- A) during the secondary assessment and during the reassessment.
- B) during the primary assessment.
- C) end of the secondary assessment.
- D) end of the primary assessment.

*Ans: A*

---

**43. You are assessing an eight-month-old whose mother states has been vomiting for two days and not eating or drinking. When assessing the pulse in an infant, which site should you check first?**

- A) The temporal artery
- B) The radial artery
- C) The femoral artery
- D) The brachial artery

*Ans: D*

---

**44. Which of the following tasks is part of a secondary assessment?**

- A) Opening the airway
- B) Checking for life-threatening bleeding
- C) Inserting an OPA
- D) Taking a blood pressure reading

*Ans: D*

---

## Section 8 — Documentation

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13 Questions

**1. Identify the two types of medical communications that are most important to inform other medical providers of a patient's condition and progress.**

- A) Verbal and nonverbal communication
- B) Oral and transmitted communication
- C) Verbal communication and sign language
- D) Oral communication and written documentation

*Ans: D*

**2. The OEC technician can verbally report to others in a timely method about an incident using the acronym "SAILER." The acronym is best represented by which one of the following?**

- A) The gender and age of the patient, the chief complaint of the patient, your location, the equipment you need, and the other resources you need
- B) The patient's name, gender, and complaints, your location, a toboggan, and airway
- C) The patient's gender and complaints, your location, oxygen, and blankets
- D) The patroller, the patient's gender and complaints, location of the patrol room and other patrollers

*Ans: A*

**3. When a patient arrives at the first-aid station, a status of the patient is given. The name of the brief oral report is called what?**

- A) Pass-off report
- B) Handoff report
- C) Tag-out report
- D) Trade-off report

*Ans: B*

**4. Which of the following documents may be used as a medical-legal record for patient care?**

- A) Field notes of the patroller, patient care report (PCR), incident report forms
- B) Supply report, annual budget, insurance carrier
- C) NSP awards, fund raising awards, snow report
- D) Patrol schedule, automobile insurance certificate, list of surrounding resorts

*Ans: A*

**5. An incident report form is provided by the area's insurance carrier. What is the purpose of the incident report form?**

- A) To collect data concerning only the patient and the circumstances surrounding an incident
- B) To communicate only medical information to other parties
- C) To collect data regarding your opinion of the incident
- D) To collect data surrounding the patient and incident, gather patient care data, and to communicate medical information to other parties

*Ans: D*

---

**6. The patient care report (PCR) is best organized and completed by using which of the following methods?**

- A) SOAP and LATHER
- B) LIES and CHEATED
- C) SOAP and CHEATED
- D) CHEATED and DISTRACTED

*Ans: C*

---

**7. Please select the most effective components that the OEC technician uses in medical documentation and that best represent the primary parts of the assessment and management process.**

- A) Chief complaint, history, examination, assessment, treatment, evaluation, and disposition
- B) Chief counsel, highest injury, achievement, trail skied, evolution, and deposition
- C) Chances taken, history, written exam, attributes, toboggan used, evacuation, and disposition
- D) Patient name, history, exam of incident, asking for advice, temperament, involvement, and distractors

*Ans: A*

---

**8. When a patient refuses care, under what part of the CHEATED acronym does that fall?**

- A) H
- B) T
- C) E
- D) D

*Ans: D*

---

**9. What part of CHEATED includes the physical exam of the patient?**

- A) C
- B) H
- C) E
- D) A

*Ans: C*

---

**10. On which form would you be most likely to use the SOAP or CHEATED acronyms?**

- A) Patient care report
- B) Annual report
- C) Handoff reports
- D) NSP application

*Ans: A*

---

**11. The PCR (patient care report) is a legal medical document completed by the OEC technician. Therefore, good written documentation is important. In the list below, choose the answer that represents the components of a good written case report.**

- A) Biased with your opinion
- B) Using only slang spoken on the street and by the skier
- C) Appropriate medical terminology
- D) Errors are not a problem.

*Ans: C*

---

**12. Rather than rewriting the whole report when an error is made, which of the following is the best method for correcting written errors on the report?**

- A) Erase the error, write the correct information, then initial, date, and time the change.
- B) Scratch the error out completely, write the correct information, then initial, date, and time the change.
- C) Use white out to cover the mistake, correct the information, then initial, date, and time the change.
- D) Draw a single line through the error, write the correct information, then initial, date, and time the change.

*Ans: D*

---

**13. What is the best possible way to add or amend information to a report that has already been submitted?**

- A) Submit the additional information as an addendum to the report then sign, date, and time the addendum.
- B) Ask for the report back so you can add to the report, then initial the changes.
- C) Once the report is submitted, you can no longer add information and should only add the information to your notes.
- D) Write the additional information on a copy of the form you kept for your records and resubmit the form with your initials and date.

*Ans: A*

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## Section 9 — Airway

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31 Questions

**1. The lower airway consists of which of the following structures?**

- A) Epiglottis, trachea, and bronchi
- B) Pharynx, bronchi, and alveoli
- C) Trachea, bronchi, and alveoli
- D) Larynx, bronchi, and alveoli

*Ans: C*

---

**2. Which of the following patients has an actual or potential critical occlusion of the upper airway?**

- A) 3-year-old with a fever and swelling of the larynx
- B) 45-year-old with spasm of the bronchioles
- C) 61-year-old with a severe cough
- D) 78-year-old with a mucus plug in a right bronchioles

*Ans: A*

---

**3. You want to open your patient's mouth to check for obstruction with the cross-finger technique. Which of the following describes the best way to accomplish the cross-finger technique?**

- A) Place your index finger and thumb between the patient's teeth and into the oropharynx. Then push the teeth apart.
- B) Place your fingers on the lower teeth and pull downwards to open the mouth.
- C) Cross your index finger and thumb on the bony part of the lower jaw. Then push up on the jaw to open the mouth.
- D) Place your thumb and index finger on the upper teeth and thumb on the lower teeth. Then spread your thumb and index finger and thumb apart opening the mouth to open the patient's mouth.

*Ans: D*

---

**4. Your patient has just had a seizure and is presenting with snoring respirations. You will open her airway with the head tilt - chin lift method. Which option correctly describes this method?**

- A) Place one hand on the forehead and one hand on the bony part of the lower jaw.
- B) Place one hand on the forehead and one hand in the mouth.
- C) Place one hand on the forehead and one hand under the back of the neck.
- D) Place your fingers under the angles of the jaw.

*Ans: A*

---

**5. You are correctly performing the jaw-thrust maneuver when you:**

- A) use the thumbs or fingers as a lever to lift the patient's mandible forward.
- B) maintain the patient's head in a neutral position and tilt the head slightly backward.
- C) open the patient's airway by slightly pushing down on the forehead and thrusting the jaw upward.
- D) place one hand on the patient's forehead and lift the jaw upward with the other hand by inserting the thumb in the mouth.

*Ans: A*

---

**6. You are caring for an unconscious patient who has vomited and appears to have a blocked airway. Using the cross- finger technique, you open the patient's mouth and can see a large piece of undigested food near the back of the throat. You would:**

- A) insert an airway.
- B) administer abdominal thrusts.
- C) perform a finger sweep.
- D) encourage the patient to cough.

*Ans: C*

---

**7. Which of the following statements shows that the OEC technician understands how to suction patients properly in a field setting?**

- A) "A rigid suction catheter is affected more by temperature than a flexible catheter, and, therefore, flexible catheters should be chilled."
- B) "To thoroughly suction a patient, I insert the rigid tip of the catheter past the base of the tongue and into the pharynx."
- C) "It is helpful to pour sterile water into a patient's mouth to liquefy vomit and make it easier to suction out."
- D) "When there is active bleeding in the mouth, I may need to use gravity and suction concurrently."

*Ans: D*

---

**8. To clear an airway of fluid and debris:**

- A) suction the airway deep by using an up and down motion.
- B) insert the tip of the suction catheter only as far as you can see.
- C) apply suction prior to placing the tip into the mouth.
- D) insert the tip of the suction catheter as deeply as possible.

*Ans: B*

---

**9. You have opened and cleared your unresponsive patient's airway, and there is no evidence of head or spinal injury. While waiting for other equipment or personnel, the easiest way to maintain this state and ensure adequate breathing is to put this patient into the recovery position. Which of the following best describes this position?**

- A) Place the patient onto his left side.
- B) Place the patient with his head down and feet up.
- C) Place the patient onto his back.
- D) Place the patient into a semi seated position.

*Ans: A*

---

**10. You have been asked to teach a new patroller how to put a patient in the recovery position. Which of the following statements about the recovery position would be correct?**

- A) "Roll the patient onto his back and then hold him there until help arrives."
- B) "After turning the patient on his side, rest the patient's head on your aid pack."
- C) "Elevate the patient's lower body so that his oral secretions will drain out easily."
- D) "Turn the patient on his side and flex his upper leg so that it anchors him on his side."

*Ans: D*

---

**11. By placing an oropharyngeal airway in a patient, you have:**

- A) protected the airway from vomit or other secretions.
- B) kept the tongue from occluding the airway.
- C) obtained a patent airway by keeping the mouth and nares from closing.
- D) minimized the risk of vomiting by closing off the esophagus.

*Ans: B*

---

**12. Which of the following statements indicates that the speaker understands how and when to use an oropharyngeal airway?**

- A) "It can be used for patients who respond to painful stimuli."
- B) "It should be sized from the tip of the nose to the angle of the jaw."
- C) "It protects the patient from aspirating vomit or other secretions."
- D) "I must watch the patient for vomiting or gagging even with the proper insertion."

*Ans: D*

---

**13. For which of the following patients is placement of an oropharyngeal airway indicated?**

- A) A responsive but confused patient with stridor respirations
- B) A patient who has vomited and responds to painful stimuli by moaning
- C) An unresponsive patient who has neither a gag reflex nor a cough reflex
- D) A patient with snoring respirations who coughs as the oral airway is placed into his mouth

*Ans: C*

---



**14. Which of the following statements indicates that the speaker knows how to properly size an oropharyngeal airway before its placement?**

- A) "The length of the airway should approximate the distance from the tip of the nose to the angle of the jaw."
- B) "To select an appropriately sized airway, you must first estimate the patient's height and weight."
- C) "The length of the airway should approximate the distance from the corner of the mouth to the angle of the jaw."
- D) "An appropriately sized oral airway can be selected by looking in the patient's mouth and estimating the length of the tongue."

*Ans: C*

---

**15. You have been ventilating the patient with an oropharyngeal airway and bag- valve mask. Suddenly the patient regains consciousness and starts to gag. Your immediate action should be to:**

- A) leave the airway in place but stop ventilations.
- B) remove the airway.
- C) suction around the airway.
- D) reassure and calm the patient.

*Ans: B*

---

**16. You appropriately size a nasopharyngeal airway by measuring the:**

- A) distance from the patient's mouth to the angle of the jaw.
- B) diameter of the patient's larger nostril.
- C) distance from the patient's nose to the earlobe.
- D) diameter of the patient's thumb.

*Ans: C*

---

**17. You are observing a rescuer insert an oropharyngeal airway into the airway of a 36-yearold male who has overdosed. Which one of the following observations indicates correct technique?**

- A) The rescuer inserts the airway into the patient's mouth with the tip pointed towards the roof of the mouth and then turns it 180 degrees once it is halfway in the mouth.
- B) The rescuer inserts the airway into the patient's mouth with the tip pointed towards the roof of the mouth and then turns it 90 degrees.
- C) The rescuer inserts the airway with the tip pointed towards the tongue until the flange of the airway is 1 cm above the patient's lips.
- D) The rescuer inserts the airway sideways into the mouth and then rotates it 180 degrees once it has reached the base of the tongue.

*Ans: A*

---

**18. You recognize that an oropharyngeal airway has been appropriately inserted when:**

- A) it cannot be dislodged by the rescuer.
- B) the flange sits about  $\frac{1}{4}$  inches from the patient's lips.
- C) vomiting is no longer occurring.
- D) air moves freely in and out of the airway.

*Ans: D*

---

**19. If a patient gags while you are inserting an oropharyngeal airway, you should remove the airway and:**

- A) have suction ready.
- B) reattempt to insert it.
- C) reattempt to insert it using a smaller airway.
- D) use a tongue depressor to better place the airway.

*Ans: A*

---

**20. You hear a candidate OEC technician explaining the use of a barrier device to another candidate. Which of the following statements would indicate that the candidate fully understands its purpose or how to use it?**

- A) "When ventilating a patient with a barrier device, watch for the rise and fall of the patient's chest."
- B) "A barrier device cannot be used as personal protective equipment."
- C) "A barrier device will keep the tongue from blocking the airway"
- D) "It is important to prefill the oxygen reservoir prior to using the barrier device."

*Ans: A*

---

**21. Which of the following statements shows an understanding by the speaker of the use of a pocket mask?**

- A) "Although a pocket mask is very effective at ventilating a patient, you are directly exposed to the patient's secretions."
- B) "A pocket mask must be securely placed over the patient's mouth and nose."
- C) "When using a pocket mask to ventilate a patient, you must make sure that the reservoir bag is in place."
- D) "A pocket mask should be used only when oxygen is connected to it."

*Ans: B*

---

**22. If while checking a size D oxygen tank you notice that the reading on the pressure regulator is 1000 psi, you should recognize that the tank is:**

- A) leaking.
- B) overfilled.
- C) half- full.
- D) almost empty.

*Ans: C*

---

**23. You are checking equipment at the beginning of your shift. The D size oxygen cylinder measures 1000 psi. You calculate that if you needed to use this tank at 15 LPM it would last:**

- A) 8.5 minutes
- B) 5 minutes
- C) 15 minutes
- D) 12 minutes

*Ans: A*

---

**24. To have a maximum amount of oxygen available for an incident with many patients, you would stock which size of tank?**

- A) D
- B) Super D
- C) E
- D) M

*Ans: D*

---

**25. What is the most common oxygen delivery device used by OEC technicians?**

- A) A nasal cannula
- B) A nonrebreather mask
- C) A simple face mask
- D) A venture mask

*Ans: B*

---

**26. Which of the following statements shows that the speaker understands how to properly store an oxygen tank after use?**

- A) "I'm going to leave the regulator set to 15 LPM so it will be ready when we need it."
- B) "I'll stand the oxygen tank on the floor beside the stretcher so everyone will know where it is."
- C) "I'll tighten the valve stem as much as I can so it won't come loose."
- D) "I'll place the oxygen tank in its protective case until we need to use it again."

*Ans: D*

---

**27. You have been asked to put a patient on a nonrebreather mask at 15 LPM. After you have finished attaching the regulator to the oxygen cylinder, your next step is to:**

- A) turn the valve stem on the top of the cylinder using a special oxygen wrench or key.
- B) attach the nonrebreather mask to the regulator and fill the reservoir.
- C) turn the oxygen control knob to 15 LPM.
- D) check the pressure gauge indicator to determine how much pressure is in the cylinder.

*Ans: A*

---

**28. Which of the following occurs during the proper application and use of a nonrebreather mask?**

- A) The mask will cover the mouth only so that air can be breathed into the nose.
- B) The flow rate should be set to 1 to 6 p.m.
- C) The oxygen reservoir should be inflated prior to application of the mask.
- D) The oxygen reservoir should completely collapse each time the patient inhales.

*Ans: C*

---

**29. Despite coaching and explaining the benefits of a nonrebreather face mask, a female patient with chest pain panics and states that she cannot tolerate the mask over her face. The more that she panics, the worse the chest pain becomes. Your best course of action is to:**

- A) remove the nonrebreather mask and apply a simple face mask.
- B) decrease the oxygen flow rate entering the nonrebreather mask.
- C) disconnect the nonrebreather and replace it with a nasal cannula.
- D) discontinue oxygen therapy and continually monitor breath sounds.

*Ans: C*

---

**30. Which of the following statements describes the correct positioning of a BVM on an adult patient's face?**

- A) The narrow part of the mask is over the bridge of the nose, and the bottom part is in the groove between the lower lip and chin.
- B) If two rescuers are using the BVM, the mask need only be placed over the mouth while the nose is pinched closed.
- C) The wide portion of the mask is at the top of the nose, and the narrow part is below the lower lip.
- D) The mask is properly positioned when the top portion lies over the bridge of the nose and the lower portion is below the chin.

*Ans: A*

---

**31. When monitoring a patient receiving oxygen through a nasal cannula, which of the following observations warrants immediate intervention?**

- A) An oxygen flow rate of 15 LPM
- B) Prongs in the nostrils that curve posteriorly into the nose
- C) Tubing that is positioned over the ears and under the chin
- D) Normal patient breathing while the nasal cannula is in place

*Ans: A*

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## Section 10 — Shock

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19 Questions

**1. Shock can best be described as:**

- A) the cells are getting glucose and other nutrients but not oxygen.
- B) the amount of oxygen reaching the cells is adequate, but carbon dioxide is not being removed.
- C) the blood contains an adequate amount of oxygen but not enough nutrients for cells to survive.
- D) inadequate tissue perfusion in which cells do not receive sufficient amounts of oxygen and nutrients.

*Ans: D*

---

**2. A patient in early onset of shock informs you that he has had severe diarrhea and vomiting over the past four days. Given this history, you would recognize the pathophysiology of the shock is probably related to:**

- A) the loss of red blood cells.
- B) decreased formed elements in the blood.
- C) stoppage of the capillaries.
- D) the loss of plasma volume.

*Ans: D*

---

**3. Which of the following lists of assessment findings indicates that a trauma patient may be in compensated shock?**

- A) Anxious, tachycardia, normal BP, pale and cool skin
- B) Slightly confused, tachycardia, hypotensive, warm and flushed skin
- C) Confused and anxious, very tachycardic, very hypotensive, cool and mottled skin
- D) Confused, very bradycardic, hypotensive, cool and cyanotic skin

*Ans: A*

---

**4. You are reassessing a patient who has sustained blunt trauma to the chest. Which one of the following reassessment findings best indicates that the patient is deteriorating and is in the decompensating phase of shock?**

- A) Strong pulse of 96 beats per minute
- B) Blood pressure of 88/50
- C) Blood oozing from an abdominal laceration
- D) Skin that is cool and dry

*Ans: B*

---

**5. In decompensated shock, the failure of body systems despite the body's attempt to oxygenate vital organs becomes apparent as:**

- A) increased blood pressure and reduced pulse rate.
- B) augmented capillary refill and increased respiratory rate.
- C) increased pulse rate and decreased blood pressure.
- D) increased respiratory rate and reduced level of consciousness.

*Ans: C*

---

**6. A patient with severe gastrointestinal bleeding is in what type of shock?**

- A) Non- Hemorrhagic
- B) Cardiogenic
- C) Hemorrhagic
- D) Distributive

*Ans: C*

---

**7. The four major types of shock are:**

- A) hemorrhagic, distributive, hypoxic, and obstructive.
- B) burn, hypovolemic, distributive, and hypoxic.
- C) hypoglycemic, obstructive, distributive, and hypovolemic.
- D) hypovolemic, cardiogenic, obstructive, and distributive.

*Ans: D*

---

**8. Which of the following statements made by a patient's family member would lead you to suspect that the patient is suffering from hypovolemic shock?**

- A) "He has had a rash for the past three days."
- B) "He cannot stop throwing up."
- C) "He has been taking an antibiotic for a chest cold."
- D) "He got up this morning and was having a hard time breathing."

*Ans: B*

---

**9. A patient in shock with abdominal pain indicates he noticed lots of blood in the toilet after having a bowel movement this morning. You would recognize the possibility of what type of shock?**

- A) Septic
- B) Hypovolemic
- C) Hypoxic
- D) Obstructive

*Ans: B*

---

**10. A patient who sustained blunt trauma to the abdominal and pelvic areas in a very serious collision is probably in which type of shock?**

- A) Obstructive
- B) Cardiogenic
- C) Distributive
- D) Hypovolemic

*Ans: D*

---

**11. Which of the following conditions is the most probable cause of cardiogenic shock?**

- A) Myocardial infarction
- B) Severe vomiting and diarrhea
- C) Gastrointestinal bleeding
- D) Systemic infection

*Ans: A*

---

**12. The underlying cause of distributive shock is:**

- A) poor fluid intake.
- B) loss of blood volume.
- C) a damaged heart that has poor contractility.
- D) dilation of blood vessels.

*Ans: D*

---

**13. Septic shock is caused by:**

- A) a failing heart.
- B) blood loss.
- C) an infection.
- D) a collapsed lung.

*Ans: C*

---

**14. Which of the following conditions could be responsible for causing obstructive shock?**

- A) Sepsis
- B) Hemorrhage
- C) Pulmonary embolism
- D) Arrhythmia

*Ans: C*

---

**15. You approach a patient and notice they have life threatening bleed. The patient appears confused, pale, and diaphoretic. Which one of the following should you do first?**

- A) Apply oxygen.
- B) Address the life threatening bleed first.
- C) Ensure that the patient's airway is open and clear.
- D) Treat the patient for shock.

*Ans: B*

---

**16. When performing a primary assessment, one of the first signs that the body may be in shock is:**

- A) skin that is warm and diaphoretic.
- B) tachycardia.
- C) hypertension.
- D) a slowed respiratory rate.

*Ans: B*

---

**17. Which of the following statements best indicates that the speaker understands the role of caring for a patient in shock in a prehospital setting?**

- A) "The job of the prehospital provider is not to treat the patient for shock and get the patient to the hospital, where treatment for shock can be started."
- B) "Because shock is a life-threatening condition, it is important to identify its exact cause so that the OEC technician can administer the proper care to correct it."
- C) "If shock is in the compensatory or early stage, it is not yet life threatening, so you can take your time to carefully assess and treat the patient."
- D) "Because shock is best treated in the hospital, one should provide care to correct problems affecting ABCDs and then rapidly transport the patient."

*Ans: D*

---

**18. A 56-year-old female has struck a tree. Your assessment reveals gurgling respirations, rapid breathing, and cool, diaphoretic skin. You also observe bruising to the chest and abdomen. Which one of the following should be your first concern?**

- A) Determine her pulse rate.
- B) Elevate her feet.
- C) Evaluate her for shock.
- D) Open and maintain her airway.

*Ans: D*

---



**19. You get a call for a "serious" injury on the expert trail. As you are skiing up to the patient, your first priority is:**

- A) looking to see how much blood is on the snow.
- B) trying to determine how many people may be injured.
- C) assessing the scene for rescuer safety.
- D) performing a primary survey.

*Ans: C*

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## Section 11 — Medical Emergencies

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33 Questions

**1. When assessing a patient with a possible stroke, the OEC technician should first work on:**

- A) identifying risk factors for stroke.
- B) determining if there is a family history of stroke.
- C) recognizing the signs of a stroke.
- D) determining the type of stroke.

*Ans: C*

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**2. Glucose:**

- A) assists the pancreas in the manufacture of insulin.
- B) is an energy source for brain cells and other cells in the body.
- C) is an essential building block for body tissues such as muscle and bone.
- D) allows the body to use insulin.

*Ans: B*

---

**3. In the mnemonic AEIOU-TIPS, the A stands for:**

- A) ADHD and alcohol.
- B) alcohol and acidosis.
- C) acidosis and anemia.
- D) abscess and acidosis.

*Ans: B*

---

**4. In the mnemonic AEIOU-TIPS, the E stands for:**

- A) epilepsy, environment, and electrolyte imbalance.
- B) edema, electrolyte imbalance, and endocarditis.
- C) epilepsy, echovirus, and environment.
- D) electrolyte imbalance, environment, and echovirus.

*Ans: A*

---

**5. The signs and symptoms of a transient ischemic attack (TIA):**

- A) are temporary and resolve within 24 hours.
- B) can persist for a few days before they resolve.
- C) are mild but permanent.
- D) are severe and permanent.

*Ans: A*

---

**6. In the mnemonic AEIOU-TIPS, the O stands for:**

- A) osteosarcoma and oxygen.
- B) oxygen and overdose.
- C) overdose and oliguria.
- D) overdose and otitis.

*Ans: B*

---

**7. In the mnemonic AEIOU-TIPS, the U stands for:**

- A) urticarial.
- B) usher syndrome.
- C) ulcers.
- D) uremia (kidney failure).

*Ans: D*

---

**8. In the mnemonic AEIOU-TIPS, the T stands for:**

- A) thyroiditis and thrombophlebitis.
- B) trauma and tumors.
- C) thrombophlebitis and tinnitus.
- D) trauma and thrombophlebitis.

*Ans: B*

---

**9. In the mnemonic AEIOU-TIPS, the first and second I stand for:**

- A) insomnia and intestine.
- B) impetigo and injury.
- C) interstitial cystitis and iodine.
- D) insulin and infection.

*Ans: D*

---

**10. In the mnemonic AEIOU-TIPS, the P stands for:**

- A) parathyroid disorders and Parkinson's disease.
- B) parathyroid disorders and psychiatric conditions.
- C) Parkinson's disease and peripheral nerve disorder.
- D) poisoning and psychiatric conditions.

*Ans: D*

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**11. In the mnemonic AEIOU-TIPS, the S stands for:**

- A) steroids, seizure, and sinusitis.
- B) seizure, stroke, and syncope.
- C) Sjogren's syndrome, seizure, and sinusitis.
- D) sickle cell anemia, Sjogren's syndrome, and steroids.

*Ans: B*

---

**12. Which of the following statements concerning the causation of seizures is correct?**

- A) A seizure occurs when the heart beats irregularly, causing a decreased amount of oxygen-rich blood to reach the brain.
- B) All seizures are caused by epilepsy, a term that describes a problem somewhere in the body.
- C) A seizure condition is a muscle problem that causes the arms and legs to jerk.
- D) A seizure occurs when there is an electrical disturbance in the brain.

*Ans: D*

---

**13. You are called to the lodge to assess a 3-year-old boy. The parents state that their son was playing with his brother and "blanked out" and "stared off into space" for several seconds. They deny any convulsing-like movement. Based on this description, you would suspect what type of seizure?**

- A) A febrile seizure
- B) A simple partial seizure
- C) A grand mal seizure
- D) An absence (petit mal) seizure

*Ans: D*

---

**14. You have been called to the lodge for a behavioral emergency. When you arrive, you find a male in his forties sitting up against a wall. He is confused and incontinent. Bystanders state that he suddenly fell to the ground and was grunting and shaking. From this description, what should you suspect first that the man has had?**

- A) A generalized seizure
- B) A diabetic reaction
- C) An absence seizure
- D) A syncopal episode

*Ans: A*

---

**15. Some patients experience a premonition indicating a seizure is about to happen known as a(n):**

- A) syncopal stage.
- B) aura.
- C) colonic phase.
- D) postictal state.

*Ans: B*

---

**16. Which of the following statements indicates that the speaker understands the danger posed by status epilepticus?**

- A) "Status epilepticus is an extremely dangerous condition because the patient can go into shock from blood loss."
- B) "Patients who do not have a history of seizures are at greater risk for status epilepticus."
- C) "The longer the seizure continues, the greater the likelihood of permanent brain damage."
- D) "Status epilepticus indicates that the patient's medications have reached toxic levels in the body."

*Ans: C*

---

**17. The primary problem in Type II diabetes is that:**

- A) sugars cannot be easily digested in the stomach and small intestine.
- B) cells exhibit resistance to insulin preventing glucose from entering the cells.
- C) insufficient insulin is produced.
- D) too much insulin is produced.

*Ans: B*

---

**18. Which of the following actions may result in hypoglycemia in a diabetes patient?**

- A) Failure to take insulin or an oral diabetes medication
- B) Lack of exercise
- C) Missing a meal when the patient has taken their insulin
- D) Overeating

*Ans: C*

---

**19. Hyperglycemia:**

- A) has a more gradual onset than hypoglycemia.
- B) is preceded by an aura, such as hallucinations or detecting unusual odors.
- C) has a more rapid onset than hypoglycemia.
- D) is more easily treated in prehospital environments than is hypoglycemia.

*Ans: A*

---

**20. Which of the following events best describes an ischemic stroke?**

- A) An artery in the brain has ruptured.
- B) Blood has collected throughout the brain tissue.
- C) A heart attack has occurred at the same time as a stroke.
- D) A blood clot has obstructed a blood vessel in the brain.

*Ans: D*

---

**21. Which of the following statements indicates that the speaker understands hemorrhagic strokes?**

- A) "They occur when clots form in blood vessels in the brain."
- B) "They occur when a blood vessel in the brain ruptures."
- C) "Their signs and symptoms generally resolve within 24 hours."
- D) "They can sometimes be treated by clotdestroying medications."

*Ans: B*

---

**22. Which of the following questions is critical for establishing a window of treatment for a patient who is weak and dizzy and may have suffered a stroke?**

- A) "Does your father have a history of heart problems or stroke?"
- B) "What time did the weakness and dizziness start?"
- C) "What medications do you take?"
- D) "Do you have any pain in your arms or legs?"

*Ans: B*

---

**23. After a delayed response, you arrive in the lodge and find a 62-year-old female is still seizing. Which one of the following should you do first?**

- A) Assess her airway and breathing.
- B) Determine if she has a seizure history.
- C) Ascertain the duration of the seizure.
- D) Move her to the aid room for further assessment and then transport her to the hospital.

*Ans: A*

---

**24. When performing a secondary assessment on a confused patient, which of the following signs is most suggestive of a seizure?**

- A) Right sided facial droop
- B) A bitten tongue
- C) Pinpoint pupils
- D) A slow heart rate

*Ans: B*

---

**25. You have assessed a patient and found a patent airway, adequate breathing, and a strong radial pulse. The patient is having no difficulty speaking and no facial droop but exhibits a slight left arm drift. You should interpret these findings as:**

- A) not suggestive of a stroke.
- B) suggestive of a stroke only if the patient's blood pressure is also elevated.
- C) suggestive of a stroke only if the patient's pupils are not equal.
- D) suggestive that a stroke may be occurring.

*Ans: D*

---

**26. Which of the following statements shows that the speaker has an accurate understanding of assessment findings related to stroke?**

- A) "For a stroke to be suspected, both arms must be equally weak."
- B) "If one arm is weak, a stroke should be suspected."
- C) "A stroke should be suspected only when one arm is paralyzed, not just weak."
- D) "If one arm is weak, stroke should be suspected only if the patient's blood pressure is also elevated."

*Ans: B*

---

**27. You have been called for a 63-year-old woman with slurred speech and right arm weakness. On arrival, the patient informs you that the slurred speech and weakness have resolved. Since your assessment reveals no deficits, you would tell the patient:**

- A) "It would be best to start taking one baby aspirin every day; this will decrease the chance of this happening again."
- B) "Sometimes seizures present this way; call your doctor in the morning to schedule an appointment."
- C) "If this happens again, wait 10 minutes to see if you are okay. If the symptoms don't go away, call 911."
- D) "You really need to be evaluated in the hospital. You could be having a stroke."

*Ans: D*

---

**28. Which of the following instructions to a possible stroke patient describes the proper procedure for assessing an arm drift?**

- A) "Hold your arms up over your head for 20 seconds."
- B) "Grab my fingers with both of your hands and squeeze as hard as you can."
- C) "Raise your hands above your head and close your eyes."
- D) "Hold your arms straight out in front of you with your palms facing up with your eyes closed."

*Ans: D*

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**29. Which letter in FAST-ED involves the eyes?**

- A) F
- B) S
- C) E
- D) D

*Ans: C*

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**30. Your assessment reveals a drooling 48-year-old male to have gurgling speech and left arm paralysis. Which of the following would be your priority?**

- A) Suction the patient's airway.
- B) Protect the patient's left arm.
- C) Determine the patient's risk factors for stroke.
- D) Monitor the patient's blood pressure.

*Ans: A*

---

**31. You have been called for a seizure emergency. You find an adult female actively seizing with a bystander attempting to place a spoon between her teeth. The bystander tells you that he is trying to keep the person from biting her tongue. Which one of the following would be an appropriate response?**

- A) Tell the bystander to continue trying to get the spoon in place.
- B) Ask the bystander to discontinue his efforts and then turn the patient onto her side to facilitate the drainage of oral secretions.
- C) Instruct the bystander to restrain the patient while you put the spoon in place.
- D) Tell the bystander that a padded tongue blade works better.

*Ans: B*

---

**32. A young boy tells you that his twin brother suffers from seizures and asks what he can do if he sees his brother convulsing. You should give him which one of the following instructions?**

- A) "Move any moveable objects and furniture away from him."
- B) "Hold him firmly against the floor until he stops seizing or until the EMTs arrive."
- C) "Insert a spoon into his mouth to keep him from swallowing his tongue."
- D) "Call EMS only if the seizure lasts more than 10 minutes or he loses control of his bladder." "When he quits seizing, give him sugar so his blood sugar is OK."

*Ans: A*

---

**33. As a general rule, a seizing patient should be transported in what position?**

- A) Supine
- B) Sitting upright
- C) On their side
- D) Supine with legs elevated

*Ans: C*

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## Section 12 — Toxicology

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18 Questions

**1. A 44-year-old patient was found at home unresponsive. Based on his clinical presentation and the fact that he was heating with a kerosene heater in an enclosed room, you suspect that he is suffering from carbon monoxide poisoning. Based on this information, which route of exposure would you recognize as responsible for the poisoning?**

- A) Transdermal route
- B) Inhalation
- C) Ingestion
- D) Absorption

*Ans: B*

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**2. Which of the following terms does not indicate a route of absorption in the body?**

- A) Ingestion
- B) Inhalation
- C) Distribution
- D) Transdermal

*Ans: C*

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**3. Distribution of a poisonous substance in the body occurs primarily through the:**

- A) lymphatic system.
- B) circulatory system.
- C) endocrine system.
- D) gastrointestinal system.

*Ans: B*

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**4. The process by which the body breaks down a substance is known as:**

- A) elimination.
- B) detoxification.
- C) metabolism.
- D) distribution.

*Ans: C*

---

**5. Substance abuse is defined as:**

- A) an intentional or unintentional use of a substance or poison that results in a medical emergency.
- B) an accidental ingestion of a toxin that interferes with normal functioning.
- C) the use of a substance or poison at least once a day.
- D) the intentional misuse of a substance that results in significant impairment.

*Ans: D*

---

**6. A poisonous substance that comes from a living organism is known as a(n):**

- A) antipsychotic.
- B) toxin.
- C) chemical.
- D) legume.

*Ans: B*

---

**7. Which of the following statements indicates that the speaker understands inhalation poisonings?**

- A) "They can occur from inhaling common household products."
- B) "They affect the lungs only and do not circulate systemically."
- C) "An inhalation emergency cannot be occurring if you don't smell anything."
- D) "Most inhalation poisonings are easily identifiable by a residue around the mouth."

*Ans: A*

---

**8. You are teaching a class about drug-related emergencies, specifically opiate overdose. Which of the following statements made by students indicates an understanding of possible signs of opiate overdose?**

- A) "Only illegal narcotics can cause hallucinations, which may make the patient violent."
- B) "When assessing a patient, you must be alert for respiratory depression."
- C) "You should always ask patients if they have pain because narcotics suppress pain."
- D) "You should always check the patient's pupils because they will be dilated in a narcotics overdose."

*Ans: B*

---

**9. A 15-year-old admits to taking "a bunch" of extra strength Tylenol tablets. Her mother states it's nothing to worry about because it was just Tylenol. Based on your training, you recognize that this ingestion could have serious consequences. It is because toxic amounts acetaminophen can cause irreversible:**

- A) liver damage.
- B) stomach damage.
- C) brain damage.
- D) hearing damage.

*Ans: A*

---

**10. A patient in the third day of withdrawal from alcohol could develop all of the following conditions except:**

- A) rash.
- B) seizure activity.
- C) hallucinations.
- D) muscle tremors.

*Ans: A*

---

**11. The most commonly abused drug in the world is:**

- A) marijuana.
- B) cocaine.
- C) amphetamine.
- D) alcohol.

*Ans: D*

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**12. The mnemonic DUMBELS and the acronym SLUDGEM can help identify specific signs and symptoms associated with:**

- A) exposure to nerve agents.
- B) ingestion of ethylene glycol.
- C) ingestion of LSD.
- D) overdoses of antipsychotic drugs.

*Ans: A*

---

**13. It is late Friday evening, and you are joining your friends at a cabin for a winter weekend trip. When you open the door, you notice how nice and warm it is inside from the wood stove. Your friends though are not very welcoming. They tell you they are not feeling well and complain of being lightheaded and nauseated. Based on their symptoms, you are concerned they may be suffering from:**

- A) carbon dioxide poisoning.
- B) alcohol toxicity.
- C) excessive heat exposure.
- D) carbon monoxide poisoning.

*Ans: D*

---

**14. Agitation, elevated heart rate, and elevated blood pressure are signs of intoxication with which of the following types of substances?**

- A) Cocaine and hallucinogens
- B) Antipsychotics and carbon monoxide
- C) Inhalants and sleeping pills
- D) Alcohol and ethylene glycol

*Ans: A*

---

**15. You are driving on the highway and see an overturned tanker. A fluid is spilling from the tanker, rapidly covering the ground around the driver's door. The driver appears to be unresponsive. The only marking on the tanker seems to be the name of the tanker company and hazard symbol. Your first step should be:**

- A) try not to step in the liquid while pulling the driver away from the truck.
- B) have someone call 911 and assess the driver's ABCDs.
- C) secure the scene; you must ensure your safety.
- D) cover your mouth and nose so that you don't breathe any fumes while you rescue the driver.

*Ans: C*

---

**16. You are visiting a friend's home and suddenly hear a child crying out in the kitchen. As you assess the scene, you note kitchen cabinets are open with bottles tossed around. The child is holding her hand to her mouth. Your friend tells you that the child must have tasted something that she didn't like. Based on your scene assessment, you would:**

- A) agree with your friend and suggest that a cookie would take the bad taste away.
- B) ask your friend if she has Ipecac so you can make the child vomit.
- C) call the poison control center and try to identify which bottle the child may have been drinking from.
- D) offer the child a glass of milk to dilute any chemical he may have ingested.

*Ans: C*

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**17. In the United States, poison-related information is available 24 hours a day by contacting the:**

- A) National Institute on Drug Abuse.
- B) Substance Abuse and Mental Health Hotline.
- C) American Association of Poison Control Centers.
- D) Chemical Emergency Center.

*Ans: C*

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**18. You are helping a friend change the battery in his truck when acid from the old battery splashes in his eye. He is complaining of pain in his right eye and is having a hard time keeping it open. He has no problems with airway, breathing, or circulation. What should you do?**

- A) Cover both his eyes and immediately drive him to the nearest Emergency Department.
- B) Place an ice pack on his eye and immediately transport him to the nearest Emergency Department.
- C) Pat the eye with a dry towel to absorb the acid.
- D) Immediately irrigate the eye with copious amounts of clean water.

*Ans: D*

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## Section 13 — Respiratory

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22 Questions

**1. The passage of oxygen and carbon dioxide through the walls of the alveolae occurs via:**

- A) diffusion.
- B) osmosis.
- C) breathing.
- D) aspiration.

*Ans: A*

---

**2. Which of the following lists best represents the correct sequence for the passage of air into the lungs once it passes the pharynx?**

- A) Bronchi, larynx, trachea, cricoid
- B) Epiglottis, esophagus, trachea, alveoli
- C) Trachea, uvula, bronchi, alveoli
- D) Larynx, trachea, bronchi, alveoli

*Ans: D*

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**3. When the diaphragm and intercostal muscles relax, which of the following events occurs?**

- A) Inhalation
- B) Release
- C) Inspiration
- D) Exhalation

*Ans: D*

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**4. The most important muscle of respiration is the:**

- A) pectoralis major.
- B) intercostal muscle.
- C) diaphragm.
- D) sternocleidomastoid.

*Ans: C*

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**5. Which of the following actions causes an individual to inhale?**

- A) The intercostal muscles relax.
- B) The chest cavity decreases in size.
- C) The diaphragm contracts and flattens.
- D) Pressure within the chest increases.

*Ans: C*

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**6. You are documenting your assessment of the patient who just left in an ambulance. The patient told you that he felt short of breath. You would document this as:**

- A) respiratory failure.
- B) arrhythmia.
- C) dyspnea.
- D) anoxia.

*Ans: C*

---

**7. The normal range for respiratory rate in children is:**

- A) 5-10 breaths per minute.
- B) 15-30 breaths per minute.
- C) 25-40 breaths per minute.
- D) 40-45 breaths per minute.

*Ans: B*

---

**8. You receive a call that there is an infant in the lodge who seems to be having difficulty breathing. As you are approaching the scene with a new candidate patroller, you ask him what the normal respiratory rate is for an infant. The candidate would be correct if he told you the rate should be:**

- A) 15-20 breaths per minute.
- B) 20-30 breaths per minute.
- C) 20- 60 breaths per minute.
- D) 15-40 breaths per minute.

*Ans: C*

---

**9. Which of the following statements about respiratory accessory muscles is not correct?**

- A) They are recruited whenever the body's oxygen demand exceeds oxygen availability.
- B) They include chest, shoulder, and abdominal muscles.
- C) They bring more oxygen into the body by helping the chest wall expand more fully.
- D) They decrease negative internal chest pressure to draw more air into the lungs.

*Ans: D*

---

**10. In an adult, the most common cause of airway obstruction is:**

- A) the tongue.
- B) a food bolus.
- C) mucous secretions.
- D) an inflamed pharynx.

*Ans: A*

---

**11. A 74-year-old male with a history of chronic emphysema is complaining of sudden onset of shortness of breath. He appears to be in acute respiratory distress. Physical exam of the anterior chest wall reveals crepitus. Based on the patient's history and physical exam, you would suspect which of the following conditions?**

- A) Spontaneous pneumothorax
- B) Congestive heart failure
- C) Acute bronchitis
- D) Carbon monoxide poisoning

*Ans: A*

---

**12. Hyperventilation syndrome is a common psychological condition that:**

- A) is always benign and resolves with no treatment.
- B) is characterized by shallow, irregular, rapid breaths.
- C) is often precipitated by exercise.
- D) results in abnormally low blood carbon dioxide levels.

*Ans: D*

---

**13. A tachypneic patient is breathing:**

- A) more slowly than normal.
- B) normally.
- C) more rapidly than normal.
- D) irregularly.

*Ans: C*

---

**14. You and your OEC candidate are evaluating a 67- year-old patient with respiratory distress. You tell the candidate to document that the patient complains of dyspnea. The candidate asks you what "dyspnea" means. You explain that dyspnea is:**

- A) a term that describes a patient who is in respiratory distress.
- B) a subjective term that means difficulty breathing.
- C) a term that describes your objective assessment of a patient's difficulty in breathing.
- D) another term for tachypnea.

*Ans: B*

---

**15. Which of the following signs would you recognize as an early sign of respiratory distress in a 7-yearold boy who complains of difficulty breathing?**

- A) Barrel chest
- B) Nasal flaring
- C) Tremors of his hands
- D) Ecchymosis on his chest

*Ans: B*

---

**16. You are having a hard time getting the medical history of a patient who is short of breath. Which one of the following pieces of information would lead you to believe that the patient has a history of a chronic lung disease such as chronic emphysema?**

- A) She takes one aspirin every day.
- B) Her chest is barrel shaped.
- C) Her respirations are tachypneic.
- D) She has a frequent cough.

*Ans: B*

---

**17. You are assessing a 24-year-old skier who is anxious, wheezing, and complaining of shortness of breath. She tells you that this sometimes happens when she is exercising. Her respirations are rapid and shallow. Based on this information, you suspect the patient is suffering from:**

- A) hyperventilation syndrome.
- B) asthma.
- C) crepitus.
- D) pneumothorax.

*Ans: B*

---

**18. Which of the following respiratory rates may signify significant respiratory issues in an adult?**

- A) Fewer than 8 respirations per minute
- B) More than 24 respirations per minute
- C) Fewer than 12 respirations per minute
- D) More than 20 respirations per minute

*Ans: A*

---

**19. You are assessing a 34- year-old man who is complaining of "not feeling well." He is alert and pale. His respirations are regular at 20 per minute. Based on your training, you recognize that his respirations are:**

- A) normal for an adult.
- B) rapid for an adult.
- C) rapid but OK because they are regular.
- D) to be bradypnea.

*Ans: A*

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**20. You are called to assist a 16-year-old female who is reportedly having trouble breathing. Your assessment reveals an anxious female with rapid respirations who is complaining of pins and needles around her lips. Based on your training you suspect that:**

- A) she is close to respiratory failure.
- B) these findings will disappear if she slows her rate of breathing because she is most likely hyperventilating.
- C) having her lie down with her feet elevated 8-12 inches will relieve the pins and needles.
- D) she is having an allergic reaction and should be treated with an epi-pen.

*Ans: B*

---

**21. Several abnormal lung sounds can be helpful in understanding what condition may be affecting a patient. One such sound is wheezing. Which of the following statements about wheezing is false?**

- A) It indicates constriction of the lower airway passages.
- B) It is typically caused by asthma.
- C) It may be heard on inhalation, exhalation, or both.
- D) It can be heard only with a stethoscope.

*Ans: D*

---

**22. During a scene size-up, which of the following observations most strongly suggests that your adult patient is suffering from an acute respiratory emergency?**

- A) The patient is holding a metered-dose inhaler.
- B) The patient is in the tripod position.
- C) The patient's respiratory rate is 20.
- D) The patient's hands are trembling.

*Ans: B*

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## Section 14 — Allergic Reactions

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17 Questions

**1. A patient informs you that any time she uses a particular soap, she experiences a mild allergic reaction. You recognize that:**

- A) the soap contains an allergen.
- B) her soap contains non-organic ingredients.
- C) the soap contains antibodies that react in her body.
- D) with continued use, the allergic reaction will diminish.

*Ans: A*

**2. Which of the following statements indicates that the speaker understands allergic reactions and anaphylaxis?**

- A) "An allergic reaction only occurs when the patient ingests or inhales an allergen; anaphylaxis occurs when the allergen is injected."
- B) "Allergic reactions are caused by pollen and food; anaphylactic reactions are caused by venoms and medications."
- C) "Anaphylaxis is a severe allergic reaction that will lead to death without emergency care."
- D) "Anaphylaxis describes a severe allergic reaction that occurs the first time a person is exposed to an allergen."

*Ans: C*

**3. You are called to the lodge to help a patient who has developed an allergic reaction after taking an antibiotic medication. You would describe this route of exposure as:**

- A) injection.
- B) contact.
- C) ingestion.
- D) inhalation.

*Ans: C*

**4. A patient can have an allergic reaction caused by intramuscular administration of a medication. This method by which an allergen enters the body is called:**

- A) inhalation.
- B) injection.
- C) ingestion.
- D) topical.

*Ans: B*

---

**5. The overreaction of the immune system to a substance that is otherwise harmless is known as:**

- A) hypersensitivity.
- B) hives.
- C) anaphylaxis.
- D) immunity.

*Ans: A*

---

**6. Which of the following statements concerning allergies is true?**

- A) Allergies affect young healthy individuals only.
- B) An allergic reaction to a substance can occur minutes, or hours, after exposure to the substance.
- C) Once hypersensitized to a substance, a given individual will experience allergic reactions to that substance randomly, not consistently.
- D) Anaphylaxis is the mildest type of allergic reaction.

*Ans: B*

---

**7. Which of the following statements from a patient suggests he may be having a mild allergic reaction?**

- A) "I have had a fever and chills for the past two days."
- B) "My heart feels as though it is skipping some beats."
- C) "I feel dizzy and weak."
- D) "I have a rash on my face and neck."

*Ans: D*

---

**8. During a primary assessment, which of the following findings would most likely lead you to suspect an allergic reaction?**

- A) Hives on the face and neck
- B) Blood pressure of 100/60 mmHg
- C) Heart rate of 55 beats per minute
- D) A complaint of dizziness

*Ans: A*

---

**9. The patient has swollen lips and hives. Which of the following scene size-up observations supports your suspicion that the patient is having an allergic reaction?**

- A) He has a glucometer among his belongings.
- B) He has a new container of antibiotics with only two pills missing.
- C) He states that several members of his family have the flu.
- D) He has a bottle of expired nitroglycerin tablets.

*Ans: B*

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**10. Which of the following foods most commonly causes an allergic reaction?**

- A) Shellfish
- B) Chocolate
- C) Peanuts
- D) Eggs

*Ans: C*

---

**11. The most severe form of an allergic reaction is called:**

- A) rash.
- B) anaphylaxis.
- C) an antigen response.
- D) rhinorrhea.

*Ans: B*

---

**12. A patient who is responsive only to painful stimuli is covered with hives. He has labored respirations and a weak and tachycardic pulse. His vital signs are pulse 128, respirations 24, and blood pressure 80/50 mmHg. Given this presentation, you would suspect:**

- A) a pulmonary embolism.
- B) an anaphylactic reaction.
- C) a stroke.
- D) a head injury.

*Ans: B*

---

**13. When the respiratory effects of an allergic reaction are obvious within a few seconds to minutes:**

- A) the reaction tends to be more severe.
- B) the reaction tends to be localized.
- C) a large amount of allergen is likely involved.
- D) epinephrine will not be effective.

*Ans: A*

---

**14. Which of the following findings is present in anaphylaxis but is not a sign of a mild allergic reaction?**

- A) Anxiousness
- B) Swelling at the site of a bee sting
- C) Hives
- D) Hypotension

*Ans: D*

---

**15. Your patient is having an allergic reaction and needs to use his epinephrine auto-injector. He is concerned that he does not recall the correct way to do it. Which of the following things would you tell him?**

- A) "Press the injector onto the outer thigh and push the tip against the thigh until the needle deploys."
- B) "Place the injector against the outer upper arm, push the tip until the needle deploys, and then push the plunger slowly to inject the medication."
- C) "Pinch the skin on the anterior thigh, press the injector tip against the skin, and wait until it activates."
- D) "Pinch the skin on the lateral thigh, press the injector against the skin, and rapidly push the plunger to inject the medication."

*Ans: A*

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**16. You are staffing a first-aid booth at a mountain bike event. A mother approaches you with her 15-year-old son and informs you that her son is severely allergic to bees. Although he hasn't been stung, she is concerned about the auto-injector he carries. The mother states that it was prescribed when her son was 13 years old. She shows you the 0.15 mg autoinjector, and you note that it expired about 12 months ago. Which of the following statements would be most appropriate for you to make?**

- A) "Call your doctor's office for a new prescription because this auto-injector is out of date."
- B) "If you go to the pharmacy where you got this, they will be able to give you a new one."
- C) "Most medications are still good for months after the expiration date. You can use this one for a little while longer."
- D) "This medication is good for today, but you should call your doctor as soon as possible to replace this."

*Ans: A*

---

**17. When administering epinephrine to an infant or child, the injector should be placed at what location?**

- A) The outer buttock
- B) The anterior thigh
- C) The upper arm
- D) The lateral thigh

*Ans: D*

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## Section 15 — Cardiac

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38 Questions

**1. The walls of the capillaries are only one cell thick. This allows the exchange of nutrients and oxygen and waste products. The capillaries serve as the bridge between:**

- A) arterioles and venules.
- B) the right atrium and the left atrium.
- C) the pulmonary vein and the pulmonary artery.
- D) arteries and veins.

*Ans: A*

---

**2. Which of the following statements concerning the flow of blood to and through the right side of the heart is correct?**

- A) Blood reaches the right atrium from the aorta, is pumped through the mitral valve into the right ventricle, and is then pumped through the pulmonary veins to the lungs.
- B) Poorly oxygenated blood reaches the right ventricle from the venae cavae, is pumped through the pulmonic valve into the pulmonary artery, and then enters the right atrium and is pumped into the lungs.
- C) Oxygenated blood reaches the right atrium from the pulmonary veins, is pumped through the tricuspid valve into the right ventricle, and is then pumped into the aorta and throughout the body.
- D) Poorly oxygenated blood reaches the right atrium from the venae cavae, is pumped through the tricuspid valve into the right ventricle, and is then pumped through the pulmonic valve and on to the lungs through the pulmonary arteries.

*Ans: D*

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**3. The left side of the heart:**

- A) receives poorly oxygenated blood from the body.
- B) pumps oxygenated blood to the body.
- C) contains the pulmonic and tricuspid valves.
- D) pumps poorly oxygenated blood to the lungs.

*Ans: B*

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**4. After jogging with your friend for about 30 minutes, she stops and asks why your heart rate increases and your heart feels like it is pumping harder after you exercise for a while. Based on your understanding of the cardiovascular system, your best response would be which of the following statements?**

- A) "You're out of shape. Those things wouldn't happen if you exercised more."
- B) "Exercise makes your body need more nutrients and oxygen. Your heart pumps harder and faster to deliver more blood containing those things to your muscles."
- C) "When you exercise, your blood pressure increases, so your heart has to work harder to overcome the higher blood pressure."
- D) "Exercise pushes more blood into your heart, so your heart has to work harder and faster to pump the blood out so you don't go into heart failure."

*Ans: B*

---

**5. Which of the following statements concerning arteries is true?**

- A) They carry blood away from the heart.
- B) They have a lower pressure than veins.
- C) They always carry oxygenated blood.
- D) They drain into the venae cava.

*Ans: A*

---

**6. Which of the following statements concerning the cardiovascular system is true?**

- A) It consists of four major components: the heart, the lungs, the blood vessels, and blood.
- B) The heart is a muscular organ located behind and to the right of the sternum.
- C) An electrical impulse that starts in the right atrium stimulates contraction of the heart muscle.
- D) The subclavian arteries supply blood to the pelvis and legs.

*Ans: C*

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**7. The largest artery in the body is the:**

- A) carotid artery.
- B) femoral artery.
- C) aorta.
- D) jugular artery.

*Ans: C*

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**8. Which of the following statements about cardiovascular disease is false?**

- A) Cardiovascular disease involves a number of diseases that affect either the heart or blood vessels.
- B) Cardiovascular disease is the leading cause of death in the United States.
- C) Most cases of cardiovascular disease are congenital and could be prevented with improved prenatal care.
- D) The underlying cause of most cases of cardiovascular disease is coronary artery disease or atherosclerosis of the coronary arteries.

*Ans: C*

---

**9. Your neighbor tells you that he is concerned about his 70-year-old wife. She is at home recovering from a heart attack that occurred about 3 weeks ago. He explains that over the last few days, her ankles and lower legs have been gradually swelling and are painful to touch. Today, she seems to be a little short of breath. You recommend that your neighbor seek immediate medical attention because these findings are suggestive of:**

- A) congestive heart failure.
- B) tension pneumothorax.
- C) an acute myocardial infarction.
- D) pericarditis.

*Ans: A*

---

**10. Which of the following statements indicates that the speaker has an understanding of chest pain and acute myocardial infarction?**

- A) "If the chest pain started during activity, it is most likely cardiac in origin. If the pain started while the patient was at rest, it is most likely not cardiac in origin."
- B) "If the patient's chest pain does not go away with nitroglycerin, you can assume that the medication has expired and is no longer effective."
- C) "If the patient experiences chest pain that radiates into the left arm, it is most likely a heart attack. If the pain radiates anywhere else, it is most likely not cardiac in origin."
- D) "Some patients experiencing acute myocardial infarction do not have actual chest pain. Instead, they may experience a sensation of 'severe heartburn.' "

*Ans: D*

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**11. A patient tells you that he has been diagnosed with coronary artery disease. You recognize this diagnosis to mean that the patient has:**

- A) a history of acute myocardial infarction.
- B) had an operation known as coronary artery bypass grafting.
- C) a failure of the heart to efficiently pump blood.
- D) narrowing of the small arteries that supply blood and oxygen to the heart.

*Ans: D*

---

**12. A patient with chest pain informs you that he has "a bad heart." You think he may have congestive heart failure. When performing your secondary assessment, which of the following signs or symptoms does not indicate that the patient is in congestive heart failure?**

- A) Swollen ankles
- B) Bradycardia
- C) Bubbling breathing noises
- D) Shortness of breath

*Ans: B*

---



**13. How do most problems with heart valves originate?**

- A) From medication errors
- B) Due to injury
- C) Congenitally
- D) Due to viral infection

*Ans: C*

---

**14. You are called to assist a 62-year-old female who is complaining of fatigue, jaw discomfort, and excessive sweating. When asked, she denies having chest pain or shortness of breath. She tells you that she has a history of hypertension and takes a blood pressure pill. She also has nitroglycerin. Which statement is inappropriate for this event?**

- A) "I am not going to attach the AED because she is conscious and has a pulse."
- B) "She is not short of breath, so we don't need to transport her."
- C) "I am going to call for ALS because she could be having a cardiac event."
- D) "I am going to check her blood pressure."

*Ans: B*

---

**15. The only reason to delay transport in case of a cardiac emergency is:**

- A) to use an AED.
- B) to take a complete history.
- C) to provide nitroglycerin.
- D) to make the patient comfortable.

*Ans: A*

---

**16. Aspirin may be used in a patient who has chest pain suggestive of a heart attack because aspirin may:**

- A) break up a blood clot that is occluding a coronary artery.
- B) relieve the headache that is caused by the administration of nitroglycerin.
- C) decrease the ability of platelets to form clots.
- D) decrease the chest pain caused by the possible heart attack.

*Ans: C*

---

**17. You are caring for a patient with chest pain who has just taken his nitroglycerin. The patient's chest pain has been alleviated. You understand that nitroglycerin is effective in treating chest pain because it:**

- A) increases the heart rate and gets more blood to the heart muscle.
- B) has a sedative effect, which makes the patient less aware of the pain.
- C) dilates the arteries and veins, which decreases the workload on the heart.
- D) causes the body to relax, decreasing the need for oxygen.

*Ans: C*

---

**18. Three commonly used cardiac medications are:**

- A) Plavix, Nitroglycerine, and Albuterol.
- B) Nitroglycerine, Coumadin, and Lasix.
- C) Lipitor, Dilantin, and Tenormin.
- D) Coumadin, Zesteril, and an Epi-pen.

*Ans: B*

---

**19. You have just arrived with an AED at a scene where fellow OEC technicians are treating a patient in cardiac arrest. EMS has been contacted and is about 1 minute away. Which of the following actions should you take?**

- A) Let the OEC technicians continue CPR and wait for ALS.
- B) Apply and use the AED.
- C) Help count compressions.
- D) Go outside to direct ALS when they arrive.

*Ans: B*

---

**20. Which of the following signs or symptoms is suggestive of right-sided congestive heart failure?**

- A) Elevated blood pressure
- B) Edema in the legs and feet
- C) A bounding pulse
- D) An irregular heart rate

*Ans: B*

---

**21. Based on these findings, you would suspect that the patient:**

- A) has an aortic dissection/aneurysm.
- B) is having a hypertensive emergency.
- C) is experiencing congestive heart failure.
- D) has pericardial tamponade.

*Ans: C*

---

**22. Which of the following statements made by a patient would cause you to suspect an abdominal aortic aneurysm?**

- A) "I have really bad belly pain that I also feel in my back."
- B) "My heart is beating so hard you can feel it on my chest."
- C) "The pain is sharp and spreads to both arms."
- D) "I seem to be vomiting up some blood."

*Ans: A*

---

**23. You are assessing a 68-year-old man with a history of abdominal aortic aneurysm. How far might pain be radiating?**

- A) Groin
- B) Knee
- C) Ankle
- D) Toes

*Ans: A*

---

**24. Hypotension, distended neck veins, and muffled or distant heart tones are hallmark signs of:**

- A) pericarditis.
- B) congestive heart failure.
- C) aortic aneurysm.
- D) pericardial tamponade.

*Ans: D*

---

**25. A 62-year-old man presents with a complaint of abdominal pain radiating to his groin. He also complains of dizziness. On examination, his abdomen is tender, and you feel a large pulsating mass. His signs and symptoms are suggestive of:**

- A) acute embolism.
- B) an abdominal aortic aneurysm.
- C) a ruptured spleen.
- D) cardiogenic shock.

*Ans: B*

---

**26. Based on her presentation, you suspect that she may have:**

- A) pulmonary embolism.
- B) pericardial effusion.
- C) instable angina.
- D) an aortic dissection/aneurysm.

*Ans: A*

---

**27. When performing CPR, your philosophy should be:**

- A) "Dont break ribs."
- B) "Push hard, push fast."
- C) "Always be careful."
- D) "Obtain informed consent."

*Ans: B*

---

**28. Which of the following is not a reason to stop CPR?**

- A) The patient has the spontaneous return of a pulse and breathing.
- B) You are ready to use an AED.
- C) It has been 25 minutes, so brain damage has occurred.
- D) Rescuers are too tired to continue administering CPR.

*Ans: C*

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**29. Blood in the right ventricle is pumped into the:**

- A) left ventricle.
- B) pulmonary artery.
- C) left atrium.
- D) aorta.

*Ans: B*

---

**30. You have been called to treat a 47-year-old man complaining of chest pain. He is alert, oriented, and complaining of pain in his chest that came on suddenly about 15 minutes ago. His skin is sweaty and cool. Which of the following actions should you take first?**

- A) Apply oxygen at 15 LPM through a nonrebreather mask.
- B) Begin CPR.
- C) Help him administer his nitroglycerin.
- D) Obtain a SAMPLE history.

*Ans: A*

---

**31. You are in the first-aid area assisting a patient who just arrived with chest pain. He relates a history of angina that usually resolves with nitroglycerin. The patient states that the pain started when he was skiing through the bumps. Over the last 20 minutes, he has taken three nitroglycerin tablets and rested, but the pain has not gone away. Your next action would be to:**

- A) try one more nitroglycerin tablet because nitroglycerin usually relieves his pain.
- B) attach an AED in case he goes into cardiac arrest.
- C) place him on high-flow oxygen and call for transfer to medical facility.
- D) offer an aspirin.

*Ans: C*

---

**32. When assisting a patient to take a nitroglycerin tablet, it is:**

- A) placed under the tongue.
- B) swallowed with only an ounce of water.
- C) chewed and then allowed to dissolve slowly.
- D) placed between the cheek and teeth.

*Ans: A*

---

**33. Which of the following medications, if taken in the last 24 hours, would be a contraindication to administering nitroglycerin?**

- A) Coumadin
- B) Aspirin
- C) Viagra
- D) Tylenol

*Ans: C*

---

**34. He is diaphoretic; his respirations are 18 per minute and adequate; his pulse is 72; and his blood pressure is 80/62 mmHg. He is on a nonrebreather mask at 15L per minute. You would:**

- A) assist him in taking his third and final nitroglycerin tablet.
- B) attach him to an AED.
- C) reassess his vital signs every 5 minutes and transfer him by EMS care.
- D) give him only half of a nitroglycerin tablet.

*Ans: C*

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**35. Which of the following actions is most important before assisting a patient to take prescribed nitroglycerin?**

- A) Determine whether the patient has any allergies.
- B) Evaluate the patient's blood pressure.
- C) Determine whether the patient has a headache.
- D) Attach the patient to an AED.

*Ans: B*

---

**36. An alert and oriented 65-year-old woman presents with shortness of breath, noisy respirations, jugular vein distention, and edema in her feet and ankles. Her pulse is 132, her respirations are 24, and her blood pressure is 86/68. Based on these findings, you would suspect that the patient:**

- A) has an aortic dissection/aneurysm.
- B) is having a hypertensive emergency.
- C) is experiencing congestive heart failure.
- D) has pericardial tamponade.

*Ans: C*

---

**37. You are assessing a 38-year-old woman who is complaining of the sudden onset of chest pain and shortness of breath. She describes the pain as sharp and states that it increases when she takes a deep breath. You note that she is breathing rapidly. Her BP is 130/82, and her pulse is 100. Based on her presentation, you suspect that she may have:**

- A) pulmonary embolism.
- B) pericardial effusion.
- C) unstable angina.
- D) an aortic dissection/aneurysm.

*Ans: A*

---

**38. Before nitroglycerin is administered, you should ensure that:**

- A) nitroglycerin is in fact prescribed for the patient.
- B) the patient rates his chest pain as at least 7 on a scale of 1 to 10.
- C) the patient's systolic blood pressure is at least 110 mmHg.
- D) the patient is standing.

*Ans: A*

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## Section 16 — Abdominal

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10 Questions

**1. When a patient can point to an exact location of pain over their abdomen, this is known as \_\_\_\_\_ pain.**

- A) diffuse
- B) parietal
- C) visceral
- D) referred

*Ans: B*

---

**2. When a patient has generalized abdominal pain and can only describe a larger area that is painful, it is known as \_\_\_\_\_ pain.**

- A) parietal
- B) rebound
- C) visceral
- D) celiac

*Ans: C*

---

**3. Which one of the following organs is not located in the abdomen?**

- A) Stomach
- B) Kidney
- C) Pancreas
- D) Small intestines

*Ans: B*

---

**4. Lower abdominal pain could indicate which of the following conditions?**

- A) An irregular heart beat
- B) An auditory tract infection
- C) A fractured rib
- D) A urinary tract infection

*Ans: D*

---

**5. A 28-year old healthy male presents with recent signs and symptoms of nausea and vomiting but very little abdominal pain. Which one of the following is your best course of action?**

- A) Suggest he follow up with his physician if the symptoms continue.
- B) Call for ALS since you suspect the patient is having a myocardial infarction.
- C) Have the patient urinate in a cup and look for blood in the urine.
- D) Suggest the patient take some over the counter analgesics to see if they help and release back to skiing.

*Ans: A*

---

**6. Which one of the following statements regarding an acute abdomen is true?**

- A) OEC technicians need to be able to pinpoint the exact source of an acute abdomen.
- B) Rarely is there nausea and vomiting accompanying an acute abdomen.
- C) OEC technicians should recognize that a patient is very ill and needs to go to the hospital.
- D) An acute abdomen rarely results in sepsis or hypovolemic shock.

*Ans: C*

---

**7. Which of the following statements regarding assessment of the abdomen is false?**

- A) Assessing abdominal complaints can be difficult because the nerve pathways for the gastrointestinal, urinary, and reproductive systems are in relatively close proximity to each other.
- B) Abdominal and pelvic organs are close to each other, and problems affecting one system or organ can seriously affect organs in another system.
- C) Assessment of the abdominal organs is straightforward because the abdominal organs are specifically located and easily palpated.
- D) One of the problems encountered while assessing abdominal complaints is a patient's reluctance to allow the exam due to embarrassment or pain.

*Ans: C*

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**8. Colic is often caused by which one of the following?**

- A) Obstruction and distention of a hollow organ
- B) Drinking too much water
- C) Contraction of the abdominal muscles from injury
- D) Passing a bloody stool

*Ans: A*

---

**9. What is typically the most comfortable position for a responsive patient who is suffering from abdominal pain?**

- A) Supine with feet elevated 8 inches
- B) Left lateral position
- C) Trendelenburg position
- D) Supine with knees slightly flexed

*Ans: D*

---

**10. When dealing with a 61-year-old female patient with acute lower right abdominal pain, which quadrant should be examined first?**

- A) Right upper
- B) Left upper
- C) Right lower
- D) Left lower

*Ans: B*

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## Section 17 — Obstetrics

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### 9 Questions

**1. The function of the uterus in pregnancy is to:**

- A) permit the excretion of wastes from the baby to the mother.
- B) protect the abdominal organs during fetal growth.
- C) manufacture specific blood cells needed for fetal survival.
- D) provide an environment for the ovum to develop into a fetus.

*Ans: D*

---

**2. Which of the following statements concerning the functions of the vagina is true?**

- A) The vagina allows the passage of a baby during delivery and the outflow of urine from the bladder.
- B) The vagina permits the outflow of menstrual blood and is the site at which a sperm fertilizes an egg.
- C) The vagina serves as a passageway for the fetus during delivery and an outlet for blood during menstruation.
- D) The vagina allows the excretion of urine from the bladder and the outflow of menstrual blood.

*Ans: C*

---

**3. Which of the following statements about the umbilical cord is true?**

- A) It connects the placenta and the mother's uterus.
- B) It detoxifies blood reaching the baby before delivery.
- C) It transmits nourishment from the placenta to the fetus.
- D) It always connects to the top side of the uterus.

*Ans: C*

---

**4. Which of the following findings should be of greatest concern when assessing a 33-year-old woman who is 8 months pregnant?**

- A) Shortness of breath when lying flat
- B) Recent onset of a fever of 99.3°F
- C) Daily contractions that are irregular and painful
- D) Vaginal bleeding and abdominal pain

*Ans: D*

---

**5. The most common cause of fetal death due to trauma to a pregnant woman's abdomen is:**

- A) abruptio placentae.
- B) a crushing injury to the fetus.
- C) penetration of the uterus.
- D) premature labor.

*Ans: A*

---

**6. In abruptio placentae, the greatest threat to the baby is:**

- A) infection.
- B) maternal shock.
- C) fetal death.
- D) pain.

*Ans: C*

---

**7. Hypotension can occur after 20 weeks of gestation when a pregnant woman is in a supine position. Which of the following actions by an OEC technician will not prevent hypotension?**

- A) Placing her in the left lateral recumbent position
- B) Elevating her right hip approximately 1-2 inches
- C) Elevating her feet 6-12 inches
- D) Tipping the spine board so that she is slightly tilted onto her left side

*Ans: C*

---

**8. The second stage of labor ends with:**

- A) contractions that become regular and occur every 3-4 minutes.
- B) delivery of the placenta.
- C) delivery of the baby.
- D) rupture of the amniotic sac.

*Ans: C*

---

**9. At what point during a delivery should you consider using suction to clear the airway of the baby?**

- A) After the baby's head has been delivered but before the torso is out of the vaginal opening and if there is obvious obstruction to spontaneous breathing
- B) Before the cord has been cut but after the baby has been assessed and determined to be stable and if there is obvious obstruction to spontaneous breathing
- C) When the baby has been delivered and has been dried and warmed
- D) When the baby's chest has been delivered but before the legs are free of the vaginal canal

*Ans: A*

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## Section 18 — Trauma – MOI

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23 Questions

**1. A heavy-set snowboarder was critically injured when he crashed into a tree on a dimly lit trail. It was reported that he appeared to be out of control while traveling at a high rate of speed down the expert trail. Which of the following factors had the greatest impact on the extent of his injuries?**

- A) The skier's weight
- B) The size of the tree he hit
- C) The dimly lit trail
- D) The speed the skier was going

*Ans: D*

---

**2. Which of the following best defines mechanism of injury?**

- A) Physical injury caused by an external force
- B) A set of problems associated with a specific source of trauma
- C) The study of movement of body segments without consideration given to its mass or the forces making it move
- D) The method that an injury occurs

*Ans: D*

---

**3. Which of the following factors does not directly relate to the severity of bodily injury?**

- A) The amount of kinetic energy absorbed
- B) The direction the kinetic energy travels
- C) The size of the injured person
- D) The density of the structures impacted

*Ans: C*

---

**4. A patient involved in an altercation was struck in the ribs with a baseball bat. Your assessment reveals intact skin with significant bruising to the right lateral chest. When palpating this area, you note instability and crepitus to the rib cage. An OEC technician would recognize a:**

- A) thoracic injury caused by penetrating trauma.
- B) chest injury caused by blunt trauma.
- C) pulmonary injury caused by penetrating trauma.
- D) chest wall injury caused by acceleration forces.

*Ans: B*

---

**5. A 49-year-old man has been stabbed in the lower right chest. After assessing him, you suspect that the knife punctured the lung and is causing internal bleeding. In this situation, the mechanism of injury would be:**

- A) a penetrating injury.
- B) an assailant with a knife.
- C) blunt trauma to the lungs.
- D) internal hemorrhage.

*Ans: A*

---

**6. Your friends are discussing a hunting accident they heard about on the news. A hunter was shot with a high-powered rifle, and the report indicated that he had damage to several internal organs. From your knowledge of mechanisms of injury, you can tell your friends that this type of injury is called a:**

- A) high-impact injury.
- B) high-velocity penetrating injury.
- C) projectile injury.
- D) blast injury.

*Ans: B*

---

**7. Your patient has been stabbed in the chest with a ski pole. The mechanism of injury is:**

- A) blunt.
- B) crush.
- C) penetrating.
- D) rotational.

*Ans: C*

---

**8. Which of the following terms is not a "mechanism of injury"?**

- A) Blunt trauma to the abdomen
- B) Rotational spiral mid-shaft tibia fracture
- C) Kidney stone
- D) Crushed foot

*Ans: C*

---

**9. High-velocity injuries and low-velocity injuries are two types of which mechanism of injury?**

- A) Blunt trauma
- B) Penetrating injury
- C) Blast injury
- D) Closed trauma

*Ans: B*

---

**10. A high-velocity penetrating injury is usually the result of:**

- A) the collision of cars traveling at greater than 65 mph.
- B) the impact of a bullet from a high-powered rifle.
- C) an out-of-control skier hitting another skier.
- D) stab wounds.

*Ans: B*

---

**11. Tissue damage caused by low-velocity injuries is usually:**

- A) not very deep.
- B) not life-threatening.
- C) limited to the path of the object.
- D) known to occur to avalanche victims.

*Ans: C*

---

**12. You are called to the vehicle maintenance department on the mountain, where a fairly large explosion has occurred. The first injured person you talk to tells you that he is having excruciating ear pain. You recognize that his ear pain may be caused by:**

- A) a crush injury.
- B) a blast injury.
- C) a penetrating injury.
- D) a blunt injury.

*Ans: B*

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**13. Dislocating a shoulder after catching a ski pole on a tree branch is an example of an injury caused by which MOI?**

- A) Rotational injury
- B) Blunt injury
- C) Whiplash injury
- D) Compressive injury

*Ans: A*

---

**14. Your patient is a skier who tells you that she caught an edge and twisted her leg. You later find out that the skier suffered from a fractured tibia and fibula. This injury was the result of which mechanism?**

- A) Rotational
- B) Blunt
- C) Crush
- D) Penetration

*Ans: A*

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**15. A rock climber falls when several boulders give way. His arm is pinned between two boulders. This MOI is best described as:**

- A) blunt.
- B) penetrating.
- C) crush.
- D) rotational.

*Ans: C*

---

**16. Your patient has been involved in a backcountry hiking accident and has multi-system trauma. It is most important to consider which of the following when making your evacuation plans.**

- A) Location of the patient's home
- B) Location of the patient's nearest family member
- C) Location of the patient's primary doctor
- D) Location of the nearest trauma center

*Ans: D*

---

**17. Trauma centers have five levels, which are based on:**

- A) proximity to ski areas.
- B) number of doctors on staff.
- C) specialization, complexity, and availability of care.
- D) insurance providers.

*Ans: C*

---

**18. Trauma centers are specifically designed to:**

- A) provide training to physicians who will work with trauma patients.
- B) manage patients with multi-system trauma.
- C) encourage patients with less urgent injuries to use lower-level facilities.
- D) be testing areas for physicians who wish to be "board certified" as trauma surgeons.

*Ans: B*

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**19. Assessment and management of the trauma patient should include:**

- A) reassessing the patient every 30 minutes.
- B) transporting trauma patients as soon as the Scene Size Up is complete.
- C) initiating CPR on all patients with chest trauma.
- D) gathering information about the mechanism of injury.

*Ans: D*

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**20. A 23-year-old skier falls about 20 feet from the chairlift. Aside from his complaints of soreness, you don't see any obvious injuries. The skier states that as long as his legs aren't broken, he's going to continue skiing. Which of the following statements would be your best response to him?**

- A) "Since this happened at a ski area, you should get evaluated. You may be able to sue."
- B) "From a fall of that height, you may have some injuries. You really should be examined."
- C) "Since you did not lose responsiveness, it's probably OK to go back to skiing. If your legs start to hurt, you should probably get checked."
- D) "You may feel OK now, but you will probably be sore later. You should go to the hospital and get some pain medication."

*Ans: B*

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**21. You arrive alone at a scene at which a 16-year-old skier has collided with a tree. He is unresponsive and has blood flowing from his ears and nose. He was not wearing a helmet. Which of the following lists reflects the most appropriate care of this patient?**

- A) Primary assessment, airway management, immobilization, transport as soon as possible
- B) Primary assessment, oxygen administration, rapid transport to the aid room, immobilization in the aid room
- C) Immobilization, transfer to a sled, completion of the assessment in the aid room
- D) Primary assessment, immobilization, bandaging of the ears to control bleeding, airway management

*Ans: A*

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**22. Your patient is an adult skier who collided with a snowboarder and was found to be unresponsive. On scene is a family member of the adult unresponsive skier. The scene is safe. After establishing the ABCDs and stabilizing the patient's head and spine, you will:**

- A) take off a ski boot and see if you can find a pulse.
- B) try to obtain a medical history.
- C) perform a head-to-toe secondary assessment.
- D) reassess the patient every 5 minutes.

*Ans: B*

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**23. The term golden hour refers to the:**

- A) time period when patient survival rates may be enhanced if critical injuries are identified and managed.
- B) amount of time you have to decide if a patient needs to be sent to a trauma hospital.
- C) amount of time it takes to stop the bleeding from a penetrating injury.
- D) hour before sunset, when most cases of on-hill trauma occur.

*Ans: A*

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## Section 19 — Soft Tissue & Burns

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60 Questions

**1. Which of the following lists correctly names the layers of the skin and immediately underlying tissue?**

- A) Dermis, muscle, epidermis
- B) Epidermis, subcutaneous, muscle
- C) Dermis, subcutaneous, muscle
- D) Subcutaneous, epidermis, dermis

*Ans: D*

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**2. What is the function of subcutaneous fat?**

- A) It produces glucose needed for energy.
- B) It has no real function and is unhealthy and detrimental to survival.
- C) It always projects underlying organs from trauma.
- D) It insulates the body and stores energy.

*Ans: D*

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**3. A patient has a laceration on his arm from a table saw. Assessment reveals dark red blood flowing slowly from the wound. You would recognize this type of bleeding as:**

- A) arterial bleeding.
- B) insignificant bleeding.
- C) venous bleeding.
- D) lymphatic bleeding.

*Ans: C*

---

**4. You are responding to a call to aid a patient who has cut himself with a knife. Reportedly, arterial bleeding is involved. Given this information, which of the following findings do you expect?**

- A) Flowing dark-red blood that is difficult to control
- B) Oozing bright- red blood that is easy to control
- C) Spurting dark-red blood that is difficult to control
- D) Bright- red blood that is spurting with each beat of the heart

*Ans: D*

---

**5. The skin performs which of the following functions?**

- A) Regulation of temperature
- B) Prevention of blood loss
- C) Elimination of carbon dioxide
- D) Dissipation of internal cold

*Ans: A*

---



**6. Which of the following injuries is most likely to result in capillary bleeding only?**

- A) An avulsion
- B) An abrasion
- C) A laceration
- D) A penetration

*Ans: B*

---

**7. You are having a difficult time controlling bleeding from a small skin avulsion on a patient's ankle. Which of the following statements made by the patient best explains why the bleeding has been so difficult to control?**

- A) "I take Coumadin (an anticoagulant) for my irregular heartbeat."
- B) "My blood pressure sometimes runs a little high."
- C) "I take steroids for my lung disease."
- D) "I drank a lot of alcohol last night."

*Ans: A*

---

**8. During your secondary assessment, a patient with a stab wound to the abdomen tells you that he is taking Coumadin, a blood thinner. Which of the following implications would taking this medication have for this patient?**

- A) The medication would not affect the patient or the care you provide.
- B) The medication would cause this patient's blood to clot more quickly, thereby minimizing any bleeding that might occur.
- C) The medication could cause the patient to bleed more profusely.
- D) The medication would reduce the patient's blood pressure, so bleeding would be less profuse.

*Ans: C*

---

**9. Assessment of a patient who fell reveals bruising to her right buttock. Her skin is intact, and she complains of tenderness in the area when you palpate it. You would recognize which of the following types of injury?**

- A) An avulsion
- B) An abrasion
- C) An open injury
- D) A contusion

*Ans: D*

---

**10. When assessing a patient, you note a bruise on the chest. Another word for documenting a bruise is:**

- A) ecchymosis.
- B) cyanosis.
- C) erythematic.
- D) hematoma.

*Ans: A*

---

**11. Which of the following phrases best describes a closed-tissue injury?**

- A) A wound that has scabbed over or healed
- B) A wound in which platelets and proteins have formed a plug
- C) A wound with no obviously visible opening, such as a needle stick
- D) A wound in which underlying tissues are damaged, but the overlying skin remains intact

*Ans: D*

---

**12. Which of the following assessment findings best indicates that a patient's injury should be classified as an abrasion?**

- A) Intact skin with ecchymosis noted; patient describes minor pain
- B) Jagged breaks in the skin; bleeding moderate but controllable
- C) Intact skin with a large accumulation of dark blood underneath
- D) Red scrapes involving the outermost layer of skin only

*Ans: D*

---

**13. A 49-year-old man was climbing on rocks when he fell backward to the ground. He presents with a two-inch linear wound on the top of his head. Bleeding has been controlled. You would document this type of injury as a(n):**

- A) avulsion.
- B) contusion.
- C) abrasion.
- D) laceration.

*Ans: D*

---

**14. You respond to a patient who has an avulsion on the left arm. When you assess this patient, you would expect to find:**

- A) a loose flap of torn skin on the left arm.
- B) abraded skin with an underlying bone fracture.
- C) bruised skin with a piece of bone protruding from the left arm.
- D) a long and deep laceration on the left arm.

*Ans: A*

---

**15. A patient has been stabbed in the chest with a 6- inch-long ice pick. When you assess the wound, you find it to be very small and insignificant in appearance, with minimal bleeding. Which of the following assumptions can you correctly make about this injury?**

- A) The wound is most likely superficial.
- B) Damage to the underlying structures is likely.
- C) Not enough information is available to make any assumptions.
- D) Because the entrance wound is small, severe bleeding is highly unlikely.

*Ans: B*

---

**16. Mechanical tattooing is defined as:**

- A) a form of body art in which dye is injected under the skin.
- B) a type of self-mutilation requiring psychiatric follow-up for the patient.
- C) the introduction of foreign debris such as dirt, gunpowder, or small rocks into the skin.
- D) multiple punctures made by a nail gun.

*Ans: C*

---

**17. Thermal burns caused by the skin directly contacting hot water or hot steam:**

- A) are the least common type of burn.
- B) are the most common type of burn.
- C) result from tanning beds.
- D) result from exposure to Gamma radiation.

*Ans: B*

---

**18. Which of the following substances cause the most serious chemical burns?**

- A) An acid with a pH of 1
- B) A chemical with a pH of 7
- C) An alkaline chemical or base with a pH of 8
- D) An acid with a pH of 6

*Ans: A*

---

**19. Exposure to direct or alternating electrical current produces severe external and internal burn injuries because:**

- A) the skin is a poor conductor with a high resistance, and blood vessels are good conductors with low resistance.
- B) both the skin and blood vessels are good conductors with low resistance.
- C) the skin is a good conductor with a high resistance, and blood vessels are good conductors with low resistance.
- D) skin is a poor conductor with a low resistance, and blood vessels are good conductors with low resistance.

*Ans: A*

---

**20. When managing electrical burns, OEC technicians should always:**

- A) use moist dressings.
- B) use antibiotic ointments.
- C) assess for signs and symptoms of internal injuries.
- D) use a topical ointment on the skin.

*Ans: C*

---

**21. Which of the following statements concerning a patient who received electrical burns to his hand after grabbing a live wire is true?**

- A) The burning will continue for hours, perhaps days.
- B) The patient will remain an electrocution hazard to rescuers for several minutes after being removed from the source of electricity.
- C) The extent of tissue damage may be much greater than it appears on the surface.
- D) Electrical burns are a minor injury.

*Ans: C*

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**22. In a hospital accident, what might cause radiation burns?**

- A) X-ray machine
- B) Steam pipes
- C) Spilled soup
- D) Exposed wires

*Ans: A*

---

**23. Severe external bleeding should be controlled during what phase of a patient assessment?**

- A) The primary assessment
- B) The scene size-up
- C) The rapid secondary survey
- D) The reassessment

*Ans: A*

---

**24. When managing a patient with soft-tissue injuries, it is essential that you:**

- A) thoroughly clean all lacerations.
- B) control any bleeding with sterile dressings only.
- C) use the appropriate personal protective equipment.
- D) cover any hematomas with an occlusive dressing.

*Ans: C*

---

**25. Which of the following sentences best describes a partial- thickness burn?**

- A) The skin is red and moist, and blisters have formed.
- B) The skin is charred or blackened and lacks sensation.
- C) The skin is red but dry, no blisters formed, and painful.
- D) The skin is white and dry with no sensation of pain.

*Ans: A*

---

**26. A burn extending through the epidermis and dermis layers into the subcutaneous layer would be classified as a:**

- A) deep partial-thickness burn (second-degree burn).
- B) superficial burn (first-degree burn).
- C) superficial partial- thickness burn (second-degree burn).
- D) full-thickness burn (third- or fourth-degree burn).

*Ans: D*

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**27. Which degree of burn causes damage to bone?**

- A) First
- B) Second
- C) Third
- D) Fourth

*Ans: D*

---

**28. Your patient is a 40-year-old man who was burned when he spilled gasoline on his pants while he was standing near the pilot light of his hot-water heater. He has partial thickness burns from his feet to just above his knees, circumferentially around both legs. According to the Rule of Nines, the man burned \_\_\_\_ of his body surface.**

- A) 9%
- B) 4.5%
- C) 36%
- D) 18%

*Ans: D*

---

**29. Hoarseness or voice changes in a burn patient should alert an OEC technician that the:**

- A) patient may be extremely anxious about the injury.
- B) patient most likely is experiencing an inhalation injury causing swelling to the airway, a true emergency.
- C) patient should be treated as a pediatric patient.
- D) patient has altered mental status.

*Ans: B*

---

**30. In the Rule of Nines, the surface of the left leg is about what percentage of the body surface?**

- A) 1%
- B) 3%
- C) 9%
- D) 18%

*Ans: D*

---

**31. Thermal burns result from:**

- A) transmitted Gamma radiation.
- B) direct contact between a heat source and the skin.
- C) indirect contact between a heat source and the skin.
- D) a chemical reaction stimulated by a heat source.

*Ans: B*

---

**32. You arrive at a scene at which a 16-year-old girl tells you she has cut her wrist with a knife. She appears to be pale, and blood is spurting from the cut. Which of the following actions should you take first after making the scene safe?**

- A) Ask her how the injury happened
- B) Apply direct pressure to the cut
- C) Apply oxygen at 15 LPM via a nonrebreather mask
- D) Obtain a blood pressure reading

*Ans: B*

---

**33. You are treating a patient who has cut herself while working in the lodge kitchen. She has a jagged laceration on her left forearm that is bleeding steadily and heavily. While attempting to control the bleeding, you should first:**

- A) tightly wrap roller gauze around a large sterile dressing placed over the injury.
- B) apply direct pressure to the elbow pressure point to slow the flow of blood into the arm.
- C) place a sterile dressing over the site and hold steady pressure on the dressing with your gloved hand.
- D) wrap the injury tightly with roller gauze and apply a tourniquet proximal to the laceration.

*Ans: C*

---

**34. You have just arrived on scene, where a young girl has pushed her arm through a plate glass window. She is responsive and has a patent airway. Bright red blood is spurting from a large laceration on her right upper arm. When your partner tells you to apply direct pressure to the laceration to prevent the further blood loss, you would:**

- A) place a tourniquet proximal to the laceration, being careful not to get blood on your hands.
- B) put gloves on before applying direct pressure over the injury site.
- C) control the bleeding by pushing the edges of the laceration together until you can put gloves on.
- D) apply direct pressure to the site with your hands until someone wearing gloves can take over.

*Ans: B*

---

**35. Which of the following statements about pressure dressings and bandages is true?**

- A) All pressure dressings must be airtight to prevent contamination.
- B) Bandages should be applied in a distal-to-proximal manner.
- C) Dressings are placed over bandages.
- D) Gauze should be placed in an open wound before a pressure dressing is applied.

*Ans: B*

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**36. For which of the following injuries may elevation be used to control bleeding?**

- A) For a laceration to the forearm
- B) For no injuries, as elevation has not been shown to help control bleeding
- C) For an abrasion on the lower leg
- D) For a puncture wound to the foot

*Ans: B*

---

**37. Which of the following actions should you take next when direct pressure and packing the wound have failed to control arterial bleeding on a patient's lower leg?**

- A) Provide oxygen at 15 LPM with a nonrebreather mask.
- B) Apply a tourniquet below the knee.
- C) Immediately transport the patient by ambulance.
- D) Apply a tourniquet above the knee and splint the extremity.

*Ans: B*

---

**38. Which of the following statements indicates that the speaker understands the application of a tourniquet?**

- A) "A tourniquet should be applied tightly enough so that arterial blood flow distal to the tourniquet is completely stopped."
- B) "If the tourniquet appears to have stopped the bleeding, you should loosen it slightly to allow a small amount of blood to perfuse the tissues."
- C) "Arteries run close to the body's surface in a joint, so the best place to apply a tourniquet is over the joint just above the artery."
- D) "A tourniquet should be applied only when directed by medical control."

*Ans: A*

---

**39. Which of the following statements about the use of a tourniquet is true?**

- A) It should be placed as distal as possible but at least several inches proximal to the wound.
- B) It can be used on most any part of the body except the neck.
- C) On an arm or leg, it should be placed 1 inch from either the elbow or knee.
- D) Tourniquets are a first resort and should be used frequently.

*Ans: A*

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**40. A wire would be a poor choice for making a tourniquet since a tourniquet should be at least how wide?**

- A) 1 inch
- B) 2 inches
- C) 3 inches
- D) 4 inches

*Ans: B*

---

**41. A young boy was out riding his bike in the neighborhood when he fell onto a stick protruding from the ground. The stick impaled him in the neck. The boy immediately pulled the stick out. Assessment reveals a gaping wound to the boy's right neck. Which of the following types of dressing would you use on this injury?**

- A) Sterile gauze carefully placed inside the open wound to control bleeding
- B) An occlusive dressing that is taped on all sides
- C) A sterile dressing placed over the wound and bandaged with roller gauze around the neck
- D) A moist sterile dressing that is taped to the neck

*Ans: B*

---

**42. Which of the following statements indicates that the speaker understands bandaging?**

- A) "Bandages are sterile gauze pads used to stop bleeding."
- B) "Bandaging material is used to secure a dressing in place."
- C) "Bandaging material must always be sterile."
- D) "After you apply a bandage, a dressing should always be put over it."

*Ans: B*

---

**43. You have applied a pressure dressing to the calf of a woman who suffered a deep laceration from a piece of broken glass. Which of the following actions should you take next?**

- A) Apply a splint
- B) Obtain a follow-up blood pressure reading
- C) Apply a tourniquet above the laceration
- D) Check CMS in the patient's foot

*Ans: D*

---



**44. Which of the following statements indicates that the speaker understands dressing and bandaging a wound?**

- A) "When applying a bandage over a dressing, apply it very tightly so the bandage won't fall off."
- B) "If bleeding from a wound cannot be stopped initially with direct pressure, apply a bandage so you can take care of other injuries."
- C) "Leave the tips of the fingers or toes exposed when bandaging an arm or a leg so you can continue to assess CMS."
- D) "Place a bandage across the joint proximal to the wound because this helps control bleeding by applying pressure to the underlying blood vessels."

*Ans: C*

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**45. The most effective method for treating a contusion is to apply a(n):**

- A) pressure dressing.
- B) tourniquet.
- C) bandage.
- D) ice pack.

*Ans: D*

---

**46. You are assessing a patient who received multiple injuries in a fight. Which of the following injuries would present the greatest danger for infection?**

- A) A contusion to the chin
- B) A closed crushing injury to the thigh
- C) An abrasion on the right side of the face
- D) A hematoma in the left foot

*Ans: C*

---

**47. Which of the following instructions is appropriate for helping a new patroller care for a patient with a very large abrasion on his arm?**

- A) "When you are done cleaning the abrasion, apply some sterile gauze soaked in sterile saline."
- B) "It is best to let the wound bleed some. This is the body's natural way of cleaning the wound."
- C) "Wash the gross contamination off, pouring on sterile saline, put on a sterile dressing, and recommend the patient go to the hospital for further care."
- D) "Do not cut away the sleeve over the top of the laceration because that will restart the bleeding."

*Ans: C*

---

**48. A patient has had part of his right thumb amputated in an accident. Friends have retrieved the thumb and wrapped it in a towel. When you arrive, you would demonstrate appropriate handling of the amputated part by:**

- A) placing the thumb in a plastic bag filled with ice.
- B) keeping the thumb in the towel.
- C) placing the thumb in a container of sterile saline or sterile water.
- D) wrapping the thumb in a moist sterile dressing, placing it in a sterile bag, and keeping it cool.

*Ans: D*

---

**49. Which of the following statements about amputations is true?**

- A) Keep the amputated part at body temperature when sending it to the hospital with the patient.
- B) A badly mangled or damaged amputated part should not be preserved or sent to the hospital because it cannot be reattached.
- C) Surgeons can reattach an amputated part only if it has been without a blood supply for less than one hour.
- D) All amputated body parts should be preserved and sent to the hospital with the patient.

*Ans: D*

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**50. Which of the following statements about impaled objects not affecting the airway is true?**

- A) Allow only the patient to pull the impaled object out because that will hurt less.
- B) The impaled object should be left in place during transport.
- C) A wound made by an impaled object always bleeds profusely.
- D) An impaled object never causes serious underlying tissue damage.

*Ans: B*

---

**51. Which of the following statements about a highpressure injection injury is false?**

- A) The presenting wound is typically large.
- B) The injury involves the introduction of a liquid or gas into the body from a pressurized source.
- C) The injury may require surgical amputation of the injured part.
- D) The injury typically involves the limbs.

*Ans: A*

---

**52. Which of the following statements about an impaled object is true?**

- A) The primary treatment goal in the field is to stabilize the object in the position found.
- B) The primary treatment in the field is to remove the object without causing further bleeding or causing as little bleeding as possible.
- C) OEC technicians may remove only objects that are impaled in an extremity.
- D) OEC technicians may remove objects that are impaled in the chest or abdomen.

*Ans: A*

---

**53. Your patient is a 25-year-old man who has a reddened area with blisters across the palm of his hand after grabbing the handle of a very hot iron skillet. Which of the following actions must be avoided in the prehospital management of this wound?**

- A) Applying a dry, sterile dressing
- B) Cooling the burn with cool water
- C) Applying an antibiotic ointment
- D) Monitor for shock

*Ans: C*

---

**54. For the OEC technician, the first priority in managing a patient with a thermal burn is:**

- A) assessing for airway compromise.
- B) completing a secondary survey.
- C) stopping the burning process to prevent further injury.
- D) applying dry sterile dressings.

*Ans: C*

---

**55. Your patient is a 35-year-old woman who spilled a cup of hot coffee on herself two hours ago. An area on her right thigh that is about twice the size of the palm of her hand is red and painful but has no blisters. Which of the following actions would be appropriate for an OEC technician to take?**

- A) Apply a lotion containing a topical anesthetic and aloe vera
- B) Apply a dry sterile dressing
- C) Apply an antibiotic ointment
- D) Apply a plastic bag full of ice to the skin of the affected area

*Ans: B*

---

**56. In treating a burn patient, you should cool and irrigate the burned tissue with:**

- A) room-temperature tap water
- B) chilled tap water for 5 minutes.
- C) tap water warmed to 102°F for 15 minutes.
- D) tap water warmed to 95°F for 10 minutes.

*Ans: A*

---

**57. Your patient is a 10-year-old boy who was exposed to a dry chemical powder and is complaining of severe pain at the site of contact on both of his hands. There is no decontamination shower on site. Which of the following actions would be the best way to manage this situation?**

- A) Brush away as much of the powder as possible and then have the patient hold his hands under running water from a faucet or garden hose.
- B) Have the fire department connect to a hydrant and spray the patient down from head to toe.
- C) Brush away as much powder as possible and then pour a bottle of sterile saline solution over the patient's hands.
- D) Brush away the powder and then bandage the patient's hands in a position of function.

*Ans: A*

---

**58. Chemical burns result from exposure to:**

- A) ultraviolet light.
- B) steam.
- C) hot light bulbs.
- D) caustic substances.

*Ans: D*

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**59. For the OEC technician, the first priority in managing a patient with an electrical burn is:**

- A) assessing for airway compromise.
- B) assessing for potential c-spine injury.
- C) ensuring the power source has been turned off.
- D) ensuring you have an AED available if needed.

*Ans: C*

---

**60. When treating a patient burned by a dry chemical, it is important to brush away any dry residue before flushing with water because:**

- A) flushing will expose other areas around the burn.
- B) some dry chemicals are activated by water.
- C) a containment area needs to be set up to catch the runoff water.
- D) flushing will cause the skin's pores to open.

*Ans: B*

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## Section 20 — Musculoskeletal

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29 Questions

**1. Which of the following functions is not one of the several functions of bones?**

- A) Providing structure and form for body tissues
- B) Protecting vital organs
- C) Responding to sensory nerve stimulation to protect skin and other tissues
- D) Producing red blood cells

*Ans: C*

---

**2. The inside layer of a joint capsule where cells make a viscous fluid for lubricating the joint is the:**

- A) synovium.
- B) cartilage.
- C) callus.
- D) fascia.

*Ans: A*

---

**3. The outermost part of a bone is a tough lining known as:**

- A) the cortex.
- B) articular cartilage.
- C) synovium.
- D) the periosteum.

*Ans: D*

---

**4. An example of a joint that permits very little movement of the bones involved is the:**

- A) hip.
- B) shoulder.
- C) acromioclavicular.
- D) elbow.

*Ans: C*

---

**5. Which of the following terms does not describe a type of joint?**

- A) Ball-and-socket
- B) Hinge
- C) Slightly moveable
- D) Post

*Ans: D*

---

**6. During your SAMPLE inquiry, a patient reports that he had surgery to repair a torn ligament. Based on your training, you know that a ligament is:**

- A) a thick muscle that surrounds a joint.
- B) a tissue that stabilizes two contiguous bone ends.
- C) a muscle that connects to a bone.
- D) connective tissue that connects muscle and bone.

*Ans: B*

---

**7. The type of muscle found in the forearm is:**

- A) skeletal muscle.
- B) smooth muscle.
- C) cardiac muscle.
- D) contractor muscle.

*Ans: A*

---

**8. The structure responsible for transmitting the force of a contracting skeletal muscle to a bone is a:**

- A) tendon.
- B) ligament.
- C) cartilage.
- D) joint.

*Ans: A*

---

**9. While you are putting wood into your wood stove, your hand comes in contact with the hot surface. Your body responds by pulling your hand away from the heat. How is this action accomplished?**

- A) One or more skeletal muscles receive a signal from the brain to contract.
- B) The smooth muscles in the arm automatically respond to heat.
- C) You use conscious thought to pull your hand from the heat.
- D) Extension of the ligaments in the elbow joint pulls the lower arm up.

*Ans: A*

---

**10. Which of the following statements about skeletal (voluntary) muscles is true?**

- A) They are mostly located on the torso of the body.
- B) They make up less mass in the body than do involuntary muscles.
- C) They have the ability to cause movement by extending.
- D) They generally connect to the skeletal system.

*Ans: D*

---

**11. A fracture of a bone will cause bleeding, and a hematoma forms around the fracture site. Over the next several weeks, this hematoma organizes into a substance called:**

- A) a callus.
- B) calcium.
- C) cartilage.
- D) a synovium.

*Ans: A*

---

**12. Which of the following statements about musculoskeletal injuries is true?**

- A) A sprain is an injury to a joint that involves the stretching or tearing of ligaments.
- B) Strains are injuries that occur to ligaments and to the joints to which they are attached.
- C) A fracture occurs when a tendon connecting a muscle to a bone is overstretched and becomes injured.
- D) A dislocation occurs when a bone that forms part of a joint is broken.

*Ans: A*

---

**13. Which of the following musculoskeletal tissues does not heal well?**

- A) Tendon
- B) Ligament
- C) Muscle
- D) Cartilage

*Ans: D*

---

**14. A patient for whom you recently provided care stops by to thank you. He reports that he was diagnosed with a strain to his lower right leg. Which of the following structures is the primary structure affected in that injury?**

- A) A joint
- B) A ligament
- C) A muscle
- D) A bone

*Ans: C*

---

**15. Your secondary assessment of a patient who is complaining of pain in his right leg reveals an opening in the skin where the fractured tibia broke through the skin and retreated back into the leg. You would recognize this injury as a(n):**

- A) open fracture.
- B) closed fracture.
- C) laceration.
- D) partial fracture.

*Ans: A*

---

**16. The finding that upon muscle contraction the active motion of a joint is reduced or lost suggests:**

- A) a dislocation.
- B) a ligament strain.
- C) torn cartilage.
- D) a ruptured tendon.

*Ans: D*

---

**17. Which of the following statements indicates that an OEC technician has a good understanding of what a joint dislocation is?**

- A) "If a joint is dislocated, the patient can move it without pain."
- B) "A dislocation is a separation or displacement of the bones of a joint."
- C) "A distal pulse is absent only in the case of a fracture."
- D) "Patients with a joint dislocation experience very little pain."

*Ans: B*

---

**18. Displacement of the bones of a joint is known as:**

- A) a dislocation.
- B) a fracture.
- C) a strain.
- D) a sprain.

*Ans: A*

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**19. The most commonly dislocated joint in the body is:**

- A) the elbow.
- B) the shoulder.
- C) the thumb.
- D) a knee.

*Ans: B*

---

**20. Your patient fell 20 feet onto his shoulder. Which of the following signs or symptoms indicates the highest probability that the humerus has been fractured?**

- A) Angulation
- B) Pain in the right upper arm
- C) Swelling in the upper arm
- D) Decreased sensation in the right hand

*Ans: A*

---



**21. Your friend tells you that he wants to try snowboarding. He is concerned about his safety and asks you which type of fracture is most common in snowboarders. You learned in your OEC training that this injury involves the:**

- A) humerus.
- B) elbow.
- C) tibia.
- D) radius.

*Ans: D*

---

**22. A common injury to the carpal bones among snowboarders who fall forward on an outstretched hand is a fractured:**

- A) scaphoid.
- B) ulna.
- C) radius.
- D) thumb.

*Ans: A*

---

**23. A snowboarder's fall over the toe side of the board onto an outstretched hand is known as the:**

- A) "Drop back."
- B) "Twist."
- C) "Dog drag."
- D) "Mousetrap."

*Ans: D*

---

**24. Which of the following statements would you make to the parent of a 10-year-old snowboarder who took a hard fall and complains of some discomfort in a wrist that is slightly swollen but has no deformity?**

- A) "It's probably just a sprain."
- B) "The child needs to be medically evaluated to ensure there is not a fracture."
- C) "This is probably a pathologic fracture."
- D) "This looks like an open fracture."

*Ans: B*

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**25. The \_\_\_\_\_ can be injured when a hip is dislocated, resulting in numbness or paralysis of a lower extremity.**

- A) cranial nerve
- B) sciatic nerve
- C) tibial nerve
- D) central nervous system

*Ans: B*

---

**26. A fracture that has three or more fragments is called a(n):**

- A) impacted fracture.
- B) comminuted fracture.
- C) oblique fracture.
- D) pathologic fracture.

*Ans: B*

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**27. The most frequent injury in skiing is a:**

- A) concussion.
- B) knee sprain.
- C) thumb strain.
- D) fractured radius.

*Ans: B*

---

**28. You respond to a 9-year-old boy who was injured in the terrain park. He is complaining of pain in his right wrist. Upon examination, you note deformity and swelling of the right wrist, pinkness of the right hand, and a strong radial pulse. Which of the following actions would be part of the proper care for this boy?**

- A) Straighten the wrist to promote blood flow to the hand.
- B) Maintain the wrist below the level of the heart to decrease swelling.
- C) Apply a cold pack to the wrist to reduce swelling.
- D) Massage the wrist gently to decrease the pain.

*Ans: C*

---

**29. You are treating a patient who is complaining of moderate pain in the right knee. The knee is swollen, discolored, and flexed about 45 degrees. CMS is intact. How would you treat this patient?**

- A) Place the patient on a long spine board and transport to the aid room, where you can examine the injury more efficiently.
- B) Straighten the injured knee and then immobilize it by attaching it to the other leg.
- C) Splint the knee in the position found before moving the patient.
- D) Apply a traction splint and straighten the knee until the pain is decreased.

*Ans: C*

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## Section 21 — Head & Spine

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33 Questions

**1. You are notified that a patient is coming to the first-aid room with bruising to the lumbar area of the back. Based on this statement, you would expect to find bruising in which area?**

- A) The upper back
- B) The neck
- C) The lower back
- D) The buttocks

*Ans: C*

---

**2. You respond to an accident at which a male patient is complaining of severe back pain. He informs you that his back pain is coming from a recent fracture of his coccyx. Based on this information, what area of the back would the pain be emanating from?**

- A) The neck
- B) The upper back
- C) The lowest part of the spine
- D) The lower back

*Ans: C*

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**3. Which of the following signs or symptoms might be evident for an injury to the autonomic nervous system?**

- A) Vomiting
- B) Elevated heart rate
- C) Inability to move facial muscles
- D) Trouble remembering one's own name

*Ans: B*

---

**4. A patient was killed immediately following a selfinflicted gunshot wound to the head. Which portion of the central nervous system was most likely damaged in order to cause the rapid death of the patient?**

- A) The cerebrum
- B) The brainstem
- C) The cerebellum
- D) The meninges

*Ans: B*

---

**5. For a person to make a fist, which of the following must occur?**

- A) The central nervous system must send a message through the spinal cord and then through the peripheral nerves.
- B) The central nervous system must send a message through the spinal cord and then through the autonomic nervous system.
- C) The peripheral nervous system must send a message to the brain and then back through the peripheral nervous system.
- D) The muscles must signal the brain to send a message through the autonomic nervous system.

*Ans: A*

---

**6. A 68-year-old female patient is complaining of a headache and generalized weakness. Her husband informs you that she was with a friend yesterday and struck her face on the dashboard during a motor vehicle collision. Her past medical history includes a stroke that caused right arm weakness and high blood pressure. When assessing this patient, which of the following findings should concern you the most?**

- A) A contusion to her left cheek
- B) Weakness in the grip of the right hand
- C) Bruising behind her left ear
- D) A complaint of pain when she moves her jaw

*Ans: C*

---

**7. Which of the following findings is most consistent with a skull fracture?**

- A) Bruising behind the ear that develops several hours after the injury
- B) Bilateral dilated pupils that do not respond to light
- C) A hematoma in the occipital area of the head
- D) Altered mental status

*Ans: A*

---

**8. A mechanical injury to the brain that results in a short-term and/or a long-term neurologic deficit is:**

- A) a repetitive head injury.
- B) a subdural hematoma.
- C) neural ischemia.
- D) a traumatic brain injury.

*Ans: D*

---

**9. A male soccer player was struck in the head with a soccer ball. Players state that he was dazed for several seconds following the impact and then asked the same questions over and over. He is currently conscious and oriented to person but is confused as to place and time. He also has a reddened area on the side of his head and face. As you proceed with your assessment, his memory continues to improve. Based on these findings, you would suspect which of the following injuries?**

- A) A concussion/TBI
- B) An open head injury
- C) An epidural hematoma
- D) A cerebral contusion

*Ans: A*

---

**10. Your patient is a young female who was thrown from her mountain bike and is now confused. Assessment findings include an open airway, adequate breathing, and a strong radial pulse. Which of the following questions would be the most important to ask her friends who were with her?**

- A) "Has she ever needed to be hospitalized before?"
- B) "Are all of her immunizations up to date?"
- C) "Is she allergic to anything?"
- D) "Did she lose responsiveness?"

*Ans: D*

---

**11. You are participating in training OEC candidates and are asked about cerebral contusions. You would explain to the class that a cerebral contusion is:**

- A) an open skull fracture with increased pressure within the skull.
- B) the formation of a pocket of blood within the brain tissue.
- C) bruising and swelling of the brain tissue.
- D) active bleeding between the brain and the skull.

*Ans: C*

---

**12. A patient involved in a motor- vehicle collision has suffered the separation of a rib from a spinal vertebra. Based on the anatomy of the spine, where has this injury occurred?**

- A) The lumbar spine
- B) The cervical spine
- C) The middle spine
- D) The thoracic spine

*Ans: D*

---

**13. A young intoxicated male patient cannot move his lower extremities after diving into the shallow end of a pool and hitting the bottom with his head. Which of the following mechanisms is most likely to be responsible for this injury?**

- A) Compression
- B) Rotation
- C) Distraction
- D) Penetration

*Ans: A*

---

**14. While performing a primary assessment on a patient who has an isolated spinal cord injury, you note that he is in severe respiratory distress and struggling to breathe. Where should you suspect that the spinal cord injury has occurred?**

- A) The diaphragm
- B) The cervical spine
- C) The thoracic spine
- D) The lumbar spine

*Ans: B*

---

**15. A primary injury to the central nervous system would be caused by:**

- A) drowning.
- B) a hematoma.
- C) a brachial plexus injury.
- D) a laceration of the spinal cord.

*Ans: D*

---

**16. Which of the following signs or symptoms best indicates that a patient has suffered an injury to the thoracic spine?**

- A) Altered mental status
- B) Paralysis of the arms
- C) Tingling in the legs
- D) Cool and diaphoretic skin

*Ans: C*

---

**17. You are assessing a 30-year-old man who has fallen about 20 feet. He is alert and oriented but states that he cannot move or feel his legs. Additionally, because his blood pressure is 82/48 mmHg, you suspect neurogenic shock. Which of the following additional assessment findings reinforces your suspicion of neurogenic shock?**

- A) Warm and dry skin
- B) Heart rate of 144
- C) Shallow rapid respirations
- D) Heart rate of 44

*Ans: D*

---

**18. You should recognize a possible spinal column injury with no spinal cord involvement when you discover which of the following assessment findings?**

- A) Tenderness to the thoracic spine with intact motor and sensory function in each extremity
- B) Intact motor ability to all four extremities with the loss of sensation to the legs
- C) Pain in the cervical spine with the loss of sensation to the right arm and leg
- D) Intact sensation to all four extremities with the loss of motor ability in the arms

*Ans: A*

---

**19. A patient involved in a motor vehicle collision has a deformity to the left side of the head. The skin overlying the deformity is still intact. Based on these assessment findings, which of the following conditions would be your greatest concern?**

- A) A possible brain injury
- B) A spinal cord injury
- C) The potential for infection of the brain
- D) Soft tissue trauma to the scalp

*Ans: A*

---

**20. An elderly patient has fallen down a flight of stairs and is complaining of neck and back pain and weakness to both legs. Your primary assessment reveals no life threats to the airway, breathing, or circulation. Manual in-line spinal stabilization is being maintained. Which of the following actions should you take next?**

- A) Apply high-flow oxygen and move the patient to a stretcher for transport.
- B) Complete a secondary assessment looking for injuries.
- C) Place an oral airway and begin assisting ventilation.
- D) Place a cervical collar on the patient and immobilize him to a long spine board.

*Ans: B*

---

**21. Which of the following statements about the care and treatment of a patient with a spinal injury in a prehospital setting is true?**

- A) Prehospital care for a patient with a spinal injury involves correcting life-threatening injuries and limiting secondary injury.
- B) It is important to identify the site of spinal injury so that proper prehospital care can be rendered.
- C) Before transporting a critically injured patient with a spinal injury, OEC technicians must perform a detailed head-to-toe neurological assessment.
- D) If a patient with a possible spinal injury is in shock, it is permissible to forego immobilization because doing this uses up time during the golden hour.

*Ans: A*

---

**22. Which description of a helmet would indicate that it needs to be removed in a prehospital setting?**

- A) Too large
- B) Too small
- C) Too far down on the neck
- D) Too thin

*Ans: A*

---

**23. Which of the following signs is a characteristic sign that may be seen during a secondary assessment of patients with spinal cord injury?**

- A) Normal skin color and complaint of headache and nausea
- B) Flushed skin color below the level of the injury and impairment or absence of sensation and movement
- C) Bruising on the back and on the abdomen at the level of the injury, with normal or impaired sensation
- D) Flushed skin color at the level of pinpoint back pain

*Ans: B*

---

**24. Which of the following assessment findings is most indicative of increasing pressure within the skull from a closed head injury?**

- A) Blood pressure of 192/82 mmHg
- B) Heart rate of 132 beats per minute
- C) Blood pressure of 110/50
- D) Pupils equal and reactive

*Ans: A*

---

**25. You are called to aid a 61-year-old man who has fallen off a second-floor lodge patio. He is unresponsive and has very slow, shallow breathing at a rate of 6 breaths per minute. When you arrive, a fellow patroller has already opened the airway and has stabilized the head. Which of the following actions should you take immediately?**

- A) Apply a cervical collar.
- B) Place the patient in the shock position.
- C) Assist ventilation with a bag-valve mask.
- D) Transfer the patient to a stretcher for emergency transport.

*Ans: C*

---

**26. A patient with a head and midline posterior neck injury was sitting found leaning against a tree. You discover that your team does not have a long backboard, so you use what alternative?**

- A) Cervical collar
- B) Whole body vacuum mattress
- C) Nonrebreather mask
- D) Towels

*Ans: B*

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**27. A 41-year-old male snowboarder struck his head on a metal rail. As you approach him, you note that he appears confused and has blood on the left side of his face and head and on his shirt. His breathing appears to be labored. Which of the following actions should you perform immediately?**

- A) Check his pupils for equality and reactivity.
- B) Apply oxygen at 15 LPM via nonrebreather mask.
- C) Apply manual in-line spinal immobilization.
- D) Expose the chest to look for possible bleeding.

*Ans: C*

---

**28. Which of the following instructions from one OEC technician to another describes the appropriate application of a cervical spine immobilization collar?**

- A) "Carefully flex his head forward a little so I can pass the collar underneath his neck."
- B) "Let's log roll the patient to one side so I can apply a cervical collar."
- C) "Keep his head in neutral position while I apply a cervical collar."
- D) "I need you to extend the patient's chin backward a little so I can fit the collar under his chin."

*Ans: C*

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**29. A properly sized cervical collar will rest on the:**

- A) spine.
- B) chin.
- C) back of the skull.
- D) shoulder.

*Ans: D*

---

**30. A 2-year-old boy fell down a flight of stairs. Which of the following instructions would you provide to other patrollers who are immobilizing the patient?**

- A) "After he is on the board, place a towel behind his head to keep his airway open."
- B) "It is better if the collar is a little too big because it will be more comfortable for him."
- C) "Let's place a folded towel under his shoulders to help maintain his head alignment."
- D) "Just apply a cervical collar and then place him directly on the stretcher."

*Ans: C*

---

**31. Which of the following statements about removing a helmet in a prehospital setting is correct?**

- A) Helmets should be removed only if they are too tight or if spinal immobilization is required.
- B) It is acceptable to leave a helmet on a patient if the patient has no airway or breathing problems.
- C) Any patient wearing a helmet should have it removed so that the airway and breathing can be properly assessed.
- D) Helmets should never be removed, so OEC technicians must be creative in working around the obstacle of a helmet while providing care.

*Ans: B*

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**32. When explaining the priorities of helmet removal to a candidate patroller, you would stress:**

- A) sliding the helmet rearward off the head so that you don't hit the patient's nose.
- B) sliding the fingers under the occiput so that the head doesn't drop backward.
- C) applying a cervical collar before removing the helmet to help stabilize the head.
- D) removing the helmet to ensure the patient's proper alignment on a backboard.

*Ans: B*

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**33. Which of the following findings indicates that a patient who received a blow to the head is suffering from something other than a concussion/TBI?**

- A) He cannot remember what happened.
- B) He asks the same questions over and over.
- C) His heart rate is 92, and his blood pressure is 144/86.
- D) His pupils are noticeably unequal.

*Ans: D*

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## Section 22 — Eye, Face & Neck

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18 Questions

**1. Which of the following structures works like a camera shutter to adjust the amount of light entering the eye?**

- A) The sclera
- B) The iris
- C) The lens
- D) The posterior chamber

*Ans: B*

---

**2. Which of the following statements concerning the eye socket or orbit is false?**

- A) It is easy to injure the orbital bones.
- B) It protects the fluid-filled eyeball.
- C) It anchors the muscles that control voluntary eye movement.
- D) It is formed by the skull and facial bones.

*Ans: A*

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**3. You are assessing a patient who was punched in the eye. You note blood in the white part of his eye. This area of the eye is known as the:**

- A) sclera.
- B) cornea.
- C) lens.
- D) retina.

*Ans: A*

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**4. The part of the eye that focuses light on the retina is called the:**

- A) vitreous humor.
- B) sclera.
- C) lens.
- D) iris.

*Ans: C*

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**5. Light stimulates nerve endings in which of the following structures to send signals to the brain via the optic nerves?**

- A) Retina
- B) Iris
- C) Lachrymal glands
- D) Lens

*Ans: A*

---

**6. A 15-year-old girl struck in the mouth with a baseball bat has lost her front teeth and is spitting a significant amount of blood. Your primary concern in treating this patient is:**

- A) saving her teeth so they can be re-implanted.
- B) checking her for a jaw fracture.
- C) watching her for seizure activity.
- D) observing her for possible airway compromise.

*Ans: D*

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**7. You are on a snowmobiling trip with friends when you see members of the lead group hollering and waving their hands up ahead. When you get to the scene, you note that the lead rider apparently did not see a chain across the road, and he struck the chain with his neck so forcefully that he was thrown backward off his machine. Someone in the group is maintaining manual in-line stabilization. When you assess the injured person, which of the following signs would indicate the injury that must be addressed and managed first?**

- A) Difficulty speaking
- B) A laceration to the forehead
- C) An open fracture of the left arm
- D) A partially avulsed ear

*Ans: A*

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**8. While running outside the lodge, a teenage girl tripped and fell. She hit her face and mouth, knocking one of her top front teeth from its socket. Although she is upset, you have assessed no threats to life, and the bleeding has been controlled. Your partner finds the tooth and asks you what to do with it. Which of the following responses would be best?**

- A) "Wrap the tooth in dry gauze and put it in a plastic bag placed in cool water."
- B) "Avoid touching the root of the tooth. We will gently irrigate it and place it back in its socket."
- C) "Scrub the base of the tooth to be sure it's clean and then we'll place it back in its socket."
- D) "Just put it in dry gauze and bring it with you. They probably won't be able to do anything with it."

*Ans: B*

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**9. You are caring for a 15-year-old male with a laceration of the left ear. Which of the following actions would you not take in dressing this injury?**

- A) Placing a bulky dressing between the ear and the scalp
- B) Securing the dressing with gauze that is wrapped securely around the head
- C) Using direct pressure to control the bleeding
- D) Wrapping gauze loosely around the head and under the opposite axilla

*Ans: D*

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**10. A mother brings her 7-year-old child to the aid room and tells you that fluid is draining from the child's right ear. She tells you that the child has had a fever since this morning and was complaining of right ear pain. Based on your OEC training, you would:**

- A) use a 4 x 4 to pack the ear and then bandage it securely using gauze around the head.
- B) put a pressure dressing over the ear to stop the drainage.
- C) have the boy lie on his right side to allow the fluid to drain.
- D) arrange for ALS transportation immediately.

*Ans: C*

---

**11. A young boy was running through the lodge with a pencil and tripped. The pencil impaled the boy's left eye and remains lodged in place. Appropriate care for the eye would include:**

- A) stabilizing the impaled object and providing immediate transport.
- B) quickly removing the pencil from the eye to prevent the loss of vision.
- C) applying ice to the affected eye to decrease pressure and reduce subsequent damage.
- D) applying gentle pressure to the pencil and the eye to keep the pencil from moving.

*Ans: A*

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**12. During an altercation, your patient was cut with a sharp knife. Your assessment reveals a laceration across the right eyelid down to the right cheek that is oozing dark red blood. It also appears that the patient's eye was cut with the knife. You would:**

- A) apply direct pressure to the eyelid and the eye to control the bleeding.
- B) apply cold packs to the eyelid to decrease swelling.
- C) clean the incisions with sterile water.
- D) cover both eyes with a sterile dressing.

*Ans: D*

---

**13. What instructions would you give to an OEC candidate who asks how to treat a patient who has visible blood in the anterior chamber of the eye?**

- A) "It is important to keep the pressure in the eye low, so we will transport this patient in a sitting position."
- B) "Apply a cold pack to the eye and transport this patient immediately."
- C) "Put a shield over the eye and bandage tightly to apply some pressure to the eye."
- D) "Put a shield over the eye and cover both eyes before immediately transporting this patient."

*Ans: D*

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**14. A patient who was hit in the face with a cloud of dust while working in an industrial setting is complaining of pain and discomfort to his left eye. While performing your assessment on the eye, you note some redness of the globe but do not see any obvious foreign object. The most appropriate care you can provide to this patient would be to:**

- A) flush eye with copious amounts of water.
- B) carefully wipe the eye with a soft piece of sterile gauze.
- C) gently massage the eye to promote tearing.
- D) have the patient keep his eye open during transport to a medical facility.

*Ans: A*

---

**15. A young girl is complaining of eye pain after having bleach thrown in her face. When assessing her eyes, you note redness and significant tearing in the right eye. You would treat this girl by:**

- A) stopping the tearing by applying gentle pressure to the eye.
- B) covering both eyes immediately.
- C) flushing the eye with sterile water.
- D) gently wiping the inner eye with pieces of sterile gauze.

*Ans: C*

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**16. You have responded to a scene at which a young man tells you he splashed a chemical into his eye. He is complaining of burning and pain in his right eye. After performing a primary assessment and finding no life-threatening conditions, you would:**

- A) obtain a set of vital signs and SAMPLE.
- B) identify the exact chemical involved so that you can get the antidote.
- C) hold the patient's injured eye open and flush it with large amounts of sterile water from the nose to the outer edge of the eye.
- D) instruct the patient to keep his eyes closed and not rub them while you arrange for emergency transport.

*Ans: C*

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**17. You respond to the vehicle maintenance department, where a middle- aged man is flushing his eyes with tap water. He tells you that acid splashed in his eyes and that they are burning. Which of the following questions is the most important one to ask right away?**

- A) "What is your past medical history?"
- B) "Are you wearing contact lenses?"
- C) "Do you take any medications?"
- D) "Have you ever had a previous eye injury?"

*Ans: B*

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**18. You are assisting another OEC technician to treat a young male with a deep laceration on his neck. Your partner appears to have controlled the bleeding. He asks you to apply an occlusive dressing. In addition to helping to control the bleeding, you recognize that another important benefit to this type of dressing on a neck wound is:**

- A) preventing a cervical collar from irritating the wound.
- B) promoting clot formation to prevent further bleeding.
- C) preventing the entry of air into the circulatory system.
- D) providing neck stabilization.

*Ans: C*

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## Section 23 — Chest Trauma

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25 Questions

**1. The thorax is enclosed by the:**

- A) ribs, sternum, thoracic spine, and diaphragm.
- B) ribs, clavicles, diaphragm, and mediastinum.
- C) thoracic vertebrae, diaphragm, mediastinum, and clavicles.
- D) clavicles, sternum, diaphragm, and thoracic spine.

*Ans: A*

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**2. The diaphragm separates the:**

- A) right lung from the left lung.
- B) lungs from the heart.
- C) thoracic cavity from the abdominal cavity.
- D) heart and lungs from the spine.

*Ans: C*

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**3. Which of the following actions occurs when the diaphragm and the intercostal muscles relax?**

- A) Inhalation
- B) Release
- C) Inspiration
- D) Exhalation

*Ans: D*

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**4. The heart and lungs are located in the:**

- A) thoracic cavity.
- B) pericardial cavity.
- C) pleural cavity.
- D) mediastinum.

*Ans: A*

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**5. Injuries at which level of the spinal cord can affect the phrenic nerves and therefore respiration?**

- A) T1-T3
- B) T4-T6
- C) C6-T1
- D) C3-C5

*Ans: D*

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**6. Which of the following statements indicates that the speaker understands chest trauma?**

- A) "Open chest injuries are more serious than closed chest injuries because air and bacteria can enter the chest through the open wound."
- B) "In general, an open chest injury caused by blunt trauma is more life threatening than a closed one due to increased blood loss."
- C) "Internal chest injuries are more difficult to assess and manage than are the more obvious external chest injuries."
- D) "Closed chest injuries are caused by penetrating trauma and cause more life-threatening injuries than do open chest injuries."

*Ans: C*

---

**7. A 39-year-old man has been stabbed once in the anterior chest. When notifying the patrol aid room by radio, you inform them that the patient has suffered what possible type of injury?**

- A) Flail chest
- B) Evisceration
- C) Lacerated spleen
- D) An open chest injury

*Ans: D*

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**8. Which of the following phrases best describes a flail chest segment?**

- A) Multiple rib fractures with bruising of the underlying lung
- B) Fractured ribs that have collapsed the underlying lung
- C) Multiple fractures to the rib cage caused by blunt trauma
- D) Two or more adjacent ribs that have been broken in two or more places

*Ans: D*

---

**9. Your patient was involved in a serious motor vehicle collision. Which of the following assessment findings best helps to determine that the patient has a flail chest segment?**

- A) Paradoxical chest wall movement
- B) Shortness of breath
- C) Pain upon inspiration
- D) Chest wall bruising

*Ans: A*

---

**10. You are managing a patient with a large flail segment in the right lateral chest. The immediate threat to this patient's life is:**

- A) blood loss.
- B) hypoxia.
- C) infection.
- D) rib fractures.

*Ans: B*

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**11. An OEC candidate asks you to explain a pneumothorax. Your response should be that it occurs when:**

- A) the lung becomes overinflated with air and then collapses.
- B) air accumulates between the inner chest wall and the outside of the lung, causing the lung to collapse.
- C) the trachea becomes obstructed, causing the lungs to collapse.
- D) air enters the lung alveoli through a traumatic opening in the chest wall.

*Ans: B*

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**12. When assessing a patient, which of the following signs or symptoms is most indicative that the patient is suffering from a tension pneumothorax?**

- A) Excessive sweating
- B) Severe heartburn
- C) Decreased breath sounds in the right lung
- D) Crepitus in the abdomen

*Ans: C*

---

**13. You have placed an occlusive dressing on a puncture wound on the right side of the chest of a 33-year-old woman. During your ongoing assessment, your primary concern is monitoring the injury for:**

- A) arterial bleeding.
- B) infection at the wound site.
- C) tension pneumothorax.
- D) ecchymosis at the wound site.

*Ans: C*

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**14. An OEC candidate indicates that she understands the difference between a pneumothorax and a tension pneumothorax when she makes which of the following statements?**

- A) "A tension pneumothorax causes blood to accumulate around the lung; a pneumothorax involves the accumulation of air only around the lung."
- B) "A tension pneumothorax can cause cardiac output to decrease; a pneumothorax does not."
- C) "A pneumothorax describes a collapsed lung; a tension pneumothorax involves both a collapsed lung and blood loss."
- D) "A pneumothorax is caused by a closed chest injury; a tension pneumothorax is caused by an open chest injury."

*Ans: B*

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**15. You are treating a 26-year-old man with a pneumothorax. The patient's breathing has been inadequate, and you have been assisting his ventilations with a bag-valve mask. The patient's condition seems to be worsening. Because you understand the pathology involved in a pneumothorax, you are most concerned that:**

- A) you are not maintaining an adequate seal around the face mask, and thus the patient is not getting enough oxygen.
- B) the pressure from the bag valve mask is causing more air to enter the pleural space and is causing a tension pneumothorax.
- C) the patient's airway is not being maintained, so you should insert an oropharyngeal airway.
- D) you may have missed an injury.

*Ans: B*

---

**16. You suspect that a trauma patient is suffering from a hemothorax to the left lung. Which of the following assessment findings would reinforce your suspicion?**

- A) Neck vein distention and absent breath sounds in the right lung
- B) Cyanosis and a blood pressure of 210/100
- C) Bradycardia and hypertension
- D) Respiratory distress and the signs and symptoms of shock

*Ans: D*

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**17. Your 35-year-old patient has received major chest trauma. He complains of shortness of breath and pain. Your assessment reveals distended neck veins and pulsus paradoxus. Based on these signs and symptoms, you believe the patient to be suffering from:**

- A) acute myocardial infarction.
- B) pericardial tamponade.
- C) pleural effusions.
- D) aortic rupture.

*Ans: B*

---

**18. You are instructing a class in Outdoor Emergency Care when a student asks you to explain the L.A.P. method of examining the thorax. Your best reply to this question would be which of the following statements?**

- A) "The LAP method divides the chest into three exam areas: the lateral, anterior, and posterior thorax."
- B) "The LAP method can be used under any conditions and in any environment."
- C) "The LAP method is a rapid assessment for deformities of the chest."
- D) "The LAP method directs you to look, auscultate, and palpate the chest."

*Ans: D*

---

**19. You are following the LAP method to examine the chest of a 38-year-old trauma patient. To assess for thoracic cage instability, you would apply moderate downward pressure on the sternum while asking the patient to take a deep breath, and then you would:**

- A) "walk" the patient's clavicles with your fingertips while the patient exhales.
- B) have the patient exhale slowly while you listen to the breath sounds.
- C) apply moderate inward pressure on the lateral walls of the rib cage and ask the patient to take a deep breath.
- D) palpate the upper thorax for subcutaneous emphysema.

*Ans: C*

---

**20. You are listening to two patrollers discuss the accident to which they just responded. Which of the following injuries would lead you to believe that one of the patients had suffered from traumatic asphyxia?**

- A) A chest wound that makes a sucking sound
- B) Abdominal bruising and distention
- C) Bluish discoloration of the neck and face
- D) A bilateral femur deformity

*Ans: C*

---

**21. You are at the scene of a shooting. Your assessment reveals a 23-year-old man who has been shot twice. The first wound is to the left lower quadrant of the abdomen and is bleeding. The second wound is to the left lateral chest and makes a sucking sound every time the patient takes a breath. The initial action of an OEC technician should be which of the following actions?**

- A) Cover the chest wound with a gauze dressing.
- B) Cover the chest wound with a gloved hand.
- C) Place direct pressure over the abdominal wound.
- D) Place the patient on high-flow oxygen using a nonrebreather mask.

*Ans: B*

---

**22. You are assisting at an accident scene and are asked to apply a dressing over a sucking chest wound on the patient's left anterior chest wall. Which of the following items would you use?**

- A) A dry sterile gauze dressing
- B) Sterile gauze soaked in sterile water
- C) A clean cravat
- D) Vaseline gauze

*Ans: D*

---

**23. You have applied a nonporous dressing to a puncture wound on a patient's chest. An OEC candidate asks you why the dressing was taped only on three sides. Your best response is that taping the dressing on three sides:**

- A) permits oxygen to enter the lungs.
- B) allows trapped air to escape upon exhalation.
- C) is less painful for the patient.
- D) decreases the chances that air will enter the chest upon exhalation.

*Ans: B*

---

**24. You treated a 19-year-old with a puncture wound to the chest by covering the wound with an impermeable dressing. Now, as you perform your ongoing assessment, you note that the patient is tachypneic and is complaining of difficulty breathing. Breath sounds on the side of the injury are also diminished. Which of the following actions should you take immediately?**

- A) Assist his breathing with a bag-valve mask.
- B) Lift a corner of the dressing from the wound for a few seconds; then reapply and check his breath sounds.
- C) Add another layer to the dressing.
- D) Insert an oropharyngeal airway.

*Ans: B*

---

**25. You are treating a 16-year-old male who was skiing out of control and ran off the trail and into the woods. When you arrive, he is alert and complaining of pain in his right lower chest. Upon examination, you note that he has impaled himself with a tree branch. He tells you he will feel better if you just remove the branch. Based on your training as an OEC technician, you would respond by making which of the following statements?**

- A) "I need to get a bulky dressing ready in case there is bleeding when I remove the branch."
- B) "Once my helpers arrive and we are ready to transport you, we can remove the branch."
- C) "We need to leave the branch in place until we get you to a hospital."
- D) "I'm not supposed to remove the branch, but you can if you think it will make you more comfortable."

*Ans: C*

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## Section 24 — Abdominal Trauma

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20 Questions

**1. The abdomen is located between the:**

- A) diaphragm and the pelvic brim.
- B) mid-chest and the level of the umbilicus.
- C) nipple line and the coccyx of the spine.
- D) base of the 5th rib and the pelvic base.

*Ans: A*

---

**2. The function of organs in the abdomen is:**

- A) only digestion.
- B) the storage of fat.
- C) the absorption of oxygen.
- D) digestion, production of white blood cells, and the filtering of blood.

*Ans: D*

---

**3. When palpating the anterior portion of a patient's abdomen, you note tenderness in the left upper quadrant. You would recognize that which organ might be involved?**

- A) The kidney
- B) The gallbladder
- C) The spleen
- D) The liver

*Ans: C*

---

**4. A patient complains of severe and sharp pain in the right shoulder when palpating the right upper abdominal quadrant. Based on this sign and location, what organ or structure is most likely involved?**

- A) Liver
- B) Kidney
- C) Spleen
- D) Appendix

*Ans: A*

---

**5. You have been called to aid a patient with generalized abdominal pain. The patient states that he recently had his gallbladder removed. You know that the gallbladder is located in the:**

- A) left upper abdominal quadrant.
- B) right upper abdominal quadrant.
- C) lower left abdominal quadrant.
- D) lower right abdominal quadrant.

*Ans: B*

---

**6. Blunt trauma to the upper right abdominal quadrant can injure the:**

- A) pancreas.
- B) spleen.
- C) liver.
- D) small intestines.

*Ans: C*

---

**7. Which of these organs is most fragile and, therefore, most likely to be injured by blunt trauma?**

- A) Spleen
- B) Heart
- C) Colon
- D) Skin

*Ans: A*

---

**8. The spleen of a patient has ruptured. Because the spleen is a solid organ, you recognize that the primary threat to life is most likely:**

- A) blood loss.
- B) pain.
- C) infection.
- D) inflammation.

*Ans: A*

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**9. The pancreas is best described as a:**

- A) nonfunctional organ.
- B) hollow organ.
- C) solid organ.
- D) fluid-filled organ.

*Ans: C*

---

**10. The most common mechanism of injury to the pancreas is:**

- A) trauma to the left flank.
- B) rotational trauma to the chest wall.
- C) direct trauma to the middle of the abdomen above the navel.
- D) trauma to the right flank.

*Ans: C*

---

**11. In which of the following injuries can the abdominal cavity migrate into the thoracic cavity?**

- A) A rib fracture
- B) An abdominal hernia
- C) An intestinal tear
- D) A tear in the diaphragm

*Ans: D*

---

**12. An injury to the abdominal wall allowing the intestines to come out of the abdominal cavity is a(n):**

- A) laceration.
- B) evisceration.
- C) contusion.
- D) abrasion.

*Ans: B*

---

**13. You would recognize which one of the following situations as an acute abdominal emergency?**

- A) A patient who complains of stomach pain and has bad-smelling gas
- B) A woman who says she drank tequila last night and now has stomach pain with diarrhea
- C) A child whose sister has hepatitis and who has just vomited for the first time
- D) A ski racer who hit a tree and has significant left flank pain prior to and during palpation

*Ans: D*

---

**14. Which of the following statements indicates that you correctly understand the goal of assessing and managing a patient who has abdominal pain?**

- A) "It is important to determine if the liver is the cause of pain because that is the most lifethreatening condition."
- B) "If the patient has tenderness upon any palpation of the abdomen, OEC technicians should assume that bleeding is the cause."
- C) "It is more important to recognize a possible abdominal injury or emergency than it is to identify the exact cause."
- D) "To best treat the patient, OEC technicians must identify the exact cause of the patient's pain."

*Ans: C*

---

**15. When taking care of a patient with an abdominal injury, your first priority is:**

- A) controlling massive bleeding.
- B) keeping the scene, patient, and all rescuers safe.
- C) achieving proper airway management.
- D) stabilizing the patient.

*Ans: B*

---



**16. A crying 42-year-old woman states that she has sharp abdominal pain that she localizes to her right lower quadrant. When assessing this patient's abdomen, you would look for distention and symmetry and then:**

- A) start the exam by palpating the abdomen in the quadrant farthest away from the one that is painful.
- B) avoid palpating the entire abdomen.
- C) start the abdominal exam by palpating the right lower quadrant.
- D) avoid palpating the right lower quadrant.

*Ans: A*

---

**17. When presented with a 6-year-old female with a stick impaled through her lower left abdominal quadrant, an OEC technician should:**

- A) stabilize the stick.
- B) push the stick through the back of the abdominal cavity.
- C) pull the stick out partway.
- D) saw the stick off, leaving some in place.

*Ans: A*

---

**18. In most cases, the position of comfort in abdominopelvic trauma is:**

- A) supine, with padding under the knees to keep them slightly bent.
- B) in a right recumbent position with the knees straight.
- C) in a left recumbent position with the back bent.
- D) supine feet elevated at least twelve inches.

*Ans: A*

---

**19. To prepare a patient with a pelvic fracture for transportation, you should:**

- A) apply a pelvic binder.
- B) place the patient on a scoop stretcher in the right recumbent position.
- C) place the patient in a supine position on a backboard.
- D) place the patient in a seated position in a toboggan.

*Ans: A*

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**20. Because pelvic fractures are often associated with heavy internal bleeding, you should expect:**

- A) to find other fractures as well.
- B) that cardiac arrest may occur.
- C) to see hypovolemic shock.
- D) to need air evacuation for the patient.

*Ans: C*

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## Section 25 — Cold Emergencies

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### 28 Questions

**1. You are assessing a conscious but confused hiker who became lost in the woods on a cold day. Your assessment shows that he has an open airway, adequate breathing, and a weak radial pulse. His skin is cold to the touch, and he is shivering. OEC technicians should recognize that the:**

- A) cold caused the patient's brain to become dysfunctional, as evidenced by the shivering.
- B) shivering actually represents a small seizure, which indicates that the brain is cold.
- C) shivering is a protective means by which the body is attempting to warm itself.
- D) shivering is the only indicator that the patient's core body temperature has fallen below 85°F.

*Ans: C*

**2. A young man who was angry with friends wandered away from a party and spent the night outside uncovered in 40-50°F. He is confused, has decreased but adequate breathing, and has a weak radial pulse. His skin is cool, and capillary refill is delayed. When looking to see if he is shivering, you note that he is not. Based on this presentation, OEC technicians can safely conclude that the:**

- A) patient is exhibiting signs indicating that he is ineffectively compensating and conserving heat.
- B) patient's body temperature is most likely dangerously low.
- C) absence of shivering indicates that body temperature is almost back to normal.
- D) cause of his confusion is not related to exposure to the low overnight temperatures.

*Ans: B*

**3. A young boy complains of pain in his fingers after spending several hours outside riding a sled in cold temperatures. After ensuring that he has no lifethreatening conditions, you turn your attention to his hands and note that his fingers are cold, and the skin is pliable to the touch. Which of the following signs or symptoms would help confirm your suspicion that he is suffering from early or superficial frostbite?**

- A) Soft and tingling skin on the fingers
- B) A history of being exposed to temperatures in the 70s
- C) Swelling of and blisters on the fingers
- D) Firm and numb cold skin on the fingers

*Ans: A*

---

**4. Which of the following statements about immersion hypothermia is true?**

- A) Death typically occurs within a few minutes of submersion in very cold water.
- B) It can take more than 30 minutes for a person who has fallen into very cold water to become hypothermic.
- C) Protective winter clothing will always protect a person from immersion hypothermia.
- D) Immersion hypothermia is most typically associated with traumatized or critically ill patients.

*Ans: B*

---

**5. The continued drop in core body temperature after removal from exposure to the cold is known as:**

- A) afterdrop.
- B) secondary hypothermia.
- C) thermoregulation.
- D) shock.

*Ans: A*

---

**6. The main cause of afterdrop is probably:**

- A) the return of cold blood from the extremities.
- B) loss of fat.
- C) remaining wet.
- D) strong winds.

*Ans: A*

---

**7. Which of the following statements concerning immersion hypothermia is true?**

- A) It occurs only when the entire body and head are submersed.
- B) Death will typically occur within a few minutes.
- C) The function of arm and leg muscles will not affect the outcome.
- D) It can take more than 30 minutes for an individual to become hypothermic.

*Ans: D*

---

**8. Which of the following instructions is most appropriate when moving a patient who has severe generalized hypothermia?**

- A) "I want everyone to take extra care in moving her very gently to the stretcher; we do not want her to go into cardiac arrest."
- B) "Be careful when moving her to the stretcher; the cold makes her very prone to bone fractures."
- C) "Let's place her on her side on the stretcher because her body temperature contraindicates placement of a nasal airway."
- D) "Let's secure her with her arms and legs extended because this position promotes the retention of body heat."

*Ans: A*

---

**9. When obtaining a history of a five-year-old boy with generalized hypothermia, which of the following questions provides the most important information?**

- A) "Are all of his childhood shots up to date?"
- B) "Does he have any medical conditions?"
- C) "When did he last eat or drink?"
- D) "Why was he outside without adult supervision?"

*Ans: B*

---

**10. A patient with moderate hypothermia will have:**

- A) a core body temperature above 95°F.
- B) a slow pulse and/or slow respirations.
- C) active shivering.
- D) a high risk of ventricular fibrillation.

*Ans: B*

---

**11. You and your friends are camping on a 20° F day. Which of the following assessment findings best illustrates that your bodies are no longer capable of compensating for the effects of the cold?**

- A) Cool and clammy skin
- B) Loss of fine motor coordination
- C) Shivering
- D) Hypertension

*Ans: B*

---

**12. Which of the following signs would you observe earliest in a hypothermic patient?**

- A) Confusion
- B) Hypotension
- C) Shivering
- D) Bradycardia

*Ans: C*

---

**13. Which of the following findings best indicates that a patient with cold skin is suffering from moderate hypothermia?**

- A) Cold waxy fingers
- B) Respirations at 28-30 per minute
- C) A heart rate of 52 beats per minute
- D) A pulse oximetry reading of 95 percent

*Ans: C*

---

**14. Which of the following findings is a sign of severe hypothermia?**

- A) Slurred speech
- B) Active shivering
- C) Coma
- D) Confusion

*Ans: C*

---

**15. You are with a candidate OEC technician who is attending to a snowboarder who has fallen. The candidate is very concerned that the snowboarder is shivering vigorously and asks you what this indicates. Which of the following statements would be your best response?**

- A) Shivering is a sign of mild hypothermia.
- B) Shivering indicates moderate hypothermia.
- C) Individuals who are shivering cannot maintain their normal body temperature.
- D) It is good to try to get a shivering patient to stop because shivering burns energy.

*Ans: A*

---

**16. A patient who is conscious and breathing has been pulled from a stream of cold water. To decrease her loss of heat via the mechanism of conduction, an OEC technician should immediately:**

- A) cover her with a blanket.
- B) remove her wet clothing then cover her with blankets.
- C) provide positive-pressure ventilation.
- D) encourage her to stop shivering.

*Ans: B*

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**17. Prevention of heat loss in the care of a patient with a cold injury should include:**

- A) carefully removing wet clothing.
- B) placing the OEC technician's jacket over the patient.
- C) rubbing cold extremities.
- D) placing the patient in a prone position.

*Ans: A*

---

**18. You have been called to aid an alert and oriented male patient whose friends state he spent several hours locked out of his house in cold temperatures. Given that the primary and secondary assessments have ruled out immediately life-threatening conditions, you have decided to rewarm the patient. Which of the following actions would be most appropriate for this patient?**

- A) Place the patient in a tub of warm water.
- B) Apply hot packs to the patient's chest, groin, and armpits.
- C) Massage the patient's arms and legs after applying warm blankets.
- D) Blow hot air from a hair dryer over the patient's arms and legs.

*Ans: B*

---

**19. You need to begin rewarming an unresponsive patient who has a core temperature of 93°F. Which of the following measures would most benefit this patient?**

- A) Give the patient warm drinks to increase internal body temperature.
- B) Gently massage the patient's arms and legs.
- C) Wrap the patient in several warm blankets.
- D) Place one extremity at a time into warm water.

*Ans: C*

---

**20. When assessing the vital signs of a severely hypothermic patient, OEC technicians should:**

- A) vigorously shake the patient to assess the true level of responsiveness.
- B) check the patient's pulse and respirations for up to one minute.
- C) check the patient's radial pulse to avoid exposing the patient's head and neck to the cold.
- D) warm their own fingers for a full minute to ensure that they can feel the patient's pulse.

*Ans: B*

---

**21. A severely hypothermic patient is in cardiac arrest. Most studies indicate that for patients who have been submerged in cold water for more than one hour, you should:**

- A) start cardiopulmonary resuscitation.
- B) focus on rewarming and rapidly transporting the patient.
- C) provide ventilations but not compressions.
- D) do nothing because there is no chance such patients will survive.

*Ans: B*

---

**22. Which of the following statements indicates that an OEC technician understands the care of a patient with a localized cold injury to the foot?**

- A) "Warming a frozen body part can cause severe pain."
- B) "If the patient has blisters on his foot, it is beneficial to break them so that the fluid inside will not freeze and cause additional injury."
- C) "Gentle massage of the foot is beneficial because it enhances the circulation of warm blood in the affected area."
- D) "You should avoid immobilizing the foot because doing so will further decrease the circulation of blood in the affected area."

*Ans: A*

---

**23. You are facilitating a talk on cold emergencies in the backcountry. You are asked when one should attempt to rewarm a hand or foot that is frostbitten. Which of the following statements would be your best response?**

- A) "You never want to rewarm a frozen body part because doing so can cause additional damage."
- B) "Attempt to rewarm a frozen body part only if the person still has some feeling in it."
- C) "Rewarm the frozen body part if the patient needs to self-evacuate."
- D) "Rewarming should take place only when there is no chance that the tissue will refreeze."

*Ans: D*

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**24. One of your friends on a hunting trip in Montana has suffered a severe localized cold injury to his hand and fingers. He does not have any lifethreatening conditions, and you elect to rewarm the affected areas. Which of the following actions is most appropriate for your friend?**

- A) Immerse the hand and fingers in water that is hotter than 120°F.
- B) Vigorously rub the hands and fingers.
- C) Slowly and gradually warm the affected hand and fingers.
- D) Thaw and rewarm the tissue as quickly as possible.

*Ans: D*

---

**25. After successfully rewarming a foot that has frozen toes, an OEC technician should:**

- A) cover the foot and toes with dry dressings.
- B) gently massage the foot and toes.
- C) ambulate the patient to ensure return of adequate motor function.
- D) place the foot below the level of the heart to enhance circulation.

*Ans: A*

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**26. A skier with a frostbite toe might experience \_\_\_\_\_ after several days or weeks.**

- A) Spontaneous rewarming
- B) Autoamputation
- C) Gradual healing
- D) Regrowth of the toe

*Ans: B*

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**27. It has taken you 45 minutes to extricate an avalanche victim. Based on your knowledge of hypothermia and avalanche victims, you know that:**

- A) hypothermia during an avalanche burial generally occurs quicker than during cold water immersion.
- B) hypothermia can occur after extrication due to a cold, windy environment.
- C) a pulseless victim can most likely be resuscitated after rewarming.
- D) a patient who is responsive is not hypothermic.

*Ans: B*

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**28. The initial priority for avalanche victims is:**

- A) warming the extremities.
- B) cervical stabilization.
- C) ensuring adequate ventilation.
- D) treating for hypothermia.

*Ans: C*

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## Section 26 — Heat Emergencies

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### 11 Questions

**1. The body cools itself primarily by:**

- A) conduction.
- B) respiration.
- C) radiation.
- D) evaporation.

*Ans: D*

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**2. You are at a football practice on a very hot summer day. A young player is panting and lying under a tree. His skin is flushed, and his teammates are continually wiping the sweat off his face, arms, and chest with towels and offering him fluids to drink.**

**Which of the following statements is most appropriate for this situation?**

- A) "Keep wiping. You are helping him retain water that his body needs."
- B) "Don't give him fluids. It will only make him sweat more."
- C) "It's OK for him to sweat. It is helping his body cool down."
- D) "Let's put his shirt back on to keep him from sweating."

*Ans: C*

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**3. The most serious of the types of heat-related illnesses is:**

- A) heat-related syncope.
- B) heat cramps.
- C) heat exhaustion.
- D) heatstroke.

*Ans: D*

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**4. The signs of heat exhaustion are:**

- A) cool and clammy skin, bradycardia, slow respirations.
- B) warm and very sweaty skin, elevated body temperature, headache.
- C) diaphoretic skin, decreased heart rate, hunger.
- D) hot dry skin, elevated heart rate, elevated body temperature.

*Ans: B*

---

**5. You recognize that heat cramps are the probable cause of a patient's problem when the patient makes which of the following statements?**

- A) "I am having a hard time moving my fingers."
- B) "I cannot remember what day it is."
- C) "I have pain in my belly and legs."
- D) "My neck feels as though it is in spasm."

*Ans: C*

---

**6. The minimum body temperature of a patient for a diagnosis of heat exhaustion is:**

- A) 99°F.
- B) 100°F.
- C) 101°F.
- D) 102°F.

*Ans: C*

---

**7. Which of the following patients would have the highest priority for transport?**

- A) A 24-year-old who is intoxicated and is complaining of pain and spasms in his legs
- B) A 34-year-old who is responsive, has cool diaphoretic skin, and is complaining of nausea
- C) A 28-year-old who is responsive, has tachycardia, and has cool, moist skin
- D) A 19-year-old who is lethargic, has hot dry skin, and has tachycardia

*Ans: D*

---

**8. You are at a park with your family when you hear a call for help. There is a teenager who is "sick." On scene, you find a 16-year-old boy who is mildly confused. Friends state that they have been playing basketball for most of the morning and afternoon. The temperature is in the 90s, and the humidity is high. The boy's airway is patent, and his breathing rate is elevated but adequate. His pulse is weak and rapid. Observation of his skin reveals it to be pale, cool, and moist. Which of the following would you do first for this patient?**

- A) Lay him in the shade in a supine position.
- B) Lay him down in a side-lying position.
- C) Administer salt tablets by mouth every 15 minutes.
- D) Get some newspaper and fan him vigorously.

*Ans: A*

---

**9. You are treating a patient with suspected heat cramps. Which of the following treatments is the usual recommended treatment?**

- A) Have the patient drink a mixture of one tablespoon of table salt in one quart of Gatorade.
- B) Have the patient drink a mixture of  $\frac{1}{4}$  to  $\frac{1}{2}$  teaspoon of table salt in a quart of cool water.
- C) Administer salt tablets every 15 minutes until the cramping subsides.
- D) Have the patient drink one quart of cold tap water every 15 minutes until he needs to urinate.

*Ans: B*

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**10. When cooling a patient with a possible heatstroke, which of the following findings would be of most concern to a rescuer?**

- A) An increase in blood pressure to 100/80
- B) A decrease in heart rate from 140 beats per minute to 120 beats per minute
- C) Observing your partner applying an ice pack to the patient's neck
- D) Observing your partner giving the patient two aspirin to bring down his temperature

*Ans: D*

---

**11. You are in the aid room with a patient who is unresponsive and has hot, dry skin. His friends state that he has been drinking and passed out in the hot sun for several hours. ALS has been called. Which of the following actions would be most beneficial to this patient now?**

- A) Prepare the patient for vomiting by placing him in a prone position.
- B) Try to give the patient sugar because he could be having a diabetic reaction.
- C) Place cold packs on the patient's groin and armpits.
- D) Gently pour cold water over the patient to rapidly cool him down.

*Ans: C*

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## Section 27 — Bites & Stings

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17 Questions

**1. You are teaching a basic OEC class and need to explain toxins. Which of the following statements is the best description of a toxin?**

- A) It is a poison made by a living creature, including plants and animals.
- B) It is a poison that comes from a chemical.
- C) It is usually transmitted by a bite or a sting.
- D) It is anything ingested that causes an anaphylactic reaction.

*Ans: A*

**2. A specific toxin or poisonous secretion of an animal that is usually transmitted by a bite or sting is known as:**

- A) venom.
- B) antigen.
- C) bullae.
- D) nettle.

*Ans: A*

**3. Plants most commonly cause toxic reactions through which of the following routes?**

- A) Ingestion and inhalation
- B) Topical contact and injection
- C) Ingestion and topical contact
- D) Inhalation and injection

*Ans: C*

**4. On a camping trip with your family, your children ask if they can pick some plants for part of their wilderness dinner. Based on your OEC training, you explain to your children that they need to be careful picking wild plants and flowers, and that:**

- A) plants that do not cause any local reaction when they are picked are safe to eat.
- B) the worst that can happen from eating a bad plant is you will get sick to your stomach.
- C) leaves may be poisonous, but most berries found in the woods are safe to eat.
- D) some plants can be very toxic, so it is best not to eat anything unless you are sure what it is.

*Ans: D*

**5. Which one of the following groups is not included in the three main groups of mushrooms that are dangerous to humans?**

- A) False morels
- B) Little brown mushrooms
- C) Amanitas
- D) Big white mushrooms

*Ans: D*

**6. It's early fall, and you have been backpacking on part of the Appalachian Trail with some friends. One member of the group comes up to you, holding some bright orange mushrooms that resemble pumpkins and asks if you think they are safe to eat. Based on your training, you would respond with which of the following statements?**

- A) "Those are very toxic and would definitely kill you if you ate them."
- B) "Those mushrooms can cause some pretty severe vomiting, diarrhea, cramps, and loss of coordination."
- C) "If you just eat the mushroom cap, you should be OK."
- D) "Most toxic mushrooms are brown, so those are probably all right to eat."

*Ans: B*

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**7. You will be taking a group of scouts on a weekend camping trip. You recognize that ticks are typically active at this time of year, so you want to teach the scouts about them. Which of the following statements about ticks is false?**

- A) Campers should check their skin regularly, especially in the evening before going to bed.
- B) A red circle with a blanched white center is characteristic of the rash caused by a tick bite.
- C) The typical disease transmitted by ticks is Lyme disease.
- D) Tick bites are painful, so you will know when you have been bitten.

*Ans: D*

---

**8. Which of the following signs and symptoms is not a characteristic of a bee, wasp, or hornet sting?**

- A) Nausea and vomiting
- B) Joint aching
- C) Pain
- D) Swelling

*Ans: A*

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**9. Most reptile-related injuries are caused by:**

- A) alligators.
- B) snakes.
- C) sharks.
- D) crocodiles.

*Ans: B*

---

**10. Marine life can cause varying degrees of injury and illness. Which of the following marine creatures can cause respiratory failure and cardiovascular collapse in humans?**

- A) Jellyfish
- B) Spiny fish
- C) Biting fish
- D) Moray eels

*Ans: A*

---

**11. Attacks by which of the following creatures do not cause both soft-tissue injuries and fractures?**

- A) Alligators
- B) Sharks
- C) Swordfish
- D) Crocodile

*Ans: C*

---

**12. You are working at the first-aid station at the local scout camp. A counselor brings in two young boys that he thinks were playing in an area where poison oak was found. Which of the following actions would not be a part of your treatment for the possible exposure to poison oak?**

- A) Putting on non-latex gloves before examining the boys
- B) Washing the boys' hands and arms with cool water and soap
- C) Applying a topical cream such as bacitracin to the boys' hands and arms
- D) Using a commercial cleanser such as Oak-N-Ivy to wash the boys' hands and arms

*Ans: C*

---

**13. Your neighbor has been cleaning up the wooded area around his yard and burning brush and weeds. He tells you that he is trying to rid the area of poison ivy. You recognize that his actions could:**

- A) cause significant respiratory problems for anyone exposed to the smoke.
- B) encourage the poison ivy to grow into any area where the smoke travels.
- C) temporarily solve his problem but will not permanently kill the poison ivy.
- D) be a great solution that you should use around your house.

*Ans: A*

---

**14. When a young child at a neighborhood cookout cries out that he was stung by a bee on his forearm, you should:**

- A) ask if anyone has an epinephrine auto-injector and assist in giving the injection.
- B) scrape the stinger off the skin with a firm, flat object such as a credit card and then apply ice to reduce the swelling and pain.
- C) immobilize the arm and place it at the level of the child's heart to prevent spread of the venom.
- D) scrub the area with soap and water to remove the stinger.

*Ans: B*

---

**15. While hiking in the Rockies, your friend is bitten on the arm by a rattlesnake. You should quickly:**

- A) cut the wound and suck out the poison.
- B) lead your friend on a hike for help.
- C) provide aspirin.
- D) immobilize the arm and place it lower than your friend's heart.

*Ans: D*

---

**16. Which of the following actions is not an appropriate treatment measure for a sting by a marine creature?**

- A) Applying a tourniquet if required for controlling life-threatening hemorrhaging
- B) Irrigating the affected area with fresh water
- C) Rubbing the affected area to remove any spines
- D) Rinsing the affected area with a vinegar solution

*Ans: C*

---

**17. When caring for someone who has been attacked by a large animal such as a moose or bear, you should assume that:**

- A) the potential for spinal injury exists.
- B) the animal probably has rabies, or it would not have attacked.
- C) unless there is a bite, there is no potential for infection.
- D) bleeding from claw injuries is the most dangerous injury.

*Ans: A*

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## Section 28 — Altitude Illness

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20 Questions

**1. The height or vertical elevation above a fixed point is known as:**

- A) ascent.
- B) elevation.
- C) altitude.
- D) ataxia.

*Ans: C*

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**2. The concentration of oxygen at sea level is approximately:**

- A) 35%.
- B) 43%.
- C) 40%.
- D) 21%.

*Ans: D*

---

**3. As altitude increases:**

- A) barometric pressure increases.
- B) the proportion of oxygen decreases.
- C) the concentration of oxygen per cubic foot decreases.
- D) air density becomes greater.

*Ans: B*

---

**4. You are teaching a class on acute mountain sickness. When one of your students asks you to explain the term acclimatization in relation to AMS, your best response would be which of the following statements?**

- A) "Physiologic changes that increase the body's ability to adjust to extreme changes in temperature."
- B) "Physiologic changes that decrease the body's need for oxygen at high altitudes."
- C) "Physiologic changes that decrease the body's need for slow ascents."
- D) "Physiologic adjustments that increase the delivery of oxygen to cells."

*Ans: D*

---

**5. Which of the following processes does not occur during acclimatization?**

- A) An increase in respiratory rate and depth
- B) Dilation of pulmonary blood vessels
- C) An increase in red blood cell production
- D) An increase in heart rate

*Ans: B*

---



**6. You are working at a mountain resort and are presented with a 50-year-old woman who states that she does not feel well. From your assessment, which of the following findings would seemingly indicate that the patient is being adversely affected by the high altitude?**

- A) Vomiting after eating large meals
- B) An elevated blood glucose level with no history of diabetes
- C) A heart rate of 62 beats per minute
- D) A complaint of feeling short of breath on exertion

*Ans: D*

---

**7. In an otherwise healthy individual, the presence of a headache and feelings of sickness at high altitude is known as:**

- A) a high-altitude migraine.
- B) acute mountain encephalitis.
- C) acute mountain sickness.
- D) high-altitude pulmonary edema.

*Ans: C*

---

**8. Labored breathing at rest and audible chest congestion herald the development of a serious, potentially life-threatening stage of what altituderelated condition?**

- A) Acute mountain sickness
- B) Peripheral edema
- C) HAPE
- D) Khumbu cough

*Ans: C*

---

**9. You are with a group of hikers on the third day of an 11,000-foot mountain ascent. One of the hikers has not been feeling well for a couple of days. Today, members of the group notice that he is having difficulty getting dressed and speaking. From your training in outdoor emergency care, you recognize that these signs and symptoms are most often associated with:**

- A) acute mountain sickness.
- B) peripheral neuropathy.
- C) frostbite.
- D) HACE.

*Ans: D*

---

**10. The most effective method for preventing highaltitude illness is to:**

- A) make gradual ascents.
- B) maintain adequate hydration and eat a highcarbohydrate diet.
- C) take medications such as Diamox.
- D) avoid drinking alcohol.

*Ans: A*

---

**11. You are asked to speak to a college group that is planning a ski trip to the Alps. In order to help them plan for reducing their risks for developing altituderelated illnesses, you suggest all of the following except:**

- A) plan a layover day at between 6,000 feet and 8,000 feet.
- B) drink 3-4 quarts of fluid a day.
- C) do a lot of heavy physical exertion early in the trip to help them acclimate.
- D) sleep at an elevation that is lower than where they will be skiing.

*Ans: C*

---

**12. Your family is planning a ski trip to a resort that is at about 10,000 feet of elevation. You suggest that you spend the first two nights of the trip at a hotel where the elevation is about 6,500 feet. When your children protest that they are in good shape and want to go immediately to the resort, you explain that you are concerned about altitude sickness and:**

- A) "Physical fitness does not necessarily prevent altitude sickness."
- B) "Just because you children are fit and don't need the gradual change doesn't mean your mom and I don't need it."
- C) "Your mom and I will exercise more before the trip so that maybe we can reduce the delay to one night."
- D) "I'll see if I can get some medication so that we don't need to stop over at the lower elevation."

*Ans: A*

---

**13. You are working at the summit of a 9,000-foot mountain. A 60-year-old woman is brought to you, complaining of headache, fatigue, and shortness of breath. You prepare to do your assessment and recognize that your goal is to:**

- A) give her some ibuprofen to ease her headache.
- B) determine whether this is an emergent condition so that you can initiate life-saving treatment.
- C) get the patient lying down to relieve her presenting symptoms.?
- D) find out if she has an inhaler that she can use to relieve her shortness of breath.

*Ans: B*

---

**14. Given that 50 percent of patients with HAPE also have symptoms of AMS, it is important that you ask patients if they have had which of the following groups of signs/symptoms?**

- A) Headache, blurred vision, peripheral edema
- B) Harsh cough, itchy skin, nausea
- C) Difficulty sleeping, vomiting, ataxia
- D) Fatigue, nausea, difficulty sleeping

*Ans: D*

---

**15. In order to assess for a key symptom of HACE, you would ask patients to:**

- A) walk a straight line heel to toe.
- B) take a deep breath and let you know if it hurts.
- C) tell you if their rings are fitting tightly.
- D) read an eye chart.

*Ans: A*

---

**16. A patient at a mountain ski resort has notable shortness of breath. She denies any past medical history and takes no medications. After applying oxygen, you realize that the fundamental treatment to helping this patient improve is to:**

- A) descend to a lower altitude.
- B) administer a respiratory medication by a metereddose inhaler.
- C) position her in a left lateral recumbent position.
- D) place her in a supine position with her legs elevated.

*Ans: A*

---

**17. You have a 32-year-old male patient who just arrived at the aid room on the top of a 9,000-foot peak. He is exhibiting signs of altitude sickness and HAPE. You recognize that the final key to a successful outcome of treatment is to:**

- A) get him to lie down so his symptoms do not worsen.
- B) start him on O2 while waiting for him to be transported.
- C) prevent further body cooling by covering him with blankets.
- D) recognize the patient's signs and descend to a lower elevation.

*Ans: D*

---

**18. Your group of climbers has reached an elevation of 8,500 feet. One of the climbers is increasingly short of breath and now has audible chest congestion. Your party has a limited supply of oxygen, which you immediately apply to the climber at 15 LPM via a nonrebreather mask. Based on your assessment that the climber has HAPE, you recognize that the next necessary treatment is:**

- A) a rapid descent of at least 1,500 to 3,000 feet.
- B) to administer diuretics such as Diamox to reduce the patient's pulmonary edema.
- C) to get the patient to a physician who can administer dexamethasone.
- D) to place the patient in the shock position.

*Ans: A*

---

**19. When treating an AMS patient, use high-flow oxygen to keep saturation at least what percentage?**

- A) 88%
- B) 94%
- C) 97%
- D) 98%

*Ans: B*

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**20. You are accompanying a team of hikers up a high mountain. The next morning, you are summoned to a tent and find one of the climbers confused and complaining of a headache. His airway is patent, and his respirations are 24 per minute. He has no medical history and was in good health until found ill this morning by his friend. Suspicious of high-altitude cerebral edema (HACE), which of the following actions should you take?**

- A) Immobilize the patient
- B) Provide high-flow oxygen
- C) Insert an oropharyngeal airway
- D) Administer oral glucose

*Ans: B*

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## Section 29 — Water Emergencies

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19 Questions

**1. The final event in the series of events in drowning is:**

- A) unresponsiveness leading to respiratory arrest.
- B) cardiac arrest.
- C) altered mental status leading to respiratory arrest.
- D) a seizure leading to unresponsiveness leading to respiratory arrest.

*Ans: B*

---

**2. The mammalian diving reflex is most prominent in:**

- A) experienced swimmers.
- B) deep-water divers.
- C) adults in warm water.
- D) young children.

*Ans: D*

---

**3. Which of these is a submersion injury?**

- A) Someone chokes on a glass of water.
- B) Someone strikes a submerged rock while airway is under water.
- C) Someone becomes hypothermic while wearing wet clothes.
- D) Someone has an allergic reaction to a drink that includes water.

*Ans: B*

---

**4. Which of the following strategies does not limit or mitigate risk in water-based activities?**

- A) Avoiding swimming alone
- B) Checking ice thickness before traversing it
- C) Avoiding alcohol consumption around water
- D) Entering swiftly moving water accompanied by a buddy

*Ans: D*

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**5. Drowning is defined as:**

- A) fluid oxidation of the lungs.
- B) death by traumatic injury to the airway.
- C) respiratory impairment due to trauma.
- D) suffocation by submersion in a liquid.

*Ans: D*

---

**6. The first thing to happen when a person drowns is usually:**

- A) laryngospasm.
- B) panic.
- C) hyperventilation.
- D) apnea.

*Ans: B*

---

**7. Trauma from a dive injury that results in tissue damage within any air-filled structure of the body is called:**

- A) internal dive trauma.
- B) tonicity.
- C) upper pressure trauma.
- D) barotrauma.

*Ans: D*

---

**8. Decompression sickness or the "bends" is a(n):**

- A) excessive O<sub>2</sub> level in the body.
- B) buildup of nitrous acid in the body.
- C) high level of carbon dioxide in the body.
- D) buildup of nitrogen bubbles in the body

*Ans: D*

---

**9. When arterial gas embolism (AGE) occurs, the gas within the lungs:**

- A) causes an open pneumothorax.
- B) causes the trachea to close.
- C) contracts, and the bronchioles start to spasm.
- D) expands causing, alveoli to rupture.

*Ans: D*

---

**10. The cause of an arterial gas embolism (AGE) is:**

- A) a rapid descent by a diver.
- B) a rapid ascent by a diver.
- C) diving below 100 feet.
- D) diving below 100 feet for longer than 30 minutes.

*Ans: B*

---

**11. Decompression sickness can cause:**

- A) swelling of the vocal cords.
- B) severe muscle and joint pain.
- C) swelling of the tongue.
- D) swelling of the cricoid cartilage

*Ans: B*

---

**12. A large nitrogen bubble can act as a(n):**

- A) carrier for glucose.
- B) embolus that blocks blood flow.
- C) oxygen inhibitor.
- D) carbon dioxide carrier.

*Ans: B*

---

**13. Squeeze is a term for excessive external pressure on various parts of the body.**

**Reverse squeeze:**

- A) is too much pressure from within a body compartment or organ.
- B) occurs when two body organs compress each other.
- C) is pressure created when holding one's breath.
- D) is pressure experienced within a hyperbaric chamber.

*Ans: A*

---

**14. Assessment of patients with water-related emergencies is not remarkably different from assessment of patients with other emergencies. OEC technicians may:**

- A) need to wear an SCBA (Scott Air-Pak).
- B) need to swim in deep water to retrieve a patient.
- C) need to don a personal floatation device.
- D) be required to wear a cold-water rescue suit and stand on a flotation ramp.

*Ans: C*

---

**15. During assessment of the water-related emergency for any submersion injury, care should be taken to:**

- A) assess the airway but not to treat the patient.
- B) assess the water temperature.
- C) protect the patient's spine to prevent additional neurological injury.
- D) use AEIOU TIPS.

*Ans: C*

---

**16. For any submersion injury that is not a lifethreatening water-related emergency, a complete secondary assessment is performed using:**

- A) DCAP-BTLS.
- B) SAMPLE.
- C) OPQRST.
- D) AVPU.

*Ans: A*

---

**17. Patients with arterial gas embolism or decompression sickness should be transported by ground to a facility that:**

- A) has advanced respiratory specialists.
- B) has a hyperbaric chamber.
- C) is nearest to the scene.
- D) is a trauma center with a rehabilitation wing.

*Ans: B*

---

**18. A patient who appears dead due to a deep coldwater drowning should be:**

- A) left at the scene for the medical examiner to pick up.
- B) transported to a medical facility while being given rescue breathing only.
- C) transported to a medical facility while being given CPR.
- D) warmed up prior to transport and then given continuing CPR.

*Ans: C*

---

**19. A patient who is conscious and breathing has been pulled from a cold stream. To decrease the patient's loss of heat via the mechanism of conduction, OEC technicians should immediately:**

- A) apply hot packs to the patient.
- B) thoroughly dry the patient and then apply a blanket.
- C) provide positive- pressure ventilation.
- D) encourage the patient to stop shivering.

*Ans: B*

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## Section 30 — Pediatrics

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35 Questions

**1. A 9-month-old boy has been stung on his tongue after sucking on a bottle that had a bee on the nipple. In comparison to the same injury in an adult, why would an OEC technician be more concerned over this child?**

- A) A child's airway has more blood vessels than an adult's, making bleeding more of a concern.
- B) Children tend to be highly allergic to bee stings, whereas the same allergy in adults is rare.
- C) A child's tongue is proportionally larger, increasing the chances of airway occlusion from even minor swelling.
- D) When injured, a child's mouth produces more saliva, making choking a major concern.

*Ans: C*

**2. Fontanel, which are openings in the skulls of newborns that allow brain expansion, close when the child is at approximately what age?**

- A) 3-4 months of age
- B) 18-20 months of age
- C) 3 years of age
- D) 1 year of age

*Ans: B*

**3. An OEC technician asks you why you should not overextend the airway when performing a head tiltchin lift maneuver on a pediatric patient. Which of the following replies would you make?**

- A) "The pediatric cervical spine is delicate and can be injured if the neck is hyperextended."
- B) "The tongue of a pediatric patient is proportionally larger and blocks the airway when the neck is hyperextended."
- C) "The cartilage of the trachea is very soft and can compress if the neck is extended too far."
- D) "The esophagus in a pediatric patient is very thick and will occlude the airway if the neck is hyperextended."

*Ans: C*

**4. You are treating a 2-year-old who fell down a flight of stairs. You are concerned about a head injury and know that toddlers have a higher risk for traumatic brain injury than adults because:**

- A) a toddler has a smaller head than an adult.
- B) during rapid deceleration, a toddler's head is propelled forward first, before the body.
- C) a toddler's balance is not fully developed.
- D) until the child is about 18 months old.

*Ans: D*

---

**5. Children exhibit great bursts of energy followed by sudden profound fatigue because:**

- A) they lose interest in their current activity.
- B) they have attention deficit disorder.
- C) they need to take a nap.
- D) they have fewer energy reserves than adults.

*Ans: D*

---

**6. You are assessing a 2-week-old baby who is sick. Assessment reveals that he has a fever and difficulty breathing. Which of the following additional assessment findings would be most concerning to you given the age of this patient?**

- A) A respiratory rate of 30 breaths per minute
- B) A pulse of 116 beats per minute
- C) Continual crying
- D) Nasal passages that are occluded by mucus

*Ans: D*

---

**7. An adolescent is defined as someone who is:**

- A) 8-16 years old.
- B) 10-16 years old.
- C) 13-18 years old.
- D) 9-18years old.

*Ans: C*

---

**8. You have been called to care for a 21-month-old girl who has been bitten by a dog. Given the patient's age, you would appropriately classify the patient as:**

- A) an infant.
- B) a school-age child.
- C) a preschooler.
- D) a toddler.

*Ans: D*

---

**9. You are reviewing psychosocial development of a preschooler in your OEC class. You recognize that "preschooler" includes children who are:**

- A) 2-3 years old.
- B) 3-5 years old.
- C) 6-8 years old.
- D) over 9 years of age.

*Ans: B*

---

**10. A mother brings her 3-year-old son into the aid room. She says he has a low-grade fever and seems to be having slight difficulty breathing. Which of the statements by the mother would make you think the child may have croup?**

- A) "Last night, he had a barking- like cough."
- B) "There are times when he continually drools."
- C) "When I turn the air conditioner on, he gets even more short of breath."
- D) "He has a rash on his chest."

*Ans: A*

---

**11. A 4- year-old girl is sitting upright on her mother's lap with her chin thrust forward. She looks lethargic and is drooling. Her airway is open, and she appears to be breathing adequately. Her mother says she has had a fever and is complaining of a sore throat. You are very concerned that she may be developing:**

- A) strep throat.
- B) meningitis.
- C) epiglottitis.
- D) pneumonia.

*Ans: C*

---

**12. Cardiac arrest in children is most commonly caused by:**

- A) respiratory failure.
- B) trauma.
- C) birth defects.
- D) SIDS.

*Ans: A*

---

**13. You have been called to assist a panicked mother who is worried about her son. When you arrive, she tells you that her 4-year-old son has been quiet all morning and napping while the rest of her children were skiing. While napping he began to "shake all over" for about 30 seconds. The child is now resting quietly and has adequate respirations. His radial pulse is strong, and his skin is very hot and moist to the touch. Based on this presentation and information, you assume that the seizure occurred secondary to:**

- A) hypoxia.
- B) a fever.
- C) hypoglycemia.
- D) altered mental status.

*Ans: B*

---

**14. Which of the following conditions is not a possible cause of seizures in young children?**

- A) Fever
- B) Hypothermia
- C) Diabetes
- D) Epilepsy

*Ans: B*

---

**15. You read in the newspaper about a 9-month-old child who died from sudden infant death syndrome (SIDS). From your training, you know that this means that the infant:**

- A) had a congenital heart defect.
- B) suffocated in his crib blankets.
- C) had a history of sleep apnea.
- D) died unexpectedly and of an undetermined cause.

*Ans: D*

---

**16. The increased pliability of the ribs of children makes them more prone to:**

- A) bruising of the lung.
- B) rib fractures.
- C) cardiac arrest.
- D) over inflation of the lungs.

*Ans: A*

---

**17. Which of the following statements about shaken baby syndrome is false?**

- A) It rarely occurs in upper-class families.
- B) It often results in a traumatic brain injury.
- C) It is a form of child abuse.
- D) Most cases are perpetrated by someone close to the child.

*Ans: A*

---

**18. Bilateral injuries, circumferential bruising, and pattern bruises are suggestive of:**

- A) shaken child syndrome.
- B) child abuse.
- C) suicidal behavior.
- D) infant neglect.

*Ans: B*

---

**19. When assessing a 3-year-old child for possible injuries after the mother's boyfriend said the child fell down a flight of stairs, which of the following findings would raise your suspicion that the child may be a victim of physical abuse?**

- A) The child cries when you palpate his arm.
- B) Several bruises are located on the child's knees and shins.
- C) The child has a broken clavicle.
- D) Circumferential bruising is apparent on the child's arm.

*Ans: D*

---

**20. Which of the following statements is most appropriate concerning dealing with caregivers and children during a medical emergency?**

- A) "I try to include caregivers in all that I do with their child so that the child and the caregivers are more comfortable."
- B) "It is best to separate caregivers from the child so that proper assessment and care can be given."
- C) "I include caregivers in the care until I get the information I need; then I remove the child and continue the assessment in the first-aid room."
- D) "I tell caregivers that everything will be okay so that they are calm, and I am better able to help their child."

*Ans: A*

---

**21. The American Academy of Pediatrics recommends the use of the Pediatric Triangle to quickly determine if a child is "sick or not sick." This method allows rescuers to quickly assess all of the following about the patient except his/her:**

- A) circulation.
- B) breathing.
- C) history of illness or injury.
- D) appearance.

*Ans: C*

---

**22. Which of the following behaviors would an OEC technician recognize as uncharacteristic of a conscious and stable 2-year-old boy who fell and hurt his hand?**

- A) He cries any time you touch him.
- B) He does not cry or protest when you take him from his mother to assess him.
- C) He becomes upset when you lift his shirt to assess his abdomen.
- D) He does not tell you where the pain is when you ask.

*Ans: B*

---

**23. While following the Pediatric Triangle of assessment, you observe an infant and note that she is paradoxically irritable. This sign is often indicative that the child is:**

- A) very ill.
- B) a victim of neglect.
- C) not seriously ill.
- D) hungry.

*Ans: A*

---

**24. When seen in a child, the "tripod" and "sniffing" positions are usually signs of:**

- A) respiratory distress.
- B) an infection of the brain.
- C) hypovolemia.
- D) a neck or spine injury.

*Ans: A*

---

**25. You are assessing an infant who has been ill and has had a fever for the past two days. As you approach the infant, you note that she has grunting respirations. Based on your OEC training, you determine that grunting is:**

- A) a soothing mechanism for a sick child.
- B) a symptom of severe respiratory disease.
- C) a symptom of significant dehydration.
- D) often normal in a child with a cold.

*Ans: B*

---

**26. You must assess the pupils of a 5-year-old boy who fell. Which of the following statements would be most appropriate for you to make before performing the assessment?**

- A) "I am going to look into your pupils with my light."
- B) "I am going to use this light to look into your eyes."
- C) "I need to test your visual acuity by shining a light in your eyes."
- D) "Open your eyes so that I can look into them."

*Ans: B*

---

**27. You are assessing a 2½-year-old child who was involved in a minor car collision. She is currently alert and oriented. While you are assessing her for possible injuries, which of the following actions would be considered most appropriate?**

- A) Assess the child, starting at her head and moving to her feet.
- B) Allow the child to hold a favorite toy during the assessment.
- C) Be stern with the child and provide strict instructions on what you expect from her.
- D) Examine possible painful sites first.

*Ans: B*

---

**28. You have been called to care for an unattended 6-year-old girl who has vomited once and is complaining of mild abdominal pain. When you are performing the physical assessment and obtaining a SAMPLE history, which of the following approaches would be considered most appropriate?**

- A) Never involve the caregiver in the SAMPLE questions.
- B) Standing above her and smiling while you ask questions
- C) Allowing her to play with your stethoscope before listening to her lungs
- D) Using "baby talk" when asking her questions about her pain

*Ans: C*

---

**29. When assessing a 3-year-old child with a respiratory illness, which of the following assessment findings would be least concerning to you?**

- A) Retractions observed above the clavicles
- B) Paradoxical breathing
- C) Patient in the tripod position
- D) A respiratory rate of 28 breaths per minute

*Ans: D*

---

**30. Which of the following instructions would you give an OEC technician who is preparing to assess a stable 9-month-old boy with a rash?**

- A) "Keep the baby calm; do not touch him during the assessment."
- B) "Start at the head and slowly work your way to the feet."
- C) "Have the mother hold him as you do the assessment."
- D) "Make sure that you do not undress the baby for the assessment."

*Ans: C*

---

**31. You are treating a child who has a minor head laceration that is bleeding profusely. Which of the following statements would show your partner that you understand bleeding in children?**

- A) "This is a minor head laceration, so we don't need to worry."
- B) "Children have a smaller blood volume than adults, so this rate of bleeding is serious."
- C) "As long as the child's vital signs are stable, we don't have to worry."
- D) "Children have great compensating mechanisms, so bleeding from the head isn't serious."

*Ans: B*

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**32. Which of the following statements regarding the treatment of a pediatric patient in a prehospital setting is true?**

- A) You must determine the exact illness in order to provide the most appropriate treatment.
- B) If the primary assessment of a sick pediatric patient reveals no deficits to the ABCDs, the patient is stable and will not deteriorate.
- C) If a child looks sick and is not getting better with care, assume that the child is getting worse.
- D) You will get early warning if the child is going into shock because a child's vital signs change early.

*Ans: C*

---

**33. You are immobilizing a 4-year-old boy on a long spine board. Which of the following actions would be appropriate when performing this intervention?**

- A) Avoid applying the chest strap across the patient's thorax.
- B) Place padding between the patient's shoulders and the spine board.
- C) Secure the patient's chest and legs to the board after the head has been secured.
- D) Place a pillow under the patient's neck to keep the airway open.

*Ans: B*

---

**34. An infant who is short of breath is alert and has adequate respirations at a rate of 54 per minute. His skin color is pink but slightly cool to the touch. When you place a pediatric mask on his face, he becomes very upset and begins to physically struggle to remove it. In this situation you would:**

- A) gently restrain the infant's hands so that he cannot remove the mask.
- B) secure the mask to the infant's face using tape.
- C) allow the mother to hold the infant and then provide blow-by oxygen therapy.
- D) omit the oxygen for now and continue to assess the infant every 5 minutes.

*Ans: C*

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**35. You are called to assist a toddler who has just had a seizure that lasted about 60 seconds. His mom reports that the child has a history of seizures. Your assessment reveals that he is now responding to painful stimuli, is breathing normally, and has a normal radial pulse. Which of the following actions would you take?**

- A) Apply oxygen via nonrebreather at 15 LPM.
- B) Place him in a recovery position and continue to assess him.
- C) Gently try to arouse him and make him more alert.
- D) Insert an oral airway.

*Ans: B*

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## Section 31 — Geriatrics

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38 Questions

**1. You have been asked to deliver a talk about the geriatric population to the new OEC class. Which of the following points would you emphasize in your presentation?**

- A) Assessment can be difficult because most individuals older than 70 suffer from dementia.
- B) Many elderly individuals have a combination of different diseases in various stages.
- C) Geriatric patients actually account for a small number of emergency medical calls and transports.
- D) Dementia is an inevitable part of aging.

*Ans: B*

**2. A new OEC technician asks you why the elderly are at higher risk for developing pneumonia than younger individuals. You inform him that the elderly are more susceptible to respiratory infections because of:**

- A) impaired swallowing of saliva.
- B) decreased alveolar gas exchange.
- C) a reduced cough reflex.
- D) hardening of the major arteries.

*Ans: C*

**3. The son of an 88- year-old patient states that he told his mother to increase the amount of an antibiotic she was taking so she would "feel better faster." This represents a problem because the elderly have:**

- A) decreased respiratory function.
- B) an increased risk for stroke.
- C) decreased liver and kidney function.
- D) an increased risk for cancer.

*Ans: C*

**4. By age 85, the brain of a typical geriatric patient can shrink by as much as 10 percent due to:**

- A) reduced use in learning.
- B) a decrease in brain cell numbers.
- C) reduced fluid volume.
- D) stroke activity.

*Ans: B*

---

**5. In a typical geriatric patient, a reduction in cerebral blood flow can:**

- A) reduce the amount of glucose and oxygen that reaches the brain.
- B) increase the patient's pulse rate.
- C) decrease the patient's blood pressure.
- D) increase the patient's respiratory rate.

*Ans: A*

---

**6. Which of the following statements shows that an OEC technician has an accurate understanding of vital signs in relation to geriatric patients?**

- A) "The typical resting heart rate in geriatric patients is less than 60 beats per minute."
- B) "Fever tends to be more common and more severe in elderly patients."
- C) "There is no difference between the vital signs of a geriatric patient and those of a young adult."
- D) "Respiratory capacity is greatly reduced in geriatric patients."

*Ans: D*

---

**7. For many geriatric patients, food is less appealing because:**

- A) they engage in less physical activity.
- B) they sleep more and eat less.
- C) their sense of smell is reduced, and they have fewer taste buds.
- D) they have less money for purchasing appealing food.

*Ans: C*

---

**8. The lung capacity in a healthy 90-year-old patient is \_\_\_\_\_ that of a 30-year-old patient.**

- A) equal to
- B) three-quarters of
- C) one-quarter of
- D) half of

*Ans: D*

---

**9. Some elderly patients have osteoporosis, a condition that results in:**

- A) decreased bone density.
- B) decreased renal output.
- C) decreased insulin production.
- D) gall stone production.

*Ans: A*

---

**10. For the same individual, on average, blood pressure at age 30 compared to blood pressure to age 70 will be:**

- A) higher.
- B) lower.
- C) the same.
- D) unrelated.

*Ans: B*

---

**11. In geriatric patients, strokes can be caused by a blockage or by:**

- A) a buildup of dopamine.
- B) a buildup of epinephrine.
- C) a rupture of a cerebral blood vessel.
- D) trauma to the cortex.

*Ans: C*

---

**12. Family members tell you that three hours ago, their 76- year-old mother suddenly became confused and had great difficulty speaking. However, within 15 minutes, she returned to normal. Based on this description, an OEC technician should be suspicious of:**

- A) a stroke.
- B) dementia.
- C) Alzheimer's disease.
- D) a transient ischemic attack.

*Ans: D*

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**13. Which of these is a common cause of COPD?**

- A) Atherosclerosis
- B) Chronic bronchitis
- C) Eczema
- D) Allergies

*Ans: B*

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**14. Over 60% of abdominal pain problems in geriatric patients require:**

- A) a trip to the emergency department.
- B) a prescription for a narcotic drug.
- C) surgical intervention.
- D) evaluation for hernia.

*Ans: C*

---

**15. Many elderly patients participate in a practice called "polypharmacy," which is the:**

- A) use of mail-order pharmacies.
- B) simultaneous taking of multiple medications.
- C) use of generic drugs.
- D) use of designer drugs.

*Ans: B*

---

**16. To identify all the drugs, prescriptions, herbal supplements, and over-the-counter drugs that a patient may be taking in combination, OEC technicians should use:**

- A) SAMPLE.
- B) OPQRST.
- C) DCAP-BTLS.
- D) AEIOU.

*Ans: A*

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**17. The class of medications that is most commonly prescribed to elderly patients is:**

- A) gastrointestinal medications.
- B) respiratory medications.
- C) neurological medications.
- D) cardiovascular medications.

*Ans: D*

---

**18. One group of drugs prescribed for geriatric patients is beta- blockers. This type of drug:**

- A) manages cardiac arrhythmias.
- B) increases a patient's pulse rate.
- C) increases a patient's cardiac preload.
- D) thins a patient's blood.

*Ans: A*

---

**19. Geriatric patients may also have a prescription for a diuretic such as HCTZ or Lasix. The role of this type of drug is to:**

- A) increase a patient's heart rate.
- B) increase a patient's respiratory rate.
- C) decrease the volume of fluid circulating in the cardiovascular system.
- D) increase the preload of the atria.

*Ans: C*

---

**20. The vital signs of elderly patients with excessive internal or external bleeding may not provide an indication of shock:**

- A) if these patients are taking beta-blockers or calcium channel blockers.
- B) because the hearts of elderly patients pump less due to inactivity.
- C) because elderly patients have the ability to compensate for bleeding.
- D) if these patients are taking blood thinners.

*Ans: A*

---

**21. Compared to younger patients, trauma in geriatric patients results in:**

- A) less-serious injuries due to the slower pace in older patients.
- B) a higher mortality rate.
- C) more bleeding.
- D) more fractures of the extremities.

*Ans: B*

---

**22. Correcting a life-threatening condition such as external bleeding in elderly patients can be more problematic because:**

- A) the skin of elderly patients does not rebound as quickly as it does in younger patients.
- B) the veins of elderly patients have weaker walls than those in younger patients.
- C) elderly patients may be on Lasix.
- D) many elder patients are taking warfarin or other blood thinners.

*Ans: D*

---

**23. Which of the following statements about traumatic injuries in the elderly is true?**

- A) Motor-vehicle collisions are responsible for over 75 percent of deaths.
- B) Head injuries are easier to detect in the elderly than in younger adults.
- C) Hyperthermia is an early sign of trauma.
- D) Injuries are most commonly caused by falls.

*Ans: D*

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**24. Falls result in \_\_\_\_% of all deaths in the geriatric population.**

- A) 5
- B) 12
- C) 40
- D) 72

*Ans: B*

---

**25. Which of the following conditions must OEC technicians consider when evaluating hypotensive geriatric patients with altered mental status who show no signs of external bleeding, chest injury, or abdominal trauma?**

- A) A hip or pelvic fracture
- B) A tib/fib fracture
- C) A genetic hypotensive condition
- D) Dehydration

*Ans: A*

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**26. An elderly woman has fallen down three steps and is complaining of back pain. After you perform a primary assessment, which of the following questions should you ask to elicit the next important piece of information?**

- A) "Did you become dizzy before you fell?"
- B) "Why didn't you use the ramp on the other side?"
- C) "Do you have a history of high blood pressure?"
- D) "Who is your family doctor?"

*Ans: A*

---

**27. Compared to younger individuals, geriatric patients who fall have a higher incidence of fractures to the \_\_\_\_\_ region of the spine.**

- A) C4-C5
- B) T1-T2
- C) L1-L2
- D) C1-C2

*Ans: D*

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**28. Which of the following statements concerning insulin pumps is false?**

- A) They have an external on/off button.
- B) They are implanted in the patient's chest.
- C) They supply insulin via a catheter.
- D) They are commonly found on a patient's belt.

*Ans: B*

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**29. An advance directive is a:**

- A) court order directing prehospital providers to administer care.
- B) directive provided by medical control.
- C) protocol directed by a paramedic on scene.
- D) legal document that provides medical direction for life-saving efforts.

*Ans: D*

---

**30. A hearing-impaired patient wearing a hearing aid is having a great deal of difficulty hearing your questions. Which of the following actions should you take first?**

- A) Ensure that their hearing aid is turned on.
- B) Increase the pitch of your voice.
- C) Write your questions on paper.
- D) Position yourself in front of the patient and shout.

*Ans: A*

---

**31. When communicating with a geriatric patient, OEC technicians should:**

- A) use layman's terms.
- B) use medical terms to describe their findings and treatment plan.
- C) use anatomical pictures and point to the affected areas as they explain.
- D) touch the patient as they explain the patient's medical issues and affected body areas.

*Ans: A*

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**32. Which of the following strategies is best for OEC technicians for communicating with a geriatric patient who has diminished eyesight?**

- A) Write your questions in large print on paper.
- B) Increase both the pitch and the volume of your voice.
- C) Stand in front of the patient and talk calmly.
- D) Obtain needed information from family members.

*Ans: C*

---

**33. An effective way to interview a geriatric patient is to:**

- A) keep repeating your question if the patient is slow to respond.
- B) use good listening skills.
- C) write your questions down on paper and show them to the patient.
- D) offer possible responses to assist the patient before she answers.

*Ans: B*

---

**34. An elderly trauma patient may struggle with mental clarity, so OEC technicians should try not to:**

- A) ask about private matters.
- B) ask open-ended questions.
- C) enlist the help of family in assessment.
- D) look for evidence of abuse.

*Ans: B*

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**35. You and your partner respond to the lodge to aid an elderly woman who is not feeling well. When you perform a secondary assessment:**

- A) you and your partner should take turns asking the patient questions.
- B) you and your partner should ignore the patient and ask the family for information.
- C) you should stand to the patient's side and raise your voice so the patient can hear you.
- D) only one of you should ask the patient questions to avoid confusion.

*Ans: D*

---

**36. You have been called to an unknown medical emergency. On scene, you are presented with an 84-year-old patient who is confused and does not obey commands. Which of the following action would be most useful in determining the patient's normal mental status?**

- A) Examine the patient's pupils.
- B) Question the patient's family members.
- C) Identify the patient's medications.
- D) Check the patient's vital signs.

*Ans: B*

---

**37. When taking the pulse of an elderly patient, it is recommended that OEC technicians:**

- A) divert the patient's attention with conversation.
- B) repeat and document the pulse rate every 2 minutes.
- C) take the pulse on both sides and compare the two rates.
- D) use electronic devices only to obtain the pulse.

*Ans: C*

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**38. You have been called by the caregiver of a 91- year-old woman with dementia. The patient is complaining of a possible fracture to her upper left arm. As you talk to the caregiver and note various bruises on the patient's body, you become suspicious of elder abuse. Which of the following actions is most appropriate at this time?**

- A) Contact the police to report your suspicion.
- B) Confront the caregiver regarding the bruises on the patient's body.
- C) Ask the patient if she is being abused.
- D) Treat the patient for a possible broken arm.

*Ans: D*

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## Section 32 — Special Populations

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### 10 Questions

**1. You are assisting a 29-year-old spinal injury paraplegic skier who lost control of his sit ski and hit a tree. Even though there are no obvious signs of injury, the patient suddenly starts to feel panicky, and you become concerned about autonomic dysreflexia. Other signs/symptoms that would support this assessment include all of the following except:**

- A) hypotension.
- B) altered mental status.
- C) headache.
- D) blurred vision.

*Ans: A*

**2. A group of ski patrollers have invited you to join their team on a bike ride to raise money for multiple sclerosis (MS) research. You recall learning about MS in your OEC class, but you ask your friend to remind you of some of the key features of this condition. Which of the following statements made by your friend would be most accurate?**

- A) "MS does not affect vision."
- B) "MS does not affect mental capacity and memory."
- C) "MS is a progressive disease that causes degeneration of both central and peripheral nerves."
- D) "Speech and mood are generally not affected by MS."

*Ans: C*

**3. What is true about a patient with a cognitive disability?**

- A) A cognitive disability can result from traumatic brain injuries.
- B) Cognitive disabilities limit the person's ability to process information.
- C) Patients with cognitive disabilities usually have problems with coordination.
- D) All of these are correct.

*Ans: D*

**4. When caring for a patient with dyslexia, it is important to remember that:**

- A) all individuals with dyslexia have below-normal intelligence.
- B) the patient may have difficulty processing new information.
- C) you should not expect any impairment in communication.
- D) many dyslexic people have spasticity.

*Ans: B*

**5. You are teaching a class on adaptive athletes and are asked to discuss Asperger syndrome. Which of the following statements is an appropriate response?**

- A) "Asperger syndrome is a condition manifested by a standard group of symptoms, including physical impairments."
- B) "Asperger syndrome is an autism spectrum disorder in which the person may be fearful in unfamiliar environments."
- C) "Asperger syndrome implies that the person has no difficulty participating in normal day-to-day activities but may have difficulty with long-range planning."
- D) "Individuals with Asperger syndrome are less likely to come in contact with rescuers than are members of the general population."

*Ans: B*

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**6. You are caring for a hearing-impaired athlete and realize that in order to communicate, you:**

- A) must know sign language or call for someone who does.
- B) can usually speak loudly enough to be understood.
- C) will need paper and a pen because most hearing-impaired individuals cannot speak.
- D) should face the person because most hearingimpaired individuals can read lips.

*Ans: D*

---

**7. It's 2 p.m. on Saturday, and you're out for a bike ride with friends. About halfway down a small hill, you see an adult and a child at the side of the road. It appears that the child may have fallen. As you approach, you ask if you can help, to which you hear a mom's thankful "yes." The child, who is about 8 years old, is sitting quietly and does not respond when you say "hi." The mother explains that her son has atypical autism. Based on this information, which of the following techniques might you use to complete your assessment?**

- A) Ask the child open-ended questions.
- B) Speak to the child loudly to get his attention.
- C) Give the child simple, one-step directions.
- D) Speak only to the child.

*Ans: C*

---

**8. When caring for patients with life-long intellectual disabilities:**

- A) use a stern, unemotional tone of voice because they may not pay attention to you.
- B) include their caretakers in the assessment because they can help with communication.
- C) provide care quickly because they will not know they are cold.
- D) talk to the patient as if they were at a sixth-grade level.

*Ans: B*

---

**9. You are assisting a 32-year-old amputee whose right leg has been amputated below the knee. The injured skier is complaining of severe pain in her right thigh, which shows evidence of a fracture. After picking up a rigid splint, what should be your next thought?**

- A) "Further amputation will probably be required."
- B) "I should use a quick splint instead."
- C) "When was the last time they ate?"
- D) "What is their temperature?"

*Ans: B*

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**10. It is a cold, snowy Saturday afternoon, and, unfortunately, the chair lift has broken, and a lift evacuation is required. You know that there is an adaptive skier group somewhere on the lift. As you prioritize the lift evacuation, you recognize that:**

- A) athletes with mobility problems are at increased risk of cold-exposure injuries.
- B) someone will need to instruct paraplegic skiers on how to detach themselves from the sit-ski so they can be lifted out for the evacuation.
- C) a blind skier's guide should be evacuated before the blind skier so that the guide can direct the blind skier down.
- D) skiers with intellectual disabilities should be left to the end because they will take a long time to understand your directions.

*Ans: A*

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## Section 33 — Behavioral Emergencies

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20 Questions

**1. Your OEC class asks you to describe the term "behavior." Which of the following statements would be the correct response?**

- A) What is acceptable behavior in one community may not be acceptable in another.
- B) Behavior pertains to how you perform a physical activity only.
- C) Meditation is the absence of behavior.
- D) Yelling is never an acceptable behavior.

Ans: A

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**2. Your first priority in managing a patient with a behavioral emergency is:**

- A) determining if the patient is a threat to himself or herself.
- B) providing the patient high- concentration oxygen.
- C) ensuring your own personal safety.
- D) assessing and managing the patient's airway.

Ans: C

---

**3. Which of the following statements about behavioral emergencies is true?**

- A) The person acts in a way that is not tolerable to those around them.
- B) The person's mental status is not altered.
- C) The person is not a danger to themselves or others.
- D) The context of the situation is not relevant.

Ans: A

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**4. Which of the following patients should an OEC technician consider a potential behavioral emergency?**

- A) A 44-year-old woman with bipolar disorder who is complaining of a fever of 99.5°F
- B) A 56-year-old man with schizophrenia who has continued taking his medications
- C) A 29-year-old man with diabetes who is combative and refusing care
- D) A 36-year-old man with a known history who tells you he is not hearing voices

Ans: C

---

**5. Two of the most common medical causes of abnormal behavior are:**

- A) exercise and hypoglycemia.
- B) hypotension and hyperthermia.
- C) hypoxia and hypoglycemia.
- D) carbon monoxide poisoning and acute myocardial infarction.

Ans: C

---

**6. Which of the following patients is exhibiting signs of psychosis?**

- A) An 18-year-old who is so anxious about peer pressure that he is unable to go to school
- B) A recently divorced 35-year-old who sleeps most of the day and is losing weight
- C) A 72-year-old who was recently diagnosed with dementia
- D) A 37-year-old who is having hallucinations

*Ans: D*

---

**7. A condition in which a person exhibits abnormal behavior but is able to function within the normal boundaries of reality is known as:**

- A) a psychosis.
- B) Schizophrenia.
- C) a mood adjustment or affective disorder.
- D) a behavioral emergency.

*Ans: C*

---

**8. You have stopped a male skier and a female skier who were racing down a closed trail. Initially, the man was calm, but suddenly he becomes angered and says that "you can't tell me what to do." When you try to talk with him, he pulls out a knife and tells you to "get away or I'll kill you." The female skier is frightened and states that her friend sometimes gets very aggressive when he drinks. You put in a call for security and additional staff, but you know it will be a few minutes before they arrive. Which of the following statements describes your best course of action in the interim?**

- A) Immediately attempt to physically restrain the man or take away his knife.
- B) Approach the man slowly and calmly and ask him to hand you the knife.
- C) Move quickly to pull the woman a safe distance away from the man.
- D) Remain a safe distance from the man and let him know you are there to help.

*Ans: D*

---

**9. You are describing techniques for remaining safe when responding to a call for a patient with a behavioral emergency. Which of the following statements is appropriate?**

- A) "You need to encounter the patient first, and then come up with your safety plan."
- B) "If a patient raises his/her voice, you should raise yours to match it. "
- C) "You should approach patients quietly and try to surprise them so they don't have time to develop a plan to attack you."
- D) "You should be sure that you have at least one escape route, and do not let the patient get between you and that way out."

*Ans: D*

---

**10. Which of the following statements shows that an OEC technician has a good understanding of assessing a patient who is having a behavioral emergency?**

- A) "Try to assess patients from a safe distance until they start to answer your questions calmly."
- B) "I avoid making eye contact because patients may perceive eye contact as a threatening gesture and respond with violence."
- C) "Touch conveys concern for a patient, so I try to touch the patient's shoulder or arm as soon as I arrive at the scene."
- D) "If a patient raises his voice to you, you must raise your voice in return. Research has shown that this helps deter violence."

*Ans: A*

---

**11. Which of the following behaviors is a sign of potential aggression in a patient?**

- A) Avoiding eye contact
- B) The clenching of both fists
- C) Refusal to answer some questions
- D) Increased sighing

*Ans: B*

---

**12. Which of the following pieces of information is most important to an OEC technician in assessing a patient with abnormal behavior?**

- A) What the patient's diagnosis is
- B) Whether or not the patient is dangerous
- C) Whether the patient is anxious or depressed
- D) What medications the patient takes

*Ans: B*

---

**13. Which of the following actions is best to keep a behavioral patient calm and to avoid provoking a violent outburst?**

- A) Inform the patient of everything that is happening and use a calm voice throughout care.
- B) Establish that you are in control of the situation and that the patient must cooperate, or you will restrain him.
- C) Inform the patient that he is responsible for all of his actions, and that if he hurts you, you will sue him.
- D) Involve as many people as possible in the situation as a "show of force."

*Ans: A*

---

**14. You and a fellow OEC technician are discussing how to talk with someone experiencing a behavioral emergency. Which of the following statements indicates that you have a good understanding of the correct approach?**

- A) "If the patient is hallucinating, you should go along with them so they won't get upset."
- B) "Don't encourage patients to talk about what is bothering them. They should only talk with a psychiatrist about that."
- C) "Demonstrate active listening skills by repeating or paraphrasing what the patient said."
- D) "If a patient yells, you should yell back so he will know you are not afraid."

*Ans: C*

---

**15. You are called to the lodge to assist a 25-yearold man who reportedly is having hallucinations. When you arrive, he is sitting quietly and appears calm. Information obtained in your SAMPLE history indicates that the patient started having hallucinations yesterday. He was diagnosed with schizophrenia a few years ago and has been prescribed two different medications. Which of the following questions is most pertinent to the situation and should be asked next?**

- A) "Do you believe that the hallucinations are real?"
- B) "Have you been taking your medications as prescribed?"
- C) "Have you been eating and drinking adequately?"
- D) "Has your doctor considered increasing the dose of your medication?"

*Ans: B*

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**16. Which of the following situations is an indication for the use of restraints?**

- A) The patient is in imminent danger of harming himself.
- B) The patient is alert, rational, and refuses medical care.
- C) The patient's friend is requesting restraints.
- D) The patient responds to the use of verbal calming skills

*Ans: A*

---

**17. When treating a patient exhibiting abnormal behavior, you should:**

- A) assume until proven otherwise that the source of the abnormal behavior is an underlying medical condition.
- B) restrain the patient as a precaution.
- C) assist the patient in taking glucose in case the patient is hyperglycemic.
- D) assume that the abnormal behavior is a manifestation of an underlying psychological problem.

*Ans: A*

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**18. Which of the following statements regarding the use of physical restraint is true?**

- A) An OEC technician must follow local protocols for restraining patients.
- B) Law enforcement personnel can restrain anyone for any reason.
- C) Once you decide to restrain a patient, use the maximum force possible.
- D) Physicians have unlimited latitude in ordering the physical restraint of a patient.

*Ans: A*

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**19. You are out with a group of friends in a local tavern. One of the patrons gets argumentative and will not calm down. By the time the police arrive, the patron has gotten combative and is breaking glass bottles. The police subdue the patron and have him restrained. As you watch all of this happening, which of the following situations would cause you to be concerned?**

- A) The police have restrained the patron in a prone position on the floor.
- B) The patron is supine on the floor, with officers holding one of his hands over his head and the other at his side.
- C) The patron is threatening to sue the officers for physically restraining him.
- D) The police say they will not release the patron until he has been seen in an emergency department.

*Ans: A*

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**20. When mechanically restraining a patient, you should:**

- A) tell the patient you will release the restraints as soon as he calms down.
- B) speak as harshly as necessary to get the patient to cooperate.
- C) use a prone position when possible to ensure an open airway.
- D) use a method consistent with your local protocols.

*Ans: D*

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## Section 34 — Special Operations

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20 Questions

**1. Special operations is a term used to denote:**

- A) infrequently performed activities that require specialized training, skills, and equipment in a remote setting.
- B) protocols OEC technicians use when working with an ambulance service in an urban setting.
- C) a federal team that is organized to handle mass casualty situations involving mountainous terrain.
- D) a specialized procedure that is used for victims a of chemical terrorism event.

*Ans: A*

**2. You overhear a group of EMTs discussing the concept of driving an ambulance with due regard for the safety of others. Which of the following statements indicates that the EMT understands the concept?**

- A) "When responding to an emergency, I drive as fast as possible so that emergency care can be delivered as quickly as possible."
- B) "When taking a patient to the hospital, I drive as quickly as possible so that I can get back into service for another call."
- C) "When approaching an intersection, I wait for all traffic to stop before proceeding through."
- D) "It is acceptable to take a nonemergency patient to the hospital using lights and siren if dispatch informs me that there is another call waiting."

*Ans: C*

**3. Which of the following statements regarding an ambulance parked at an incident is true?**

- A) If a law enforcement vehicle is on the scene, the ambulance should be parked right beside it.
- B) When walking from the scene to the parked ambulance, you should walk on the side of the road closest to traffic so you can be more easily seen.
- C) Ambulance warning lights should be turned off so they will not blind oncoming drivers.
- D) The area around the ambulance and the accident scene is known as the ambulance operation zone.

*Ans: D*

**4. Which of the following tasks is not a task an OEC technician should be ready to perform in assisting an EMT in the back of an ambulance?**

- A) Medication administration
- B) Putting a cervical collar on a patient
- C) Hemorrhage control
- D) Splinting

*Ans: A*

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**5. You and your friend arrive at an accident scene where a compact car has rolled down an embankment. Based on your OEC training, you realize that before anyone enters the vehicle, it should be stabilized. Vehicle stabilization includes all of the following actions except:**

- A) shutting off the engine.
- B) having several men hold the vehicle to prevent it from moving.
- C) disconnecting the battery cables.
- D) engaging the parking brake.

*Ans: B*

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**6. Which organization helps create Community Emergency Response Teams?**

- A) The Federal Emergency Management Agency
- B) The National Incident Management System
- C) The Incident Command System
- D) The Medical Reserve Corps

*Ans: C*

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**7. Organized under the DHHS, community-based volunteer medical professionals who have agreed to help in the event of a disaster are known as the:**

- A) National Response Framework.
- B) Emergency Support Annexes.
- C) Medical Reserve Corps.
- D) Disaster Medical Assistance Team.

*Ans: C*

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**8. HAZWOPER training is required by the federal government for anyone who works in an environment in which uncontrolled hazardous materials may be encountered. Which is the correct word for a letter in HAZWOPER?**

- A) W: water
- B) E: energy
- C) R: rescue
- D) O: operations

*Ans: D*

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**9. Which of the following statements regarding the NFPA safety diamond is true?**

- A) The NFPA safety diamond tells you whether the fuel in a fire is paper, liquid, gas, or wood.
- B) The NFPA safety diamond must be displayed on any building or vehicle that contains hazardous materials.
- C) The number of NFPA safety diamonds on a vehicle indicates the toxicity of the hazardous material.
- D) The NFPA safety diamond identifies the known safety hazards of a given material.

*Ans: D*

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**10. Which of the following statements concerning HAZWOPER training is false?**

- A) Training includes the types of hazardous materials present and their risks to the rescuers and the public.
- B) All EMS personnel are trained in all aspects of a HazMat-related incident.
- C) Training generally includes classroom instruction and practical training.
- D) Annual refresher training is generally required.

*Ans: B*

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**11. In a hazmat incident, a boundary that is set up to prevent members of the public from entering the incident is called a(n):**

- A) incident perimeter.
- B) isolation perimeter.
- C) decontamination line.
- D) hazard boundary line.

*Ans: B*

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**12. The contaminated area in a hazardous incident is known as the:**

- A) contaminated zone.
- B) decontamination zone.
- C) hot zone.
- D) cold zone.

*Ans: C*

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**13. In a HazMat incident, the transition area in which decontamination occurs is known as the:**

- A) warm zone.
- B) treatment area.
- C) transport zone.
- D) cold zone.

*Ans: A*

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**14. You are instructing an OEC class on the handling of a hazmat incident. You are asked where OEC technicians would be assigned if they were assisting at a hazmat scene. Which of the following statements is the appropriate response?**

- A) "Because of your training in this class, you could be assigned to any of the hazard control zones."
- B) "OEC technicians do not have the skills required to assist in a HazMat incident."
- C) "Because of your OEC training, you would probably be assigned to the cold zone."
- D) "OEC technicians could assist inside of the isolation perimeter."

*Ans: C*

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**15. Toxic nerve agents:**

- A) overstimulate the nerve endings and the central nervous system.
- B) block stimulation of sensory nerve endings only.
- C) decrease production of chemical transmitters.
- D) block stimulation of both sensory and motor nerve endings.

*Ans: A*

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**16. In the event that OEC technicians are exposed to an organophosphate or a chemical nerve agent, they may need to self-administer the nerve agent antidote. Which of the following statements regarding the self-administration of a nerve agent antidote is false?**

- A) "If severe symptoms are present, three atropine auto-injectors and three 2-PAM CI injectors are administered in rapid succession."
- B) "The medication is administered in a way that is similar to that for an Epi-Pen."
- C) "If the nerve agent has been ingested, exposure may continue for some time, and a relapse is a possibility."
- D) "Depending on the symptoms observed, the initial treatment will either be an atropine autoinjector or a 2-PAM CI injector."

*Ans: D*

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**17. Which of the following activities is not considered a special operation?**

- A) A vehicle extrication
- B) Toboggan transportation
- C) A low-angle rescue
- D) A lift evacuation

*Ans: B*

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**18. Search and rescue operations consist of which of the following five basic tasks?**

- A) Information gathering, grid assignments, search, medical treatment, and evacuation
- B) Mobilization, intelligence gathering, containment, search, and rescue
- C) Notification of rescue personnel, perimeter marking, grid assignments, search & rescue, and reporting
- D) Incident Commander assigned, team leaders and teams assigned, grids determined, search & rescue, and reporting

*Ans: B*

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**19. Logistics, one of the four basic functions of an organized avalanche rescue, is primarily concerned with:**

- A) getting the rescuers to the scene.
- B) arranging for extrication of the victims.
- C) caring for the rescuers in the field.
- D) keeping a log of the search's activities.

*Ans: C*

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**20. Which of the following lists correctly places the tasks of fire ground operations in order of priority?**

- A) Size-up, confinement, control, ventilation, rescue/life safety, accountability, property conservation
- B) Confinement, size-up, control, property conservation, rescue/life safety, accountability, ventilation
- C) Size-up, accountability, rescue/life safety, confinement, control, ventilation, property conservation
- D) Accountability, rescue/life safety, confinement, control, size-up, property conservation, ventilation

*Ans: C*

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